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Modeling the dynamic soil-structure interaction for large civil engineering structures

Impact of the wave regime on the global response

Non-confidential report

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Résumé

Le développement des méthodes numériques ainsi que des ordinateurs au cours des deux dernières décennies ont permis aux chercheurs de se confronter à des problèmes complexes dans l'étude de l'interaction sol-structure (ISS). Dans ce contexte, des études récentes ont mis en lumière l'importance de modéliser un régime d'ondes 3D et complet pour déterminer la réponse d'une structure, par opposition à la pratique courante qui décompose l'excitation sismique en des ondes de volume d'incidence verticale. Dans ce cadre, l'Agence Nationale pour la Recherche finance un projet nommé IM-SURF dont l'objectif est de déterminer la réponse d'une structure soumise à des ondes de surface.

Le but de ce projet de fin d'études (PFE) est d'implémenter une chaîne de calcul basée sur les méthodes numériques développées au cours des cinquante dernières années, afin de résoudre des problèmes d'ISS non-linéaires de manière efficace. Cette méthodologie permet de mener des études paramétriques sur des problèmes réalistes d'ISS avec un champ d'onde complet. En particulier, nous chercherons à quantifier l'interaction cinématique vis-à-vis du mouvement de tangage et à fournir une interprétation de ce phénomène physique.

Nous avons ainsi développé une méthodologie basée sur la Thin Layer Method (TLM) et sur la Domain Reduction Method (DRM) pour résoudre des problèmes d'ISS non-linéaires complexes. La TLM permet de calculer la réponse en champ libre du domaine de sol. Après convolution avec un signal de référence, nous extrayons la réponse aux nœuds de la couche de la DRM et injectons ce mouvement dans un modèle aux éléments finis. Grâce à l'association de ces deux méthodes, nous sommes capables de calculer la réponse d'une structure, avec un régime d'onde complet, y compris en la présence de fortes non-linéarités. Pour démontrer l'efficacité de notre méthodologie, nous étudions en profondeur le mouvement de tangage causé par des ondes de Rayleigh et par des ondes *SV* d'incidence verticale pour des structures idéales superficielles et enterrées. Nous étudions la réponse en fonction d'un rapport caractéristique (la longueur d'onde divisée par la longueur de la structure) et de l'enfoncement relatif. Nous montrons qu'il existe deux régimes pour la réponse : transmission de la rotation à la structure depuis l'excitation sismique et atténuation par interaction cinématique. Ensuite, nous comparons les contributions relatives de chaque type d'onde au mouvement de tangage. Nous montrons que les rotations induites par les ondes de Rayleigh ne sont pas négligeables *a priori*. Nous menons enfin une étude simple du décollement d'une structure.

Mots-clés : Onde de surface, ISS non-linéaire, mouvement de tangage, structures enterrées, modélisation aux éléments finis

Abstract

The development of numerical methods and hardware in the past two decades has allowed researchers to address complex problems in the study of soil-structure interaction (SSI). In this context, recent studies have demonstrated the importance of modeling the full 3-D wave regime to determine the response of a structure, in opposition to the current practice of decomposing the seismic excitation in equivalent vertically incident body waves. As a part of this research effort, the French research agency (ANR) is funding a project named IM-SURF whose objective is to assess the response of a structure subjected to surface waves.

The goal of this internship is to implement a calculation methodology, based on the numerical methods developed over the past fifty years, to solve realistic SSI problems in an efficient way. Using this methodology, one may study the response of a structure subjected to a full wave regime with a parametric study. In particular, in this internship, we seek to provide a quantification of the kinematic interaction in terms of the rocking motion and a clear interpretation of this physical phenomenon.

We developed a methodology based on the Thin Layer Method (TLM) and the Domain Reduction Method (DRM) to solve complex nonlinear SSI problems. The TLM allows one to compute the response of the soil domain in the free-field, thanks to an analysis in the frequency domain. Then, after convolution with a reference signal, we extract the response at the DRM nodes and inject the movement into a finite element model. Thanks to the association of these two methods, we are able to compute the response of a structure, with a full 3-D wave regime, even in the presence of strong nonlinearities. To demonstrate the comprehensiveness of our methodology, we study in depth the rocking motion (i.e. the rotation around the axis orthogonal to the propagation plane), caused by Rayleigh waves and vertically incident *SV* waves on superficial and embedded ideal structures. We study the response as a function of a characteristic ratio (the wavelength divided by the length of the structure) and the relative embedment. We show that there are two regimes for the response: transmission of the rotation to the structure by the seismic excitation and attenuation by kinematic interaction. Then, we compare the contribution of each wave type to the rocking motion. We show that the rocking motion caused by Rayleigh waves is not negligible *a priori*. We also provide a simple case study of the uplifting of a structure.

Keywords: Surface waves, nonlinear SSI, rocking motion, embedded structures, finite element method

Table of contents

Table of contents

List of tables

List of figures

List of acronyms

ANR : Agence Nationale de la Recherche

BEM : Boundary Element Method

BRGM : Bureau de Recherches Géologiques et Minières

DRM : Domain Reduction Method

ENPC : École Nationale des Ponts et Chaussées

ENSTA : École Nationale des Sciences et Techniques Appliquées

FE : Finite Element

FEA : Finite Element Analysis

FEM : Finite Element Method

FT : Fourier Transform

GCC : Génie Civil et Construction

GDS : Géodynamique & Structure

IAEA : International Atomic Energy Agency

PFE : Projet de Fin d'Études

SSI : Soil Structure Interaction

TLM: Thin Layer Method

USGS: United States Geological Survey

List of appendices

1. Introduction

1.1. Scientific and technical context

The design of large civil engineering structures, which are subject to strict safety requirements, such as nuclear installations, must take into account the effect of potential earthquakes. To accurately design these structures, one needs to solve the dynamic soil structure interaction (SSI) problem, namely assessing the way the structure interacts with its foundation domain during the seismic excitation. This problem is complex because the soil, the nature of the earthquake and the structure all participate in it and because these three components all interact with one another. Even though this study must be done for each combination of structure, foundation and soil, the general problem of SSI has broader interests. Indeed, understanding the salient features of dynamic SSI from a theoretical point of view, allows one to provide guidelines and implement regulations (in Europe, Eurocode 8 is the standard reference for conventional structures). To tackle the SSI problem, the ordinary approach is to focus on the effect of body waves (*P* waves and *S* waves). However, surface waves, which are generated by the reflection of body waves at the free surface, and/or at interfaces between soil layers, may also act onto the structure, not only in the form of imposed translations but also in the form of imposed rotations. This latter aspect has been hardly studied from a theoretical perspective and thus has led to few implementations in engineering guidelines. In this context, the project IM-SURF, funded by the French Research Agency (ANR: Agence Nationale pour la Recherche), aims to study the influence of the wave regime on the SSI problem and in particular, the effect of surface waves on structural response. This project spans over four years and started in December 2024. It covers many aspects from the geological scale to the structural scale. Several research partners participate in this project : GDS (coordinator), the École Centrale-Supélec, the Bureau de Recherches Géologiques et Minières (BRGM), the École Nationale des Sciences et Techniques Appliquées (ENSTA) and UC Davis. The scope of this internship (PFE: projet de fin d’études) is part of the ANR research project IM-SURF.

1.2. Problem and objectives

Solving the SSI problem and determining the response of a structure may be understood in two different ways. First, from a theoretical perspective, solving the SSI problem would mean assessing the impact of each parameter (e.g. wave regime, soil constitutive law…) and explain the physical phenomena at play. Second, from an engineering perspective, the goal is to estimate the response of a structure with a given set of parameters. This implies implementing efficient numerical methods or providing general guidelines. However, as mentioned previously, the SSI problem is very broad. Therefore, we do not seek to cover in depth all the aspects of this problem. We rather try to develop a comprehensive methodology that allows one to study every aspect of the real-life problem. In other words, we seek to implement a methodology that allows one to gather hindsight on the physical phenomena and then regroup these results to provide guidelines for the engineering practice. To this end, parametric studies are the major tool used in uniting the two perspectives as they allow one to understand the impact of a given parameter and also to draw abacuses. With such a methodology, we hope to pave the way for a more comprehensive approach to SSI problems that account for all the physical phenomena at play, in real-life conditions.

In order to demonstrate the efficiency and the flexibility of the calculation methodology, we tackle one aspect of the kinematic interaction problem under seismic solicitation. When a wave propagates in a soil domain and encounters a structure, the presence of the structure disrupts the wave propagation. In turn, the structure undergoes a certain movement. The impact of body waves on the translational components of the structure is thoroughly understood. However, the impact of body waves and surface waves on the rotation component (more precisely the "rocking motion") has been less investigated. Thus, we address the problem of determining the kinematic interaction for a structure subjected to a full wave regime with two objectives in mind: giving a coherent physical explanation of this phenomenon and providing easy-to-use engineering guidelines.

We use previous works and discussions as a starting point. The methodology guideline for SSI, published by the International Atomic Energy Agency (IAEA), which we may also call the "Tecdoc" for simplicity, predicts qualitatively different behavior of the response depending on the characteristics of the surface wave and of the structure (IAEA, 2022). When the wavelength varies, the foundation enters different "regimes" as an effect of the kinematic interaction. We will try to verify these predictions and to quantify the rotations caused by surface waves. In the common practice, rotations may be neglected if they are less than approximately 1 mrad, in other words if the vertical displacement of one end is less than 10 cm over 100 m, although this value strongly depends on the context. Thus, this first step of the study will determine if the rotations caused by surface waves are negligible during the design of structures. To achieve this, we lead a parametric study on the characteristics of the structure and the seismic wave, using the calculation methodology we implemented. The idea is to run multiple numerical experiments and to interpret the results to better understand how SSI problems work with surface waves. Numerical experiments rely on the use of the finite element method (FEM).

From this starting point, we then generalize our results to the problem of kinematic interaction in general. To this end, we introduce canonical curves for the effects of pure elastic kinematic interaction. We then repeat the same analysis for vertical SV waves and compare the results with Rayleigh (R) waves. As a last demonstration of the versatility of our methodology, we also present some nonlinear calculations. The fruitful discussions that will arise from all these results we obtain will prove the relevance and the broad extension of our methodology.

To summarize, this PFE comprises two important steps:

a. The implementation of a complete calculation methodology allowing for the consideration of full wave regimes including surface wave components for SSI calculations.

b. The application of this calculation methodology to the problem of kinematic interaction. Thanks to these calculations we are able to determine the response of a rigid massless raft in terms of the rocking motion. This step also comprises the understanding and the interpretation of the "Tecdoc" predictions in view of the obtained results for R waves.

1.3. Outline

In this report, we will first present the theoretical notions that are used in the development of our numerical models. On the one hand, they deal with the propagation of seismic waves; on the other hand, with the numerical methods used to simulate the SSI problem. In this first section, we will present the theory of Rayleigh waves and the numerical method called Domain Reduction

Method (DRM). Then, we will explain the methodology of our calculations and how this calculation approach is implemented. To validate this calculation approach, we will also compare the results we get in simple cases with the analytical ones from the theoretical equations. We will also present the methodology to lead parametric studies. Next, we will study the kinematic interaction for Rayleigh waves. We will interpret the results we obtain and compare them with the predictions of the "Tecdoc". Finally, we will study the kinematic interaction for SV waves.

In the appendices are given several details on results used in this report. Appendix 1 is also dedicated to the styling convention used for scalar, vectors and tensors and to the system of axes used.

2. Seismic wave modeling: from the geological scale to the structural scale

In this section, after recalling some general aspects of seismic wave propagation, we will establish the equations for the Rayleigh wave, which will later be a tool to validate our calculation methodology. In an effort to keep this report concise, we choose to only derive the equations for the Rayleigh waves. The derivation of the equations for body waves may be found in numerous textbooks. Then, we will explain how to transit from the geological scale to the geotechnical scale to model the SSI problem.

2.1. General aspects of seismic wave propagation

2.1.1. Presentation of the different scales

When an earthquake occurs, there exists different scales to study the propagation.

The *geological scale* embraces the propagation in the whole soil domain, from the fault to the free surface. The geological scale may span over hundreds of kilometers. Thus, it is not suited for finite element analysis (FEA) and for the determination of stresses and strains in a structure. On the contrary, it focuses on the propagation depending on the geological context (sedimentary basin, hard soil…).

The *geotechnical scale* focuses on an area close to the structure (less than a kilometer around it). It comprises the major features of the structure, the foundations and the soil in a nearby region. This scale is suited for the FEA, even though it requires some special techniques to address the fact that the soil is an unbounded domain. We study the SSI problem at this scale.

The *structural scale* allows one to study the stresses inside the structural elements (beams, floorings…). This scale is also suited for a FEA, but it lies outside the scope of this report.

2.1.2. Equation of propagation of seismic wave

Throughout the PFE, we assume that all materials are isotropic. At the geological scale, we also assume that the soil is linear elastic. We use the summation convention on repeated indices. We also use the notation $(x_1, x_2, x_3) = (x, y, z)$ to simplify the expression of partial derivatives.

Under these hypotheses, the relation between the Cauchy stress tensor $\boldsymbol{\sigma}$ and the displacement field $\mathbf{u}$ is the following:

$$\boldsymbol{\sigma} = \lambda\, \mathrm{tr}(\boldsymbol{\epsilon}) I + 2\mu \boldsymbol{\epsilon} \tag{2-1}$$

Where λ and μ are the two Lamé coefficients, $\boldsymbol{\epsilon}$ is the strain tensor (equal to the symmetric part of the gradient of $\mathbf{u}$) and $\boldsymbol{I}$ is the identity tensor. This equation is called Hooke’s law.

By writing the equilibrium between the different forces, we formulate the equation of elastodynamics:

$$\frac{\partial \sigma_{ij}}{\partial x_j} + \rho b_i = \rho \frac{\partial^2 u_i}{\partial t^2} \tag{2-2}$$

Where (b_i) is the vector of body forces. We assume it is zero in the following.

To establish the wave propagation equation, we follow Semblat and Pecker (2009) and Andersen (2006).

We introduce two tensors:

- The volumetric strain: $\Delta(\mathbf{x}, t) = \frac{\partial u_k(\mathbf{x},t)}{\partial x_k}$
- The infinitesimal rotation tensor: $W_{ij}(\mathbf{x}, t) = \frac{1}{2}\left(\frac{\partial u_i(\mathbf{x},t)}{\partial x_j} - \frac{\partial u_j(\mathbf{x},t)}{\partial x_i}\right)$

In the following, we drop the $(\mathbf{x}, t)$ when there is no ambiguity.

By taking the divergence of the equation of equilibrium, we get the equation of propagation for dilatation waves:

$$(\lambda + 2\mu)\frac{\partial^2 \Delta}{\partial x_i \partial x_i} = \rho \frac{\partial^2 \Delta}{\partial t^2} \tag{2-3}$$

These waves propagate at speed $c_P = \sqrt{\frac{\lambda+2\mu}{\rho}}$ and are called primary (*P*) waves.

We write $w_i = \frac{1}{2}\epsilon_{ijk}W_{kj}$ where ϵ_{ijk} is the Levi-Civita permutation symbol. By applying the rotational operator to the equation of equilibrium, we obtain the equation of propagation for shear waves:

$$\frac{\partial^2 w_i}{\partial x_j \partial x_j} = \frac{\rho}{\mu}\frac{\partial^2 w_i}{\partial t^2} \tag{2-4}$$

Shear waves propagate at velocity $c_S = \sqrt{\frac{\mu}{\rho}}$. These waves are called secondary (*S*) waves. Secondary waves can be split into *SH* waves (horizontal propagation plane) and *SV* waves (vertical propagation plane).

Thus, we always have the following inequality: $c_S < c_P$.

N.B.: The vertical propagation plane is defined by a vertical vector and the wavenumber vector of body waves. The horizontal propagation plane is defined by the wavenumber vector of body waves and a horizontal vector orthogonal to the vertical propagation plane (off-plane vector in the case where the problem is invariant in the third direction, which is the case in our study). The propagation plane is represented on Figure 1. When the problem has two vertical planes of symmetry and when the body waves have a vertical incidence, there is no difference between SH and SV waves. However, in general, SH waves and SV waves are decoupled. SH waves are correlated to L waves (Love waves), while SV are correlated to R waves. Since R waves and SV waves always have the same propagation plane, we will keep this convention and call SV wave a vertically incident S wave (which generates no vertical displacements) when it has the same propagation plane as the Rayleigh waves we study.

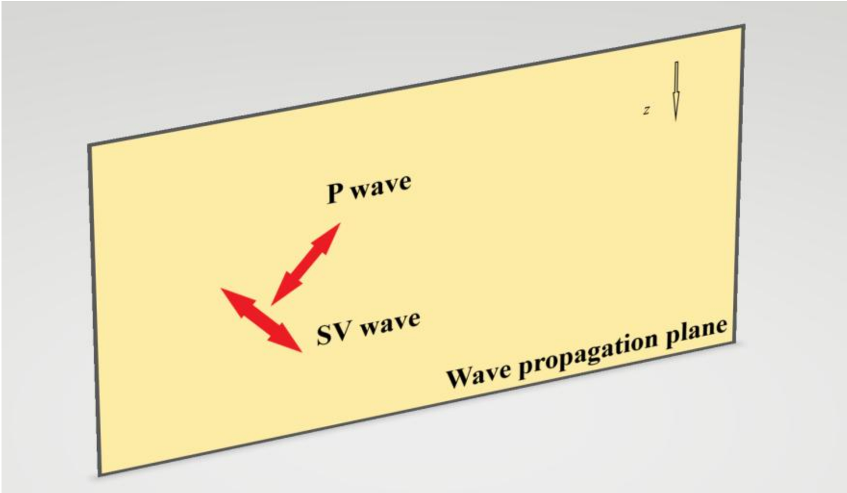

Figure 1 : Propagation plane for body waves.

2.1.3. Physical phenomena at play and surface waves

Just like in any other wave propagation problem, we may define the impedance matrix $\boldsymbol{Z}$ to characterize the properties of the soil domain. Let Σ be a surface of the soil domain on which the vector of tractions $\mathbf{p}$ is applied. The impedance matrix connects the vector of displacements $\mathbf{v}$ to the tractions $\mathbf{p}$. Formally, we may write:

$$\mathbf{p}(\mathbf{x}, t) = \boldsymbol{Z}.\mathbf{v}(\mathbf{x}, t) \quad \mathbf{x} \in \Sigma \tag{2-5}$$

The coefficients of the impedance matrix depend on the soil characteristics (mass, stiffness and damping). Therefore, when the mass, the stiffness or the damping changes at an interface, the coefficients of $\boldsymbol{Z}$ also change.

When an incident wave encounters a change in $\boldsymbol{Z}$ two phenomena happen:

- Refraction of the incident wave.
- Reflection of the incident wave.

The angles of the incident, refracted and reflected waves are linked by Snell-Descartes laws.

Thus, for a stratified soil, refraction and reflection occur at each interface between the soil layers. This is also true when body waves reach the free surface. In particular, surface waves are generated by body waves when they get reflected at the free surface or at the interface between two layers. The two main types of surface waves are Love waves (generated by *SH* waves) and Rayleigh waves (generated by *P* and *SV* waves). In this report, we focus on the effects of Rayleigh waves.

The importance of these considerations will be made clearer later, when we establish the FEM model.

The propagation of seismic waves is shown schematically on Figure 2.

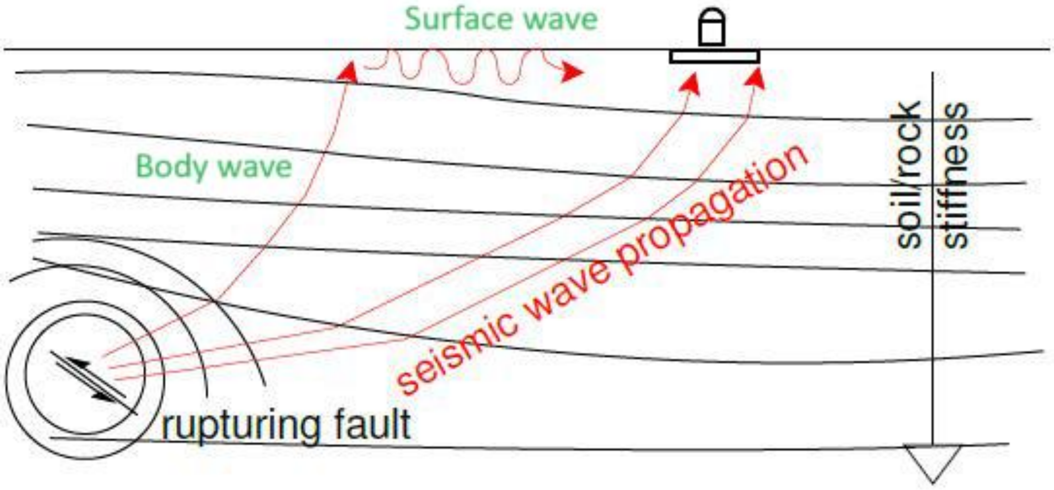

Figure 2 : Propagation of seismic waves- Figure adapted from IAEA (2022).

2.1.4. Influence on the structure

In a general manner, the SSI can be split into two components:

- Kinematic interaction: the presence of foundation elements within the soil creates a contrast in rigidity and impacts the wave regime in the near field zone.
- Inertial interaction: the wave regime induces a vibration in the structure (governed by the mass, stiffness and damping characteristics of the structure) which in turn generates stresses on the foundations.

In the following, we focus on kinematic interaction.

Let λ_R be the wavelength of the surface wave and L_f the length of the foundation. The "Tecdoc" predicts the response of a superficial structure subjected to a surface wave depending on the ratio ${\lambda_R}/{L_f}$. We quote paragraph 7.3.3.2 from the "Tecdoc":

"Most of the time, vertical motions are a result of Rayleigh surface waves; therefore, it is important to analyse vertical motions and decide if modelling as 1C is appropriate. To this end, the wavelength of surface waves plays an important role. If the Rayleigh surface wavelength (which features both horizontal and vertical components) is longer than 12 times the dimension of the object, then object rotations due to differential vertical displacements at object ends are fairly small and the object moves up and down as if excited with a vertical wave. This is shown in [Figure 3], as the upper case. In this case, it is appropriate to use 3x1-D modelling even with nonlinear/inelastic models. On the other hand, if the wave is shorter than 12 object dimensions, then vertical motions are gradually replaced by object rotations, while vertical motions are reduced. The lower left corner of [Figure 3] shows a limiting case where the seismic wave is 4 times longer than object dimension, which results in minimal vertical motions of the object, and maximum rotations, due to differential motions of object ends. For shorter surface waves, as shown in the lower right of [Figure 3], waves might not excite any significant dynamic behaviour of the object

(except local deformation) as their wavelengths are shorter than twice the object length."

In paragraph 7.3.3.2, the "Tecdoc" aims to compare the different modelling techniques for wave regimes, namely 3C, 3x1C and 1C (where C stands for component). In comparison, our objectives are slightly different: we aim to accurately assess the kinematic interaction for structures subject to surface waves. Therefore, we find the following rewording helpful in understanding our approach:

- $\lambda_R / L_f > 12$: the characteristic length of the spatial variation of the wave is very large. The seismic excitation only causes vertical displacements and rotations are negligible. We may model the wave regime as an equivalent vertical *P* wave.
- $\lambda_R / L_f < 4$: The wavelength is short relative to the foundation length. Thus, the surface wave does not induce any global movement but only local deformations.
- $\lambda_R / L_f \approx 4$: When the wavelength and the length of the foundation are in such proportions, we may imagine that one end of the foundation is located on a node while the other one is located on an antinode of the wavelength. This leads to a phenomenon of amplification that makes the foundation swing. The rotations generated by the surface wave are no longer negligible and the modeling by an equivalent vertical *P* wave is no longer possible. Thus, we expect a peak in terms of rotations when the characteristic ratio λ_R / L_f is close to 4.

These different cases are illustrated on Figure 3.

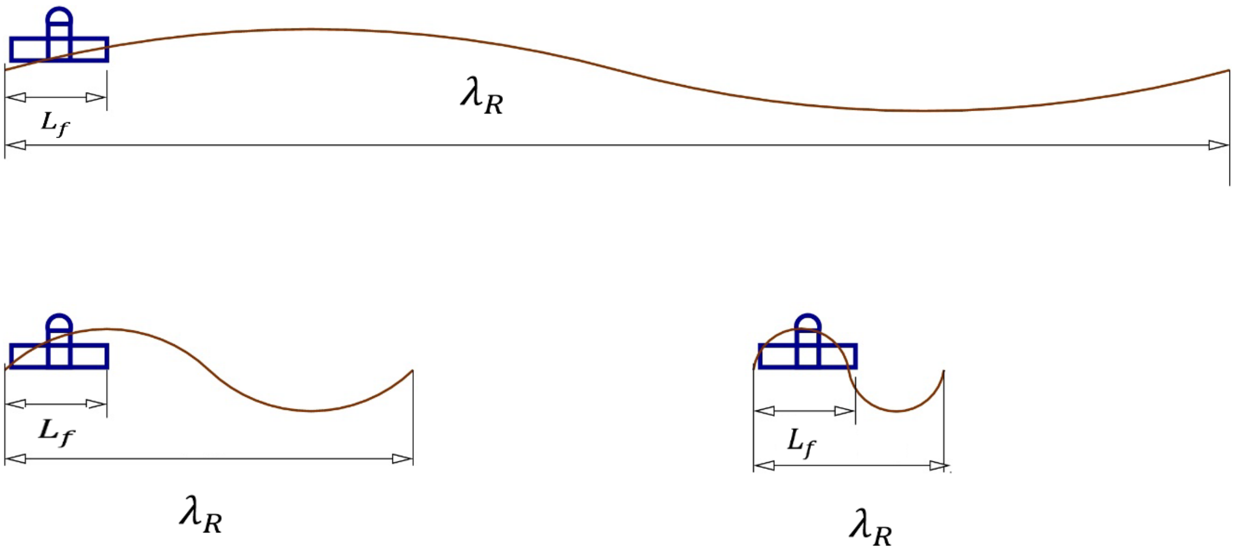

Figure 3 : Prediction of the response of the structure based on a characteristic ratio - Figure extracted from IAEA (2022).

Because of the amplification phenomenon predicted by the "Tecdoc", we anticipate that large structures (whose length are approximately 100 m) may be more sensitive to the effects of surface waves (see also Table 1). Thus, these large foundations will undergo stronger rotations that might not have been anticipated by engineering guidelines. It is therefore important to quantify such an amplification effect to provide appropriate engineering guidelines if necessary.

As we may see on Figure 3, the vertical amplitude of the surface wave and the length of the raft are kept constant. The only parameter that varies is the wavelength of the surface wave. In order to make meaningful comparisons with the "Tecdoc", we will follow the same approach in our parametric study.

As an example, we give in Table 1 the value of the length L_f that corresponds to a characteristic ratio $\lambda_R/L_f = 4$ (named "Amplification L_f"), depending on different soil characteristics. The frequency of the Rayleigh wave is assumed to be $f_R = 0{,}5$ Hz. The two first rows are arbitrary values for the wave propagation velocities (dynamic soil properties) that aim to illustrate our point. The mechanical characteristics of the last row are taken from Wang *et al.* (2021).

$(c_S\ ;c_P)$ [m/s]	λ_R [m]	"Amplification L_f" [m]
(250 ; 500)	466	117
(300 ; 600)	560	140
(500 ; 817)	911	228

Table 1 : Values of λ_R and L_f that leads to a phenomenon of amplification.

From Table 1, we notice right away that potential large rotations as anticipated by the "Tecdoc" could only occur for soft soils combined with very large foundations. This situation might happen when trying to distribute the weight of a structure over a large area of soft soil. For medium or hard soils however, this situation is improbable because it would require extremely large foundations. Moreover, we may show (Semblat and Pecker, 2009) that the amplitude of displacements of body waves is inversely proportional to their velocity. Thus, the stiffer the soil is, the smaller the amplitudes of body waves are. We shall discuss this point extensively in the conclusion of this study.

2.2. Theoretical equations for the Rayleigh wave

2.2.1. In a uniform half space

In this paragraph, we will derive the theoretical equations for the Rayleigh wave in a uniform half space. We follow the book by Semblat and Pecker (2009) and Andersen (2006). The context of the problem is represented on Figure 4.

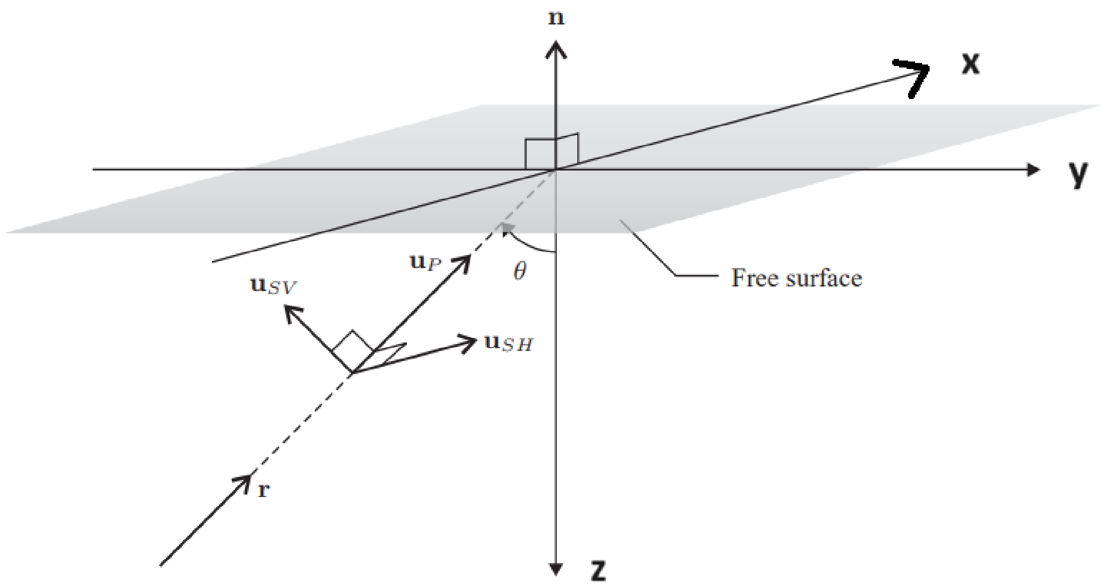

Figure 4 : Propagation of seismic waves in a uniform half space – Figure adapted from Andersen (2006).

Since Rayleigh waves are generated by the reflection of *P* and *SV* waves, it is relevant to decompose the equation of elastodynamics in a way that splits the contribution of the two types of body wave. The Helmholtz theorem allows us to do so. It states that a vector field (provided with some convergence conditions that are respected here) can be decomposed into a divergence-free part ($\boldsymbol{\psi}$) and a rotation-free part (ϕ). Let us apply this theorem to the displacement field $\mathbf{u}$:

$$u_i = \frac{\partial \phi}{\partial x_i} + \epsilon_{ijk} \frac{\partial \psi_k}{\partial x_j} \qquad (2\text{-}6)$$

By substituting this decomposition into the equation of elastodynamics, we obtain the four following scalar equations (one scalar equation for ϕ and three scalar equations for $\boldsymbol{\psi}$):

$$\frac{\partial^2 \phi}{\partial x_j \partial x_j} = \frac{1}{c_P^2} \frac{\partial^2 \phi}{\partial t^2}, \qquad \frac{\partial^2 \psi_i}{\partial x_j \partial x_j} = \frac{1}{c_S^2} \frac{\partial^2 \psi_i}{\partial t^2} \qquad (2\text{-}7)$$

This is another way of formulating the wave propagation equation; this time in terms of Helmholtz potentials ϕ and $\boldsymbol{\psi}$ rather than in terms of volumetric strain Δ and rotation $\boldsymbol{W}$.

To establish the equations, we look for a surface wave propagating in direction x (horizontal) and which has components in the x and z (vertical) directions. In other words, we look for necessary conditions that the solution must satisfy to be an actual surface wave. We also assume that the propagation term and the amplitudes term are decoupled.

To solve this problem, we consider a monochromatic wave. Formally, we write:

$$\phi(x, z, t) = \Phi(\mathrm{z}) e^{\mathrm{i}(k_R x - \omega t)} \qquad \psi_y(x, z, t) = \Psi_\mathrm{y}(\mathrm{z}) e^{\mathrm{i}(k_R x - \omega t)} \qquad (2\text{-}8)$$

Where ω is the angular frequency of the surface wave, c_R its speed and $k_R = \frac{\omega}{c_R}$ is the wave number. We introduce the reduced wave numbers $\gamma_P = k_R^2(1 - \frac{c_R^2}{c_P^2})$ and $\gamma_S = k_R^2(1 - \frac{c_R^2}{c_S^2})$. The equations of propagation can be rewritten:

$$\frac{\partial^2 \Phi}{\partial z^2} - \gamma_P^2 \Phi = 0 \qquad \frac{\partial^2 \Psi_y}{\partial z^2} - \gamma_S^2 \Psi_y = 0 \tag{2-9}$$

The solution of these two ordinary differential equations is a linear combination of real exponentials (we assume that the reduced wavenumbers are positive otherwise the solution of the two previous equations would not be a surface wave). At infinity (when $z \to \infty$), the displacement field must remain finite (it should be zero if there is no source), therefore the positive exponential must be zero. The Helmholtz potentials now reduce to:

$$\phi(x,z,t) = A_P e^{-\gamma_P z} e^{\mathrm{i}(k_R x - \omega t)} \qquad \psi_y(x,z,t) = A_S e^{-\gamma_S z} e^{\mathrm{i}(k_R x - \omega t)} \tag{2-10}$$

Inserting these results into the definition of the Helmholtz potentials, we obtain the following expressions for the displacement field:

$$\begin{cases} u_x = (\gamma_S A_S e^{-\gamma_S z} + \mathrm{i} k_R A_P e^{-\gamma_P z})\, e^{\mathrm{i}(k_R x - \omega t)} \\ u_z = -(\gamma_P A_P e^{-\gamma_P z} - \mathrm{i} k_R A_S e^{-\gamma_S z})\, e^{\mathrm{i}(k_R x - \omega t)} \end{cases} \tag{2-11}$$

From these relations, we can derive the stresses thanks to Hooke's law. The free surface condition at $z = 0$ allows us to write the following system of equations:

$$\begin{cases} (\gamma_S^2 + k_R^2) A_P - 2\mathrm{i}\gamma_S k_R A_S = 0 \\ 2\mathrm{i}\gamma_P k_R A_P + (\gamma_S^2 + k_R^2) A_S = 0 \end{cases} \tag{2-12}$$

This system yields a non-zero solution only if its determinant is zero. Thus, we obtain the so-called Rayleigh wave frequency relation:

$$(\gamma_S^2 + k_R^2)^2 - 4\gamma_P \gamma_S k_R^2 = 0 \tag{2-13}$$

The components of the displacement field become:

$$\begin{cases} u_x = \mathrm{i} A_P \left(k_R e^{-\gamma_P z} - \dfrac{\gamma_S^2 + k_R^2}{2 k_R} e^{-\gamma_S z} \right) e^{\mathrm{i}(k_R x - \omega t)} \\ u_z = A_P \left(-\gamma_P e^{-\gamma_P z} - \dfrac{\gamma_S^2 + k_R^2}{2 \gamma_S} e^{-\gamma_S z} \right) e^{\mathrm{i}(k_R x - \omega t)} \end{cases} \tag{2-14}$$

We notice that the phase shift between the horizontal and the vertical component is equal to $\frac{\pi}{2}$. In particular, for $z < 0{,}19\ \lambda_R$ (where λ_R is the wavelength of the Rayleigh wave), the movement of the particles is retrograde. This means that the elliptical particle motion near the surface is opposite to the direction of wave propagation: if the wave propagates towards the positive x-axis, the motion of the particles at the surface is counterclockwise.

The velocity of the Rayleigh wave c_R is determined thanks to the Rayleigh wave frequency number equation. We insert the definition of γ_P, γ_S and k_R into it and after some rearrangements, we obtain:

$$\left(\frac{c_R}{c_S}\right)^6 - 8\left(\frac{c_R}{c_S}\right)^4 + (24 - 16\alpha^{-2})\left(\frac{c_R}{c_S}\right)^2 - 16(1 - \alpha^{-2}) = 0 \tag{2-15}$$

Where $\alpha = \frac{c_P}{c_S}$. There always exists a solution to this equation with $c_R < c_S$. As expected, c_R only depends on the mechanical characteristics of the soil domain.

Following Andersen (2006), the variation of the amplitudes of the displacement field along the depth is given on Figure 5. We implement the previous theoretical equations in Python and obtain the results presented on Figure 6. To compare the two graphs, we overlay some values of the graph presented by Andersen (2006) in the form of red crosses on our numerical implementation. The graphs obtained with Python are in excellent agreement with the theoretical values extracted from Andersen (2006). The superscript 0 refers to the free surface.

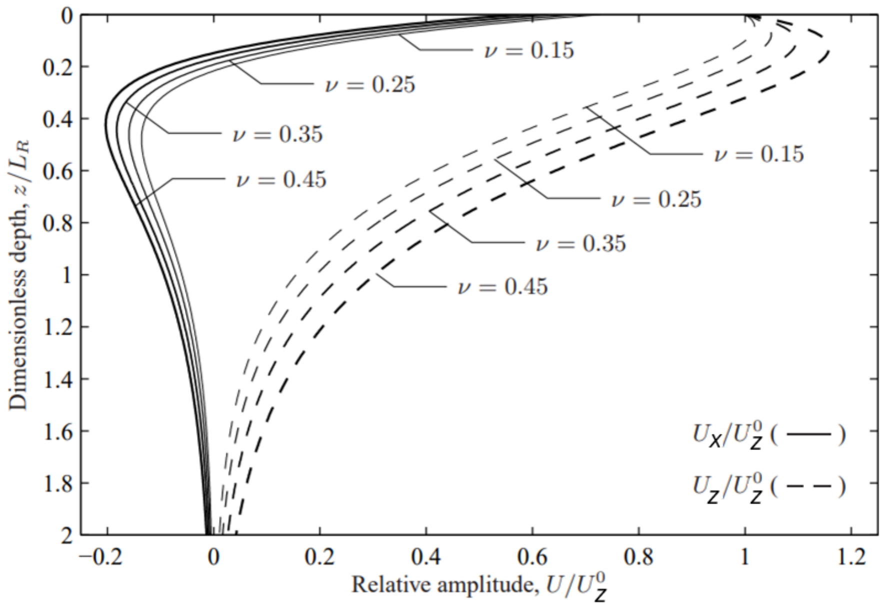

Figure 5 : Amplitudes of the Rayleigh displacement field - Figure adapted from Andersen (2006).

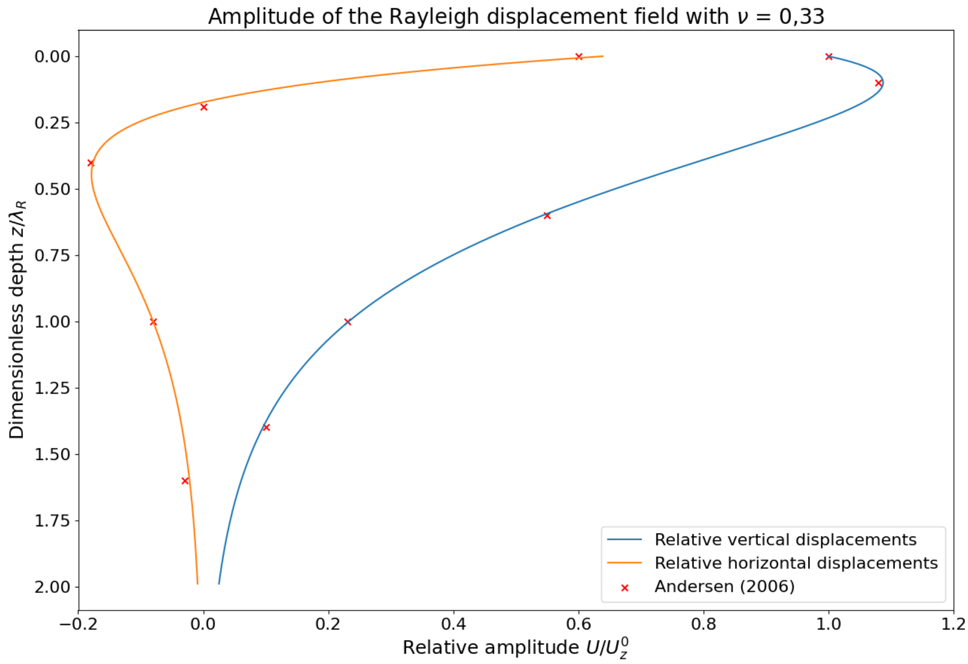

Figure 6 : Theoretical equations implemented in Python.

In nature, the frequency of Rayleigh waves ranges from 0 Hz to 2 Hz. This observation will prove important when we analyze the results of the numerical simulations.

2.2.2. In a horizontally layered medium

In the case of a horizontally layered medium, the above equations are no longer valid. Continuity conditions between the stresses and the displacements at the interface between two layers need to be respected. Therefore, other techniques are used to compute the solution. Some rely on the Helmholtz decomposition of the displacement field to infer transfer matrix between layers (see Haskell (1953) and Kausel and Roësset (1981)). These methods are exact in the sense that they do not entail any discretization of the medium (Kausel, 2018). However, they might not be suited for numerical analysis as they make use of heavy mathematical operations. Other methods rely on a vertical discretization of the soil domain (see Lysmer (1970) and Waas (1972)). These methods are better suited than the previous ones for numerical analysis. In particular, the method developed by Waas (1972) is called the Thin Layer Method (TLM). This method is implemented is the software SASSI (Ostadan, 2006) that we will use later in this study. It is important to note that the TLM works in the frequency domain.

2.3. Reducing the problem to the geotechnical scale

Our objective is to study the displacement field at the geotechnical scale that results from a realistic seismic motion at the geological scale. If we ran the full calculations on the geological scale (incorporating in the same model a detailed representation of the geotechnical features), the simulations would be too heavy and too costly to compute. Therefore, we need to find a method to reduce the dimensions of the problem, allowing for an efficient (from a computation point of view) passage from the geological to the geotechnical scale. Such a method exists and is called the Domain Reduction Method (DRM). In this paragraph, we will present this method and then discuss the interest of coupling it with the TLM.

2.3.1. Presentation of the domain reduction method

The main idea of the DRM is to split the soil domain into a small region of interest Ω (geotechnical scale) and a broader one Ω^+ in which the earthquake propagates (geological scale). The domain Ω has local features that may impact the response of the structure. For example, we might want to study the amplification phenomenon caused by a sedimentary basin or the effect of inelastic soil behavior around the structure. These studies are led at the geotechnical scale and a geological model would not be suitable for including these (relatively) small features. Thus, we need to find a method to go from the geological scale to the geotechnical scale. The problem then becomes determining the forces that one needs to apply at the frontier Γ between the two domains to replicate a realistic motion at the geotechnical scale. Let us consider that a fault in Ω^+ generates the forces $\mathbf{p}_e$. The indices e, b and i refer to exterior domain Ω^+, boundary Γ and interior domain Ω respectively. The problem is illustrated on Figure 7.

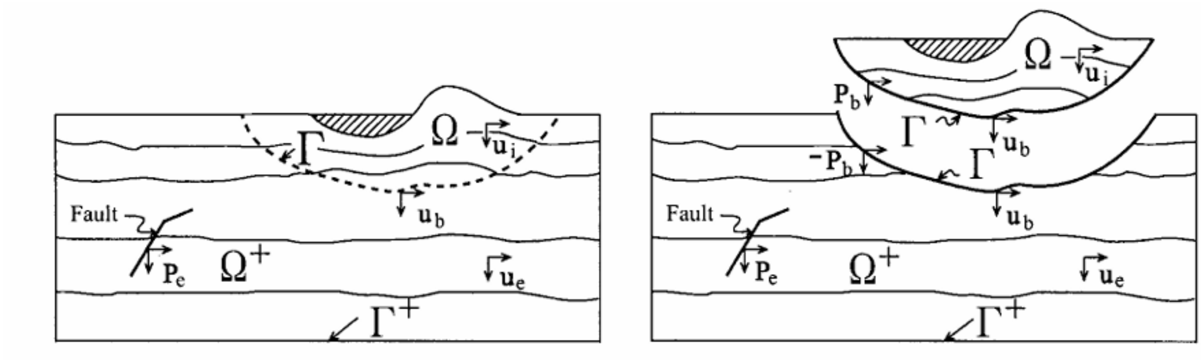

Figure 7 : Partition of the soil domain in the DRM - Figure extracted from Bielak *et al.* (2003).

The equations of motions in the partitioned soil domain are:

in Ω :

$$\begin{bmatrix} \boldsymbol{M}_{ii}^{\Omega} & \boldsymbol{M}_{ib}^{\Omega} \\ \boldsymbol{M}_{bi}^{\Omega} & \boldsymbol{M}_{bb}^{\Omega} \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_i \\ \ddot{\mathbf{u}}_b \end{pmatrix} + \begin{bmatrix} \boldsymbol{K}_{ii}^{\Omega} & \boldsymbol{K}_{ib}^{\Omega} \\ \boldsymbol{K}_{bi}^{\Omega} & \boldsymbol{K}_{bb}^{\Omega} \end{bmatrix} \begin{pmatrix} \mathbf{u}_i \\ \mathbf{u}_b \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{p}_b \end{pmatrix} \tag{2-16}$$

in Ω^+ :

$$\begin{bmatrix} \boldsymbol{M}_{bb}^{\Omega^+} & \boldsymbol{M}_{be}^{\Omega^+} \\ \boldsymbol{M}_{eb}^{\Omega^+} & \boldsymbol{M}_{ee}^{\Omega^+} \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_b \\ \ddot{\mathbf{u}}_e \end{pmatrix} + \begin{bmatrix} \boldsymbol{K}_{bb}^{\Omega^+} & \boldsymbol{K}_{be}^{\Omega^+} \\ \boldsymbol{K}_{eb}^{\Omega^+} & \boldsymbol{K}_{ee}^{\Omega^+} \end{bmatrix} \begin{pmatrix} \mathbf{u}_b \\ \mathbf{u}_e \end{pmatrix} = \begin{pmatrix} -\mathbf{p}_b \\ \mathbf{p}_e \end{pmatrix} \tag{2-17}$$

To determine what forces one needs to apply on contour Γ to obtain a realistic motion in Ω, we first compute the free-field response. We consider a simplified domain Ω_0 in which the local features of interest are removed. The idea is to solve a geological propagation problem whose components are not too small relative to its size. For example, if we wanted to study the amplification effect of a local sedimentary basin, we would remove it from the geological model at this stage of the analysis. We assume that this simplification does not impact the nature of the forces generated by the fault (which is a very reasonable approximation considering the scale of the problem). This procedure is illustrated on Figure 8.

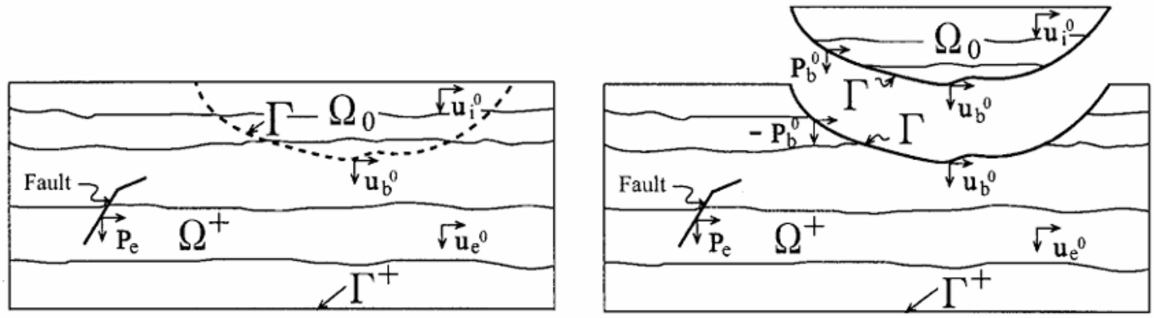

Figure 8 : Simplification of local features - Figure extracted from Bielak *et al.* (2003).

The superscript 0 refers to the free-field problem. The equation of motion for the free-field problem becomes:

$$\begin{bmatrix} \boldsymbol{M}_{bb}^{\Omega^+} & \boldsymbol{M}_{be}^{\Omega^+} \\ \boldsymbol{M}_{eb}^{\Omega^+} & \boldsymbol{M}_{ee}^{\Omega^+} \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_b^0 \\ \ddot{\mathbf{u}}_e^0 \end{pmatrix} + \begin{bmatrix} \boldsymbol{K}_{bb}^{\Omega^+} & \boldsymbol{K}_{be}^{\Omega^+} \\ \boldsymbol{K}_{eb}^{\Omega^+} & \boldsymbol{K}_{ee}^{\Omega^+} \end{bmatrix} \begin{pmatrix} \mathbf{u}_b^0 \\ \mathbf{u}_e^0 \end{pmatrix} = \begin{pmatrix} -\mathbf{p}_b^0 \\ \mathbf{p}_e \end{pmatrix} \tag{2-18}$$

At this stage, we obtain the forces at the fault $\mathbf{p}_e$ in terms of the free-field response:

$$\boldsymbol{M}_{eb}^{\Omega^+} \ddot{\mathbf{u}}_b^0 + \boldsymbol{M}_{ee}^{\Omega^+} \ddot{\mathbf{u}}_e^0 + \boldsymbol{K}_{eb}^{\Omega^+} \mathbf{u}_b^0 + \boldsymbol{K}_{ee}^{\Omega^+} \mathbf{u}_e^0 = \mathbf{p}_e \tag{2-19}$$

However, this solution includes the terms $\boldsymbol{M}_{ee}^{\Omega^+} \ddot{\mathbf{u}}_e^0$ and $\boldsymbol{K}_{ee}^{\Omega^+} \mathbf{u}_e^0$. They require that the free-field response be stored on the entire soil domain which is not efficient numerically. To overcome this difficulty, we decompose the total displacement as the sum of the free-field response and of a residual displacement $\mathbf{w}_e$: $\mathbf{u}_e = \mathbf{u}_e^0 + \mathbf{w}_e$. We substitute the expressions of $\mathbf{p}_e$ and $\mathbf{u}_e$ in the equations of motion (2-16) and (2-17) and we obtain:

$$\begin{bmatrix} \boldsymbol{M}_{ii}^{\Omega} & \boldsymbol{M}_{ib}^{\Omega} & \mathbf{0} \\ \boldsymbol{M}_{bi}^{\Omega} & \boldsymbol{M}_{bb}^{\Omega} + \boldsymbol{M}_{bb}^{\Omega^+} & \boldsymbol{M}_{be}^{\Omega^+} \\ \mathbf{0} & \boldsymbol{M}_{eb}^{\Omega^+} & \boldsymbol{M}_{ee}^{\Omega^+} \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_i \\ \ddot{\mathbf{u}}_b \\ \ddot{\mathbf{w}}_e \end{pmatrix} + \begin{bmatrix} \boldsymbol{K}_{ii}^{\Omega} & \boldsymbol{K}_{ib}^{\Omega} & \mathbf{0} \\ \boldsymbol{K}_{bi}^{\Omega} & \boldsymbol{K}_{bb}^{\Omega} + \boldsymbol{K}_{bb}^{\Omega^+} & \boldsymbol{K}_{be}^{\Omega^+} \\ \mathbf{0} & \boldsymbol{K}_{eb}^{\Omega^+} & \boldsymbol{K}_{ee}^{\Omega^+} \end{bmatrix} \begin{pmatrix} \mathbf{u}_i \\ \mathbf{u}_b \\ \mathbf{w}_e \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ -\boldsymbol{M}_{be}^{\Omega^+} \ddot{\mathbf{u}}_e^0 - \boldsymbol{K}_{be}^{\Omega^+} \mathbf{u}_e^0 \\ \boldsymbol{M}_{eb}^{\Omega^+} \ddot{\mathbf{u}}_b^0 + \boldsymbol{K}_{eb}^{\Omega^+} \mathbf{u}_b^0 \end{pmatrix} \tag{2-20}$$

Thus, the effective nodal forces are:

$$\mathbf{P}^{eff} = \begin{pmatrix} \mathbf{p}_i^{eff} \\ \mathbf{p}_b^{eff} \\ \mathbf{p}_e^{eff} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ -\boldsymbol{M}_{be}^{\Omega^+}\ddot{\mathbf{u}}_e^0 - \boldsymbol{K}_{be}^{\Omega^+}\mathbf{u}_e^0 \\ \boldsymbol{M}_{eb}^{\Omega^+}\ddot{\mathbf{u}}_b^0 + \boldsymbol{K}_{eb}^{\Omega^+}\mathbf{u}_b^0 \end{pmatrix} \tag{2-21}$$

This solution only includes crossed stiffness and mass terms. Numerically, this means that they vanish everywhere except on a one-element wide contour Γ_e around Ω. Therefore, we do not need to store the response over the whole soil domain. Figure 9 illustrates where the effective nodal forces need to be applied.

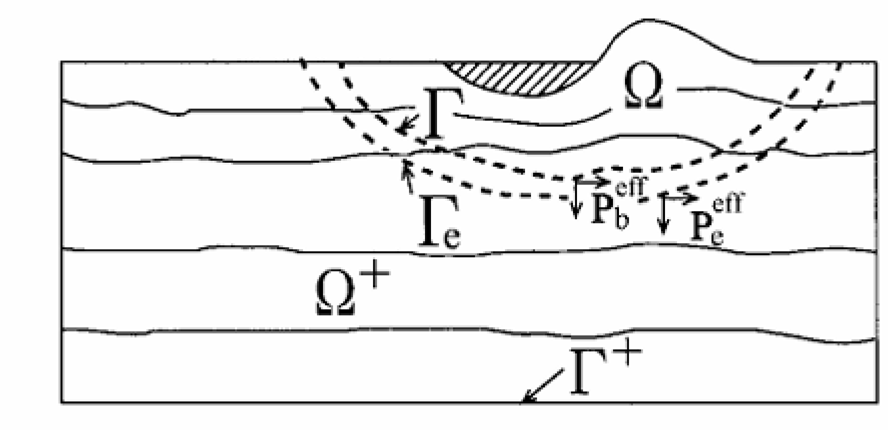

Figure 9 : Contour on which to apply the effective nodal forces - Figure extracted from Bielak *et al.* (2003).

The effective nodal forces only use cross-terms for stiffness and mass matrices. Moreover, the effective nodal forces are formulated without accounting for any source of dissipation. This leads to two major conditions when using the FEM:

a. The DRM layer and the soil around it should be linear elastic with no damping.
b. The DRM layer needs to be exactly one finite element wide.

To conclude, we can introduce a realistic seismic motion into a domain of interest by computing the free-field response. The motion is then produced by equivalent nodal forces. This method is available in software Real-ESSI (Jeremić *et al.*, 2021), which is the software we are going to use to run time-domain computations.

2.3.2. Interest in coupling the TLM with the DRM

2.3.2.1. Efficient generation of full 3D wave regimes

Collecting data of real earthquakes and generating realistic seismic motions are difficult problems that lie outside the scope of this study. However, the coupling of the TLM and the DRM is an efficient way of generating realistic ground motion while having complete control over the wave regime. In this implementation, we use the free-field response computed thanks to the TLM to introduce it into a FEM model through the DRM. In other words, the TLM step replaces the free-field analysis required in the geological scale to define input data for the DRM layer. This implementation will be presented extensively in the following sections. The advantages of this coupling are the following:

- Complete control over the wave regime: the TLM works by solving an eigenvalue problem formulated for a specific type of wave. Therefore, the type of wave (R, S or P), its reference layer and its inclination (for body waves) are input parameters for the computation of the free-field response.
- Efficiency: the free-field response is computed in the frequency domain without making use of the FEM. Thus, calculations are really fast and need only to be done once.
- 3D calculations: contrary to the classical deconvolution / convolution scheme over a 1D soil column, this coupling allows us to work with a full 3D wave regime. The importance of working with a full 3D wave regime has been highlighted in recent work (for example, *cf.* Abell *et al.*, 2018).

This methodology has also been used by Nguyen *et al.* (2022), though with different objectives than ours. Their objective was to compare the results they obtained with known solutions for a basin, a valley, a stratified soil domain and a 12-story building subjected to a R wave. In comparison, our objective is to pave the way for inelastic computations with a full 3D wave regime and to quantify the kinematic interaction of a structure subjected to a R wave.

Since the TLM works in the frequency domain and the DRM in the time domain, one needs to perform convolution of the TLM solution with a reference signal before being able to use it to define effective nodal forces for the DRM. Details on the convolution by a reference signal will be given later on, in section 3.4.

2.3.2.2. Exploitation of real seismograms

To some extent, the research project IM-SURF lies in the continuation of the research project MODULATE, which has also been funded by the ANR. The research project MODULATE aimed at "developing a methodology based on the physics of surface waves, to describe the evolution of the spectral content of the ground motion at a site located in a sedimentary basin." (MODULATE, 2019). Whereas MODULATE focused on identifying surface waves, IM-SURF aims at assessing the response of structures subjected to these surface waves.

To identify surface waves, an original methodology was developed by Meza-Fajardo *et al.* (2015). The methodology relies on a correlation analysis in the time-frequency domain. It accounts for two physical phenomena:

a. P-SV-R waves are decoupled from SH-L waves.
b. The phase shift between the horizontal and vertical components of R waves is $\pi/2$.

This methodology allowed the creation of a large database which comprises numerous time histories of real earthquakes along with the extracted surface waves.

The convolution technique in SASSI to obtain the free-field response may prove very powerful if coupled with this database. Indeed, one could precisely determine the impact of surface waves only on the response of the structure. This possibility goes far beyond a mere speculative interest as it might help assess if engineering guidelines that only account for the effect of body waves provide sufficient safety margins relative to the impact of surface waves.

In the following, when we want to run computations with realistic values, we use the time-history of the 1971 San Fernando earthquake. This earthquake occurred on the 9th of February in Southern California and its magnitude was $M_w = 6{,}6$ (Stover and Coffman, 1992). This event is part of the database of MODULATE. The time history of this event is presented in Appendix 6.

3. Numerical solution of the SSI problem: from the wave regime to the response of the structure

This section is dedicated to explaining the many distinct aspects of the methodology we developed. After presenting the main steps, we will dwell on every step and the questions they arise.

3.1. Methodology overview

We organize the calculation methodology in the following steps and give the software used in parathesis:

- Definition of the geometrical, mechanical and numerical parameters (Excel);
- Meshing of the FE model (Coreform Cubit);
- Computation of the free-field response (SASSI);
- Simulating the SSI problem (Real-ESSI);
- Post-processing the results (Paraview).

We also use Python scripts extensively to create bridges between the different steps.

To lead the study in an efficient manner, we adopt the following criteria when implementing the numerical solution:

- As many geometrical and mechanical properties of the problem as possible need to be parameters that can be adjusted easily.
- Running a parametric study should require few actions from the user.
- The whole implementation must be rather quick, relative to the available computational resources.

3.2. Parametrization

3.2.1. Tool used to input parameters

To define the parameters, we use an Excel workbook which comprises the following sheets:

- "Soil profile": defines the number of soil layers and their thickness.
- "Materials": defines the mechanical characteristics for the structure and the soil as well as their mass densities. The properties of the structure are Young's modulus E and Poisson's coefficient ν, whereas for the soil they are the propagation velocities of the body waves ($c_P; c_S$).
- "Geometry": defines all the dimensions of the model (length and height of the soil domain and of the foundation, thickness of absorbing layers, dimensions of the foundation and embedment, mesh size…).

- "SASSI": defines the required parameters for the computation of the free-field response in SASSI, namely the time and frequency discretization and the number of frequencies to compute in the frequency domain.
- "Wave": defines the parameters of the convolution signal.
- "Damping": defines parameters for the damping of soil layers.
- "Numerical parameters": defines the time discretization for the time domain analysis, the time window of the simulation and the transient algorithm used for the integration of the equations of motions.
- "Parametric study": allows the user to quickly generate the input files for a parametric study over a selected parameter. Examples of parameters over which such a study may be performed are the embedment of the foundation, its volume mass, the frequency of the convolution signal, etc.

On each sheet, there is a button linked to a corresponding Python script that generates the desired input files. A view of every sheet is provided in Appendix 2.

3.2.2. Key hypotheses

To study the kinematic interaction from a theoretical perspective, we make the following hypotheses:

- The raft has no mass and has infinite stiffness.
- The soil is uniform and linear elastic.
- The contact between the soil and the raft is perfect.

Numerically, these hypotheses translate into:

- The volume mass of the raft is $\rho = 0$ kg/m^3 and its Young's modulus is taken equal to $3{,}5.10^6$ GPa.
- The soil has sufficient depth ($h_s = 300$ m) and has uniform speed velocities for S waves and P waves: $(c_S\,; c_P) = (250\,; 500)$ m/s. Its mass density is $\rho_1 = 2100$ kg/m^3.
- The raft and the soil share the same nodes at the interface between them.

The methodology we developed allows one to address more complex problems. For example, one may consider gravity effects, the soil may be nonlinear etc. However, we make these hypotheses to accurately understand the effects of pure kinematic interaction.

N.B.: The fact that the mass density of the raft ρ is set exactly equal to 0 might raise concern regarding the convergence during the numerical integration. A brief explanation of why this works is presented in Appendix 3. Furthermore, the mechanical parameters for the raft have been tested on simple cases to make sure the algorithm converges properly.

3.2.3. Values of mechanical and geometrical parameters

Values of mechanical and geometrical parameters that complete the model are listed below. When relevant, some of these values will be justified further in this report. Parameters that do not appear in the list will be presented in the next paragraphs as they require more details.

- The length of the soil domain is $L_s = 600$ m.
- Poisson's coefficient of the raft is $\nu = 0$.
- The embedment of the foundation is $e_f = 0$ m.
- The thickness of the DRM layer is $e_{DRM} = 3$ m.
- The thickness of each of the three lateral absorbing layers is $e_{al,lat} = 30$ m.
- The depth of each of the three lower absorbing layers is $e_{al,inf} = 25$ m.

N.B.: The thickness e_{DRM} has to be chosen such that the DRM layer is one-element wide. This is the case in all our simulations.

3.3. Finite element model

3.3.1. Brief presentation of Coreform Cubit

Coreform Cubit is a meshing software that allows one to mesh complex geometry thanks to an adaptative mesh algorithm. The software can take augmented text files as input ("journal" files) and generate meshes in many different formats (such as Exodus and LS-DYNA for example). We use Abaqus-type mesh files as output ("inp" files) since they are easy-to-read augmented text files. Thus, it is easy to operate on these files with Python scripts. They can be efficiently converted to other formats that are not available in Cubit but that are used in Real-ESSI and SASSI. This feature is convenient for our purpose of running many simulations with as few inputs from the user as possible.

3.3.2. General overview of the model

A view of a typical finite element model is given on Figure 10.

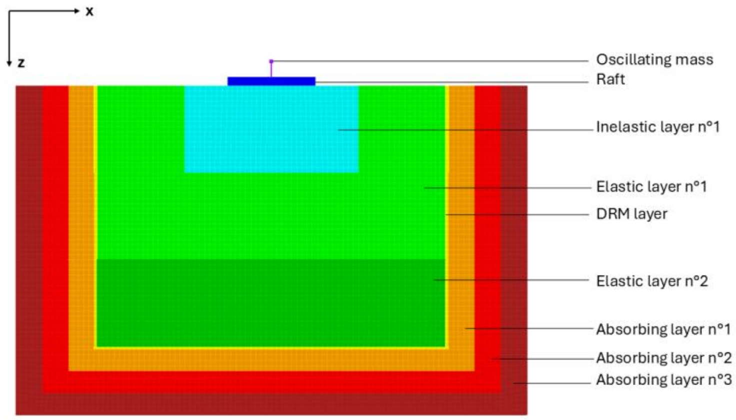

Figure 10 : Finite element model for time domain analysis.

The different components of the model are the following (the corresponding color on Figure 10 are given in parentheses):

- Horizontally layered elastic linear soil domain (green and dark green).
- Horizontally layered inelastic / nonlinear soil domain (light blue). As we explained in paragraph 2.3.1, this area should not come too close to the DRM layer as the soil has to remain elastic around it.
- Raft (dark blue).
- Structure (purple). It is modeled as a mass positioned at a height h_p from the raft.
- DRM layer (yellow). It is one finite element wide (see paragraph 2.3.1).
- Absorbing layers of increasing damping (orange, red and dark red).

Each component of the model and the hypotheses made about them will be presented in the following paragraphs.

The purpose of this report is to study the kinematic interaction solely, from a theoretical perspective. Thus, we wish that no second order effect impact the rotation of the structure. This is the reason why we do not include the superstructure (purple) in parametric studies. This option is left available for further development. Moreover, we assume for the time being that the soil remains elastic (on Figure 10 the light blue area is replaced by green).

From now on, the terms raft, foundation and structure all refer to the dark blue element on Figure 10 without the purple one.

3.3.3. Number of dimensions and boundary conditions

In order to save computation time, we only simulate the problem in the direction of the propagation of the Rayleigh wave within the plane xz. Thus, the model is "pseudo-2D", in other words it is comprised of a single row of volume elements (in the out-of-plane direction) whose thickness is 1 m in the y direction (see Figure 46 in Appendix 1). This does not add any approximation since the problem is invariant in the out-of-plane direction.

We need to choose boundary conditions that are coherent with the "pseudo-2D" modeling. Therefore, we block the orthogonal displacements on the lateral and inferior boundaries of the model (fixed roller support) and block the out-of-plane displacements for all nodes.

3.3.4. Mesh properties

The finite element model is based on 3D 8-node bricks. This element is represented on Figure 11.

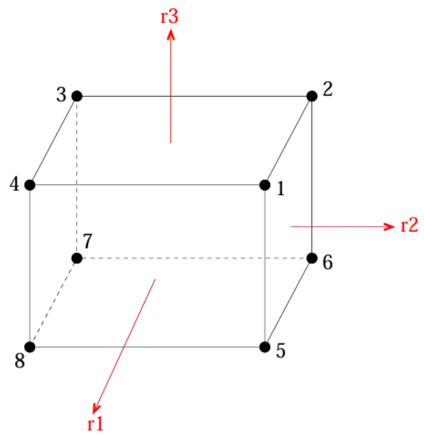

Figure 11 : 8-node brick element - Figure extracted from Jeremić *et al.* (2021).

To choose the element size m_s, we follow the standard criterion which states that each wavelength must be represented by at least 8 finite elements. Thus, the maximum frequency that may be introduced into the model is

$$f_{max} = \frac{c_R}{8m_s} \tag{3-1}$$

We choose $m_s = 6$ m. We recall that the soil mechanical properties are given by $(c_S\,; c_P) = (250\,; 500)$ m/s. We can infer the velocity of Rayleigh wave: $c_R = 233$ m/s. The maximum frequency covered by the mesh is $f_{max} = 5{,}2$ Hz. Considering that typical frequencies of Rayleigh waves are hardly above 1 Hz, we find this maximum frequency to be sufficient.

The value of m_s is also greater than the thickness of the DRM layer and the thickness of the model in the *z* direction. Therefore, we guarantee that the DRM layer will be one element wide and that the whole model will also be one element wide in the out-of-plane direction (making the model actually "pseudo-2D"). For these reasons, we retain this value.

With these parameters, the model comprises roughly 18 000 nodes and 9 000 elements.

The nodes have only three degrees of freedom (the three translations). Thus, rotations are not directly accessible and need to be computed during the post-processing phase. We shall explain later in this section how we calculate rotations (see paragraph 3.8.1).

3.4. Numerical computation of the free-field response

3.4.1. Brief presentation of SASSI

SASSI (System Analysis for Soil-Structure Interaction) is a software that solves linear elastic SSI problems in the frequency domain. It was originally developed by Prof. Lysmer at the University of California, Berkeley. SASSI uses substructuring methods to first compute the response of the site with a TLM and boundary element method analysis. Then, it computes the response of the structure with a FEA. The connection between the two types of models is made through "interaction nodes".

3.4.2. Overview of the methodology to compute the free-field response

Below are listed the steps we follow to compute the free-field response and inject it into the FEM model that will be solved in the time domain with Real-ESSI:

1. Retrieving the time and frequency discretization parameters from the Excel sheet.
2. Retrieving the position of the DRM nodes from the meshed model and inputting them into SASSI as interaction nodes.
3. Choosing the reference time signal for convolution.
4. Solving the free-field problem in the frequency domain.
5. Determining the solution in the time domain.
6. Encoding the results in a readable file by Real-ESSI.

3.4.3. Technical presentation of the free-field response computation in SASSI

In SASSI, the free-field response is computed by implementing the following steps:

1. Solution of the eigenvalue problem relative to a certain wave type for a given range of frequencies.
2. Determination of the eigenmodes and the spectra in the frequency domain for each node.
3. Fourier transformation of the reference signal.
4. Convolution of the transformed signal with the frequency spectra.
5. Inverse Fourier transformation to return to the time domain.

SASSI uses a control point and a control direction as reference to determine all the displacements in the model for a given wave type (P, S, R). For a Rayleigh wave, we choose the control point as the interior node at the top left extremity of the DRM layer. The control direction is z and the wave propagates along the x axis. Thus, this control point directly introduces the control motion in the FEM model. The amplitude of the vertical displacement is kept constant during the parametric study.

Since the user can only specify one control direction, it is best to compute each wave type separately and then to superpose the results at the very end (because the soil is assumed to be linear elastic). This way, one has complete control over the reference signal for each wave type. We only work with Rayleigh waves for now, but this remark will be useful later on.

It is worth noting that the propagation of the wave is accounted for by a phase shift in the frequency domain depending on the distance between the control point and the point of interest.

3.4.4. Parameters chosen in SASSI

Below are listed the parameters we use in SASSI:

- Number of frequencies computed: $NFREQ = 50$.
- Number of points in frequency domain $NFFT = 512$.
- Time step $dt = 0{,}02$ s. This time step is the same as the one we will use for the time domain computation.

The relation between the time discretization dt and frequency discretization df is:

$$\mathrm{d}f = \frac{1}{NFFT \times \mathrm{d}t} \tag{3-2}$$

For the given parameters, we obtain $\mathrm{d}f = 0{,}098$ Hz. Thus, the frequency range covered by this analysis is 0,098 Hz to 4,88 Hz. The corresponding total simulation time in the time domain is $t_f = NFFT \times \mathrm{d}t = 10{,}24$ s.

N.B.: $NFFT$ has to be a power of 2 since SASSI uses the fast Fourier transform algorithm. Also, t_f has to be greater than the simulation time given as input in Real-ESSI.

3.4.5. A Ricker wavelet as a reference signal

To study the SSI problem in the time domain, we decided to use a Ricker wavelet as the reference signal for convolution. The displacement as a function of time for the Ricker wavelet is:

$$u_0(t) = A(2\pi^2 f^2 (t - t_0)^2 - 1)e^{-\pi^2 f^2 (t - t_0)^2} \tag{3-3}$$

Where A is a multiplicative factor that allows us to control the amplitude of the wavelet, f is the frequency for which the maximum of amplitude occurs in the frequency domain and t_0 is the time at which the maximum is reached.

The distance between the two minor peaks is (Semblat and Pecker, 2009):

$$t_b = \frac{\sqrt{6}}{\pi f} \tag{3-4}$$

The properties of the Ricker wavelet are illustrated on Figure 12**.**

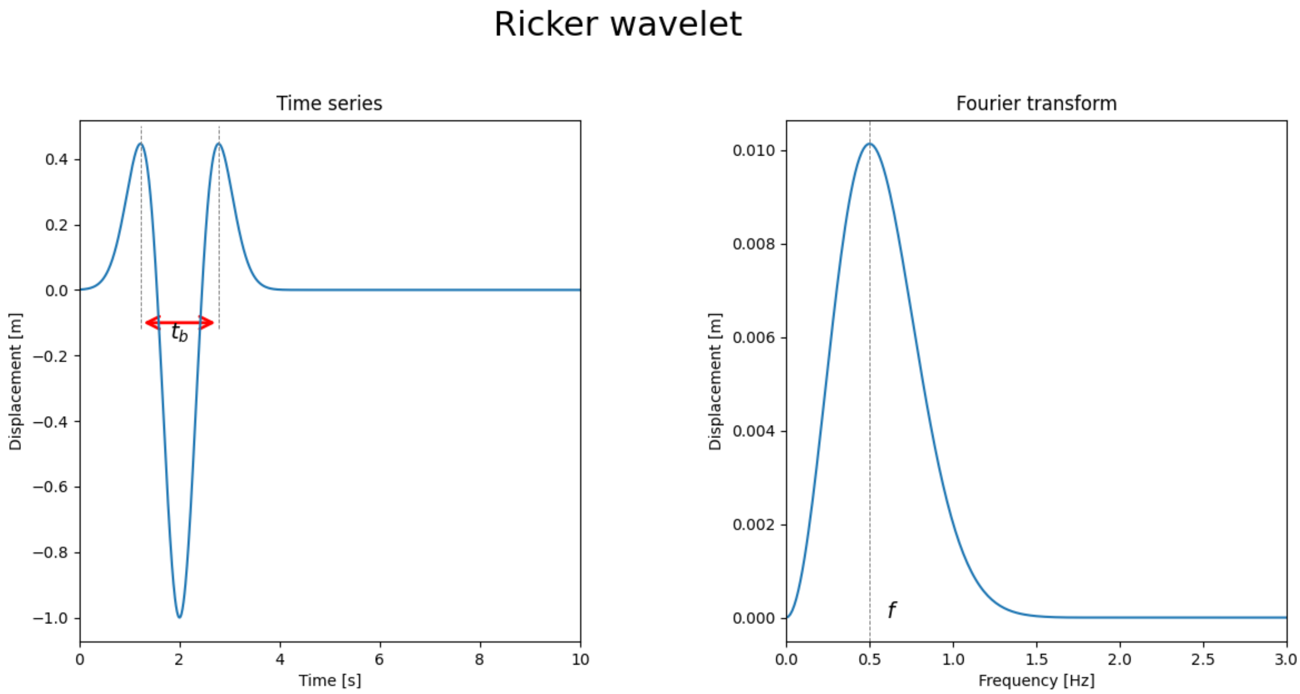

Figure 12 : Properties of the Ricker wavelet.

It is interesting to calculate the maximum vertical velocity of the particles subjected to a Ricker wavelet. Details on this solution are provided in Appendix 4. It can be written in the form:

$$v_{0,max} = A \times K \times f \qquad (3\text{-}5)$$

Where K is a positive constant.

Since the reference time signal is convolved with the propagating wave, this means from a geometrical perspective that $\tan(v_{0,max}/c_R)$ is the maximum slope to which the foundation is subjected. For small amplitudes, which is the case here, $\tan(v_{0,max}/c_R) \approx v_{0,max}/c_R = A \times K \times f \,/\, c_R$. Thus, the maximum slope is a linear function of the frequency. This result will be useful later for the interpretation of the results.

We choose the following parameters:

- Predominant frequency of Rayleigh wave: $f \in [0{,}4\,;2{,}6]$ Hz.
- Time of peak displacement: $t_0 = 2$ s.
- Amplitude: $A = 0{,}05$ m.

With these parameters, the breadth of the wavelet lies in $t_b \in [0{,}3\,;1{,}4]$ s. The magnitude of the amplitude A is inspired by the database of the research project MODULATE. The time of peak displacement is set so that the structure goes through the four steps described later on, in paragraph 3.5.3. The frequency range is set to have interesting values of the characteristic ratio λ_R / L_f (see Table 2 in paragraph 4.3.1). Thus, for this theoretical study, the frequencies used might be greater than the ones found in nature. This point will be discussed in the conclusion.

3.5. Simulation of the SSI problem in time domain

This section aims to present how the SSI problem has been simulated in the time domain.

3.5.1. Brief presentation of Real-ESSI

Real-ESSI is a computer program developed by Prof. Jeremić and his team at UC Davis (see Jeremić *et al.*, 2021). The objective of the program is to run realistic simulations of SSI problems in time domain. To this end, many nonlinear components are available in the software (material constitutive laws and contact elements). Real-ESSI was developed for Linux distributions. Also, features for parallel computing are available in Real-ESSI. Thus, if one wants to run large parameter studies, where models have strong nonlinearities, Real-ESSI might prove very efficient.

In this software, a modular DRM method is also implemented, that allows the user to input body waves (through the matrix formalism of Haskell (1953)) or to input customized signals. In this study, we use our own signals, since they are the ones we obtain from the analyses with SASSI.

3.5.2. Damping

3.5.2.1. Rayleigh damping

To simulate damping in the FE model solved with Real-ESSI, we use Rayleigh damping. In a general manner, Rayleigh damping may be written in the form:

$$\boldsymbol{C} = \alpha \boldsymbol{M} + \beta \boldsymbol{K} \tag{3-6}$$

Let us consider a dynamic mode ω_i. By definition, we have the equation $\boldsymbol{K} = \omega_i^2 \boldsymbol{M}$. Thus, the damping ratio of the i^{th} mode may be written:

$$\xi_i = \frac{\alpha}{2\omega_i} + \frac{\beta}{2}\omega_i \tag{3-7}$$

Therefore, if we set the value of ξ for two different modes i and j, we may find the value of α and β by solving the following matrix system:

$$\frac{1}{2}\begin{bmatrix} \frac{1}{\omega_i} & \omega_i \\ \frac{1}{\omega_j} & \omega_j \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \xi_i \\ \xi_j \end{pmatrix} \tag{3-8}$$

We set $f_i = 0{,}5$ Hz and $f_j = 5$ Hz (where $f_k = 2\pi/\omega_k$ is the frequency of the k^{th} mode).

To control the intensity of the damping, we then multiply α and β by the same factor c_{ref} . From now on, when we mention the intensity of the damping or give numerical values, we refer to this coefficient. The damping in the FEM model (referred to by index FEM) is modeled using the following equation:

$$\begin{cases} \alpha_{FEM} = \alpha \times c_{ref} \\ \beta_{FEM} = \beta \times c_{ref} \\ \boldsymbol{C}_{FEM} = \alpha_{FEM}\boldsymbol{M} + \beta_{FEM}\boldsymbol{K} \end{cases} \quad (3\text{-}9)$$

We are aware of the problems that may arise from the use of Rayleigh damping (see for example Hall (2006), Charney (2008) and Alipour and Zareian (2008)). The main problem comes from an overestimation of the damping when modeling plastic materials if the initial elastic stiffness matrix is used to compute the Rayleigh damping. Indeed, the stiffness decreases as the plastic deformations increase and the rigidity matrix is overestimated. However, the soil remains in the linear domain for this study, so this phenomenon is not an issue here.

3.5.2.2. Absorbing layers

Since all degrees of freedom are blocked for the nodes at the edges of the FE model, spurious reflections may occur. If this happens, the reflected wave might return into the soil domain and interfere with the wave regime under scrutiny. To avoid this problem, we surround the soil domain with three absorbing layers of increasing damping. These layers are placed in the outer soil domain with respect to the DRM layer.

As mentioned before, a difference of impedance between two contiguous materials may also generate spurious reflections. The reflection coefficient increases with the difference of impedance. This is the reason why we use three absorbing layers with gradually increasing damping. The damping as a function of depth is represented on Figure 13.

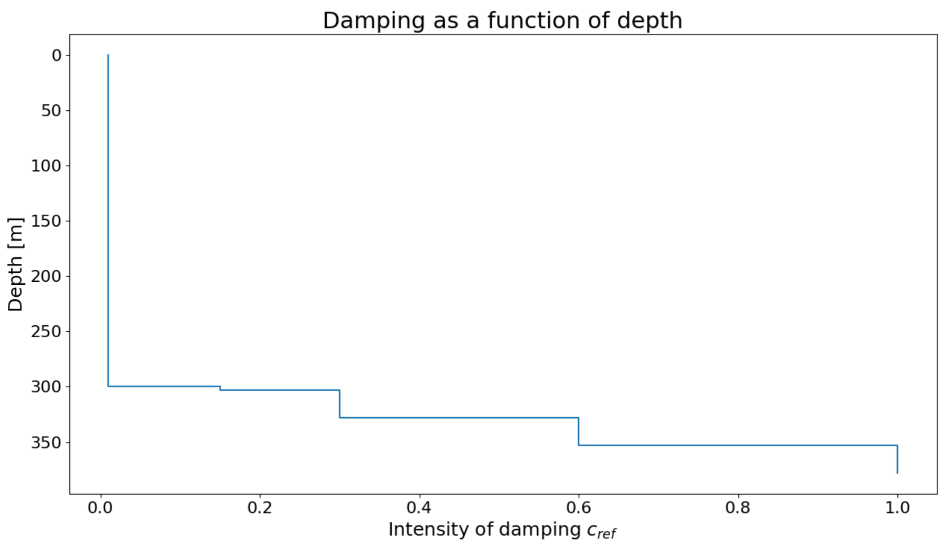

Figure 13 : Damping intensity in the FE model.

Furthermore, in the general case, the absorbing layers have varying mechanical properties along the depth so that they have the same one as the soil layer they are in contact with. This is not apparent on Figure 10 because we colored the elements to highlight their functions but we show on Figure 14 all the blocks used to mesh the model (N.B.: the mesh is symmetrical). The color of these blocks does not correspond to their mechanical properties. They only show how the mesh is built to ensure that the model is symmetrical and that all layers have the right properties, when we assign the properties to each block in the FE program.

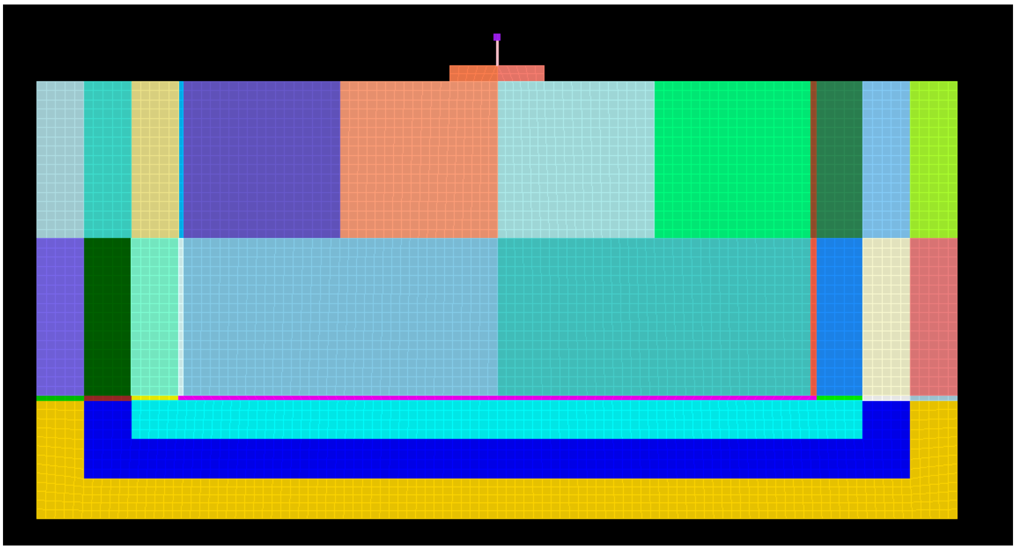

Figure 14 : Blocks used to mesh the FE model.

In this study, we assume that the soil is uniform. Thus, we only use models where the mechanical properties of the absorbing layers are uniform.

3.5.2.3. Damping parameters in the model

Our main interest is to study the kinematic interaction. Therefore, we want the motion injected at the DRM layer to be as similar as possible to the one that acts onto the foundation. This way, the quantification of the kinematic interaction (namely the ratio of rotation in the free-field by the one of the foundation) will not be affected by soil damping. For this reason, we work with low values of damping. In the soil domain, we set c_{ref} equal to 1%.

In addition, the value of parameter γ in the Newmark algorithm is set to 0,5, so there is no numerical dissipation (see next paragraph).

These parameters have been tested on simple cases and give satisfying results. In general, spurious reflections cannot be totally wiped out but they remain of low amplitude compared to all the wave signals at stake.

3.5.3. Transient numerical integration

To solve the transient problem, we use the average accelerations method, in other words a Newmark integration scheme with parameters $\gamma = 0{,}5$ and $\beta = 0{,}25$.

In general, the Newmark integration scheme is defined by the following equations:

$$\begin{cases} x_t = x_{t-\Delta t} + \Delta t \dot{x}_{t-\Delta t} + \dfrac{\Delta t^2}{2}[(1-2\beta)\ddot{x}_{t-\Delta t} + 2\beta \ddot{x}_t] \\ \dot{x}_t = \dot{x}_{t-\Delta t} + \Delta t[(1-\gamma)\ddot{x}_{t-\Delta t} + \gamma \ddot{x}_t] \end{cases} \quad (3\text{-}10)$$

The algorithm of average accelerations is implicit, unconditionally stable and does not induce any numerical dissipation.

More details on this integration scheme and its implementation in Real-ESSI are given in Appendix 3.

The total duration of the simulation is $t_f = 8$ s and the time step is $\mathrm{d}t = 0{,}02$ s. This leads to 400 time steps. These parameters were chosen in order to have four phases in time domain:

1. Complete rest of the whole domain.
2. Introduction of the wave into the soil domain and propagation towards the foundation.
3. Forced excitation of the foundation.
4. Free oscillation of the foundation.

Thus, the simulation approaches real conditions.

3.6. Comparison between the two software

As it appears from the previous presentations, Real-ESSI and SASSI are two very different software, that use different methods to solve dynamic SSI problems. Real-ESSI runs general calculations in the time domain with the FEM whereas SASSI runs linear elastic computations in the frequency domain thanks to a substructuring method.

However, one might not replace the other and we think it is fruitful to use both for the following reasons:

- As mentioned in paragraph 2.3.2, the coupling between the TLM and the DRM allows one to rapidly generate full 3D wave regimes.
- For linear elastic problems, comparing the results obtained in the time domain and in the frequency domain might be used as a validation of our methodology. By comparing the results of the two software, we expect to obtain similar results. (N.B.: the treatment of the damping is different between the two software. To ensure that the comparison remains relevant, we make sure that the damping values are close to 0).
- The computations with Real-ESSI allow us to easily generate animations of the dynamic response to check our implementation (see next paragraph).
- Once our methodology has been fully validated and has provided interesting results, we may address nonlinear problems in the future. In nature, the problems of SSI are rarely linear: uplift, nonlinear soil, etc. Thus, it is really important to provide a time domain methodology that can handle nonlinear aspects of the real SSI problem, while being at the same time in accordance with the frequency domain analysis for linear elastic problems.

3.7. Post-processing the results

To post-process the results, we use a combination of Python and Paraview.

3.7.1. Brief presentation of Paraview

Paraview is an open-source software designed to plot scalar and tensor fields on a given geometry. Thus, it has a wide range of applications. In particular, one may use it to plot the results of FEM analyses. The coupling with Real-ESSI allows the user to plot the displacement field, the stress field, the strain field, amongst others.

Two useful representations of vector fields are available in Paraview:

- *WarpByVector*: The components of the displacement field are used to scale the plotting of the model in the corresponding directions. A scale factor may amplify the magnitude of the vectors used for scaling. It allows the user to obtain an amplified visualization of the deformed mesh. This is helpful to check if the waves are injected properly and to understand how the structure responds to the seismic excitation.
- *Glyph*: In this animation, the model is not scaled but the vector corresponding to the displacement field are shown. It allows the user to track the trajectory of a given particle. This is helpful to check if the Rayleigh wave has been properly injected. Indeed, we expect a retrograde motion for shallow particles.

3.7.2. Overview of the post-processed results

Let L_s be the length of the soil domain and L_f be the length of the foundation. The origin of the coordinate system is the center of the foundation. Seven points function as "sensors" during the simulation. One is located at the top left corner of the DRM interior layer (the one adjacent to the soil domain). The position of the six other ones is represented on Figure 15.

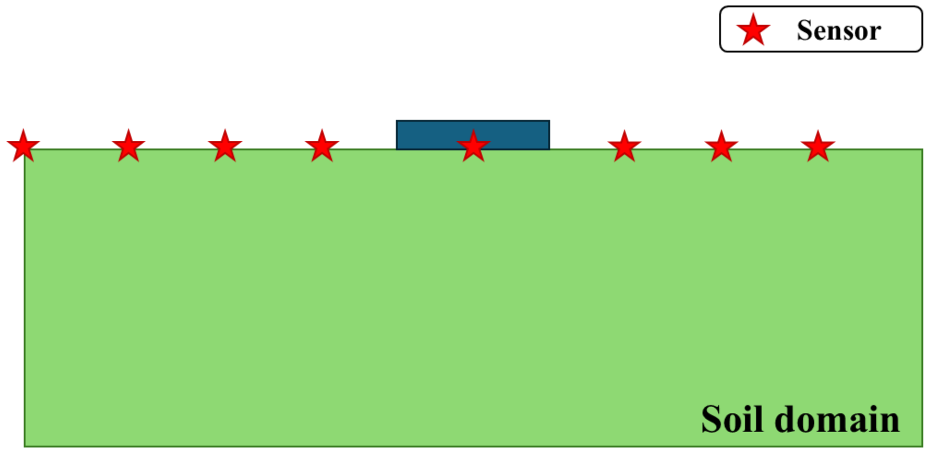

Figure 15 : Position of the sensors on the free surface.

At these sensors, we extract in the form of a graph and of text files:

- The time series and Fourier transform (FT) of the horizontal displacement
- The time series and Fourier transform (FT) of the vertical displacement

- The time series and Fourier transform (FT) of the rotation

We also extract the rotations of the foundation in time and in frequency domain (see next paragraph). This data will be used for the parametric study described in the next section.

Additionally, for each simulation, we generate animations using the *WarpByVector* and *Glyph* representations of the displacement field.

N.B.: The positions given for the sensors might not lie exactly on a node. During the pre-processing phase, we make sure to select the node which is closest to the given position.

3.8. Other methodological aspects

3.8.1. Computation of the rotation

Since the foundation is rigid, we can calculate its rotation from the displacements of its ends. Indices g and d refer to the left and right end of the foundation respectively ("gauche" and "droite" in French). The situation is shown on Figure 16.

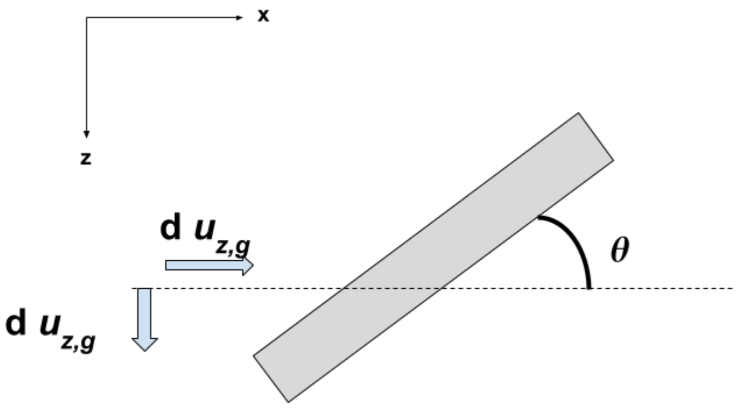

Figure 16 : Rotation of a rigid foundation.

The rotation θ of the raft is:

$$\theta = \tan^{-1}\left(\frac{u_{z,g} - u_{z,d}}{L_f - u_{x,g} + u_{x,d}}\right) \quad (3\text{-}11)$$

For sensors, we use the same formula over a finite element. The two ends are two adjacent nodes. In the formula, we replace L_f by the length of the finite element. This calculation is valid only if the finite element is small enough.

3.8.2. Normalization of the quantities involved

Since we lead a parametric study, we try to find dimensionless groups that impact the physical phenomenon of SSI (this is the idea behind Vaschy-Buckingham theorem). This helps understand the physical phenomena from a general perspective. Furthermore, from an engineering perspective, a relevant normalization is what makes the results easy to use for designing structures.

3.8.2.1. Normalization of the length parameters

In the spirit of the "Tecdoc", a natural dimensionless group would be λ_R/L_f . However, since we use a Ricker wavelet as a reference signal, the Rayleigh wave is not monochromatic. To overcome this problem, we retain the value $\lambda_R = c_R/f$, where f is characteristic the frequency of the Ricker wavelet for which the maximum of amplitude is reached in the frequency domain.

Vertically propagating SV waves do not "see" the length of the raft L_f but rather the embedment e_f. Thus, the appropriate normalization is λ_R/e_f. In our study however, we keep the relative embedment constant so both length parameters could work for the normalization.

3.8.2.2. Normalization of the rotation for Rayleigh waves

The problem of normalizing the rotation is more difficult as many options are available. Namely, the following options exist:

a. Normalize the rotation by the free-field rotation, before any propagation in the FEM domain. In practice, this rotation is retrieved from the sensor located at the DRM (see previous paragraph).

b. Normalize the rotation by the free-field rotation in a free-field model, without the foundation. The rotation is retrieved at the point where the center of the foundation would have been in the complete model.

c. Normalize the rotation by the theoretical maximum slope of the convolved signal.

In this study, we retain options a and b. They may be respectively interpreted as a:

a. Theoretical approach (normalization a): the objective is to assess the influence of various parameters on the kinematic interaction. Thus, the free-field rotation is measured in an elastic linear domain, before its propagation. This approach allows one to run parametric study over certain parameters to quantify their impact.

b. Engineering approach (normalization b): the objective is to estimate the rotations of the foundation from a given time history (or a spectrum), given by regulations. Thus, the free-field rotation is measured at the center of the FE model, after it has propagated within the model. This approach allows one to draw graphs of the kinematic interaction coefficient and provide engineering guidelines.

We also plot the absolute rotation, without any normalization. These approaches will be further compared in paragraph 4.3.3.

3.8.2.3. Normalization of the rotation for SV waves

Since vertically incident SV waves generate no rotation at the free surface, it is difficult to find a way of normalizing the results. Amongst others, Dominguez (1978) uses the equivalent vertical displacement that the wave would generate at one end of the foundation if it rotates around its center (in our case the equivalent vertical displacement would be written $u_{eq,z} = \theta L_f/2$). Then this displacement is normalized by horizontal displacement caused by the SV wave in the free field.

Even though this normalization does not really emphasize the importance of rotations as such, we decided to use it nonetheless because it allows one to plot the response of the foundation in the form of a canonical curve (see paragraph 4.4). Using this approach, the curve we obtain is satisfactory when the relative embedment is kept constant, which is the same for Rayleigh waves. These *a posteriori* considerations will become clearer in section 4. Furthermore, in accordance with our objectives, this normalization would be easy to use in the engineering practice. However, it is physically very different from the one used for Rayleigh waves, as no reference rotation motion has been considered.

3.8.3. Necessity of using a reference time signal for convolution

One may wonder why we convolve the free-field response in the frequency domain with a reference signal (which is a Ricker wavelet) and not introduce a monochromatic wave right away. First of all, the use of a wavelet is more realistic than a monochromatic wave and is better suited for a time domain analysis (a mere computation in the frequency domain would have been enough otherwise). Furthermore, the implementation of such a methodology paves the way for the use of realistic time signals retrieved from databases and seismological stations.

Nevertheless, the use of Ricker wavelet is not of pure speculative interest. Recall the phases of the simulation described in paragraph 3.5.3. Erratic rotations appear at the transition between one phase to another. This is a significant issue since the rotations are the quantity we want to measure precisely. Figure 17 shows the time series of the rotation of the foundation for a monochromatic Rayleigh wave. The erratic rotations are circled in red.

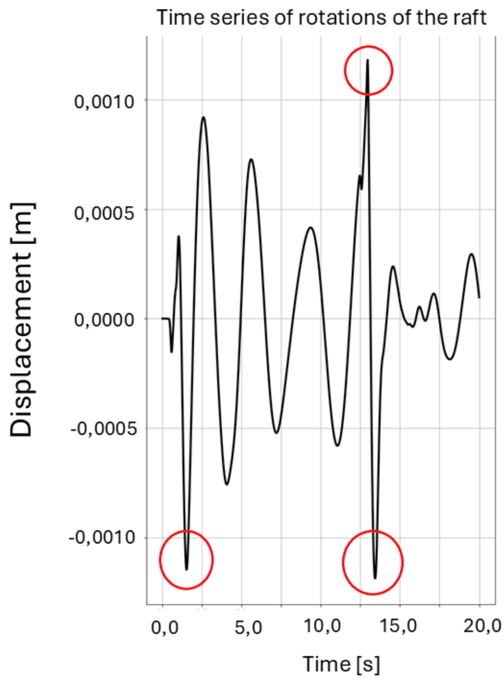

Figure 17 : Erratic rotations at the transitions between phases during the time domain simulation.

This problem disappears when a Ricker wavelet is used instead. See Figure 18.

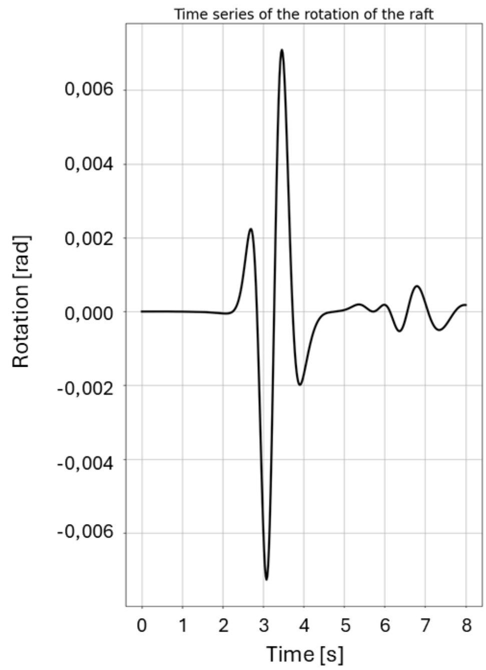

Figure 18 : Rotation of the foundation when the reference motion is convolved with a Ricker wavelet.

This phenomenon is likely linked to the regularity of the functions considered. If the seismic excitation is harmonic, we need to stop it when the signal is 0 at the DRM layer to ensure continuity. However, at this moment the time derivative of the signal is 1. Thus, the derivative of the signal is discontinuous between the excitation phase and the free oscillation phase. However, the derivative of the Ricker wavelet approaches zero when the wavelet approaches zero (as well as all the higher order derivatives). Therefore, continuity is smoothly ensured at the transition between the excitation phase and the rest phase.

3.9. User manual

The calculation methodology is designed to require as few user actions as possible. Therefore the "*Parameter study*" sheet in the Excel workbook can generate all the input files for a parametric study with one click on a button and place them in a given folder. Then the user can just copy/paste the folder on the machine where Real-ESSI is installed. Finally, running the script *etude.sh* ("study" in French) will launch the calculations and create all the post-processed material described above.

With the parameters described in this section, and for ten values of the parameter under scrutiny, the total simulation time (pre-processing and post-processing included) amounts to 10 hours approximately. Also, 10 GB of disk space are required. Further computational aspects will be discussed in next paragraph.

3.10. Computational aspects

3.10.1. Specifications of the computers used

Since the FE models we studied remain rather small, we developed the calculation methodology and ran our calculations on a standard computer. The computer is operating on Windows 10. It has 8 GB of RAM and its CPU is an Intel i5 with 4 cores (one thread per core) and with a frequency of 3,5 GHz.

Since Real-ESSI has been developed for Linux distributions, we use a virtual machine whose host is the standard computer. The virtual machine is operating on Ubuntu 22.04. We allocate 4 GB RAM and one core of the CPU to the virtual machine.

3.10.2. Duration of the calculations

On Figure 28, it took roughly 10 hours to generate the orange curve (Real-ESSI, time domain analysis) and 1 minute to generate the blue one (SASSI, frequency domain). The time domain analysis uses 10 GB of storage and the frequency domain analysis 60 MB of storage.

These enormous differences may be explained by the following considerations:

- The frequencies are computed once and for all in SASSI. The movement is then convolved to generate the full-time history of the response. On the other hand, Real-ESSI computes the whole propagation problem for each time step.
- SASSI is not a FEM program. Therefore, it does not compute the solution at thousands of nodes. We noticed that increasing the number of interaction nodes in SASSI greatly increases the computation time and the storage needed. The superficial foundation only requires 30 interactions nodes. However, 410 interactions nodes take 12 minutes to compute and use more than 1,2 GB of storage (which is less efficient than Real-ESSI if we think in terms of the number of nodes).

During this internship, the calculations with Real-ESSI have been run in series. To reduce the computation time, one could use the parallel computation algorithm provided in Real-ESSI. Parallel computing is not available in SASSI.

It is also important to keep in mind that SASSI only runs elastic linear computations. If one wants to model any nonlinear elements, a time domain analysis is necessary. This is the reason why we developed this methodology in the time domain rather than in the frequency domain. The main objective of this internship was to develop a complete calculation approach. A frequency domain analysis proved useful to check our results for linear elastic cases (see also paragraph 3.6).

3.11. Validation of the implementation

3.11.1. Validation of SASSI

To check our implementation and whether SASSI provides the expected results, we run a simple free-field analysis and compare it with the theoretical equations for the Rayleigh waves in a uniform half space. Results are shown on Figure 19. The agreement between the theoretical equations and the results from SASSI is excellent.

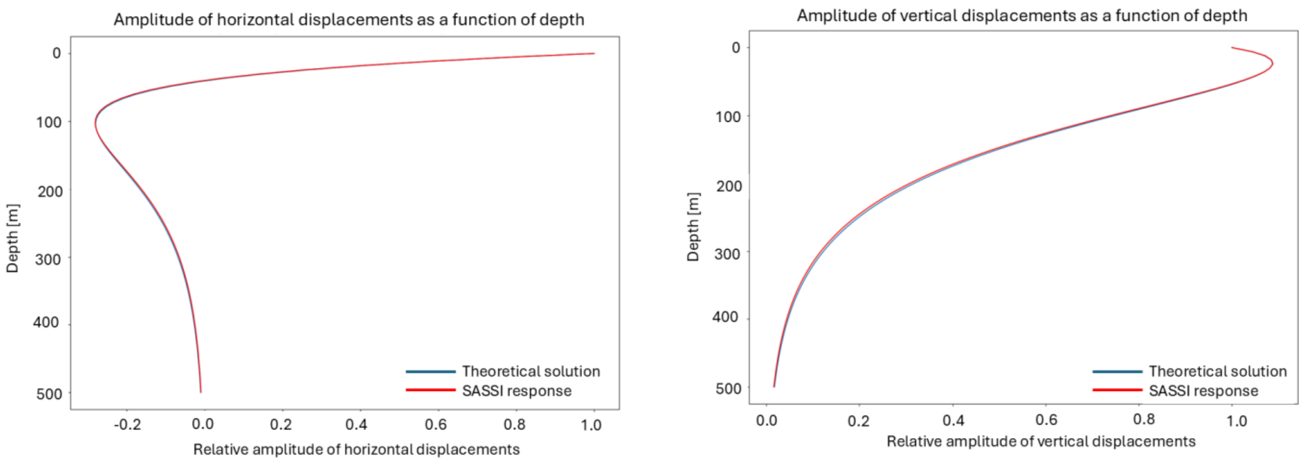

Figure 19 : Comparison between theoretical displacements and the solution obtained with SASSI.

3.11.2. Other validations for the FE model

The formula to compute the rotation of the raft has been evaluated in a simple case by comparison with the results given by a beam element. We placed a rigid beam at the free surface and introduced a Rayleigh wave in the soil domain. The results obtained by retrieving the rotation from the degree of freedom of the beam or by computing it from the displacements of its ends were almost identical, the difference being less than 5 %.

Recall that the raft is modeled by 8-Node-Brick elements in three dimensions (see Figure 11). Since the foundation is rigid and massless, the arbitrary height of the raft chosen to generate the FE model should not impact the response of the structure. Thus, we need to make sure that this theoretical consideration is respected during the numerical analysis. To this end, we compare the results obtained for different heights and also for shell elements in place of the volume elements. We obtain similar results; in other words if we overlay the response of these different models, the curves coincide. For simplicity, we chose to use volume elements for the study of rafts subjected to Rayleigh waves. In Coreform Cubit, it is more convenient to only mesh volume elements (otherwise overconstraints may appear at the interfaces between the elements and need a case-by-case correction, which is not suited for our goal to run extensive parametric studies). This choice also ensures a smooth transition to the numerical study of buried structures, which will be modeled with volume elements.

We also compare the results we obtain in terms of the vertical displacements with the ones presented by Wolf and Obernhuber (1980). There is good agreement between the two.

N.B.: To keep this report concise, we do not further dwell on the validation considerations. We mention them to show that our methodology is robust and does not depend on implementation choices.

4. Application of the complete methodology to the problem of quantifying the kinematic interaction

This section describes the results we can obtain thanks to the methodology we have developed. It provides a detailed solution for a given case but is also appropriate for parametric studies. The flexibility and the completeness of our methodology will be apparent in this section.

4.1. Importance of the wave regime

Abell *et al.* (2018) and Wang *et al.* (2021), amongst others, have highlighted the importance of modeling the full 3D wave regime to properly study the seismic response of structures, especially in the presence of significant wave components. In this study, it is interesting to verify if this observation also applies to Rayleigh waves, namely if the behavior of the structure could be reproduced by introducing body waves instead of a surface wave. In particular, for low frequencies (i.e. long wavelengths), the "Tecdoc" states that a surface wave can be modelled by an equivalent P wave.

To run this analysis, we follow this methodology:

1. Inject a Rayleigh wave in the soil domain without any structure and retrieve the free field response.
2. Decompose the total displacement field at the ground surface into a P wave for vertical displacements and a S wave for horizontal displacements. We consider that these body waves propagate vertically along the soil profile.
3. Inject these two waves into the model with the structure.
4. Extract the rotations of the structure and compare with the results from the previous paragraph.

When running this analysis with SASSI, we observe no rotation for the foundation (the order of magnitude is 10^{-13} rad). Thus, we conclude that modeling the Rayleigh wave by equivalent body waves does not account for the real behavior of the structure.

When running this analysis with Real-ESSI, we observe some rotations (the order of magnitude is 10^{-4} rad). After examining the post-processed videos of the simulations, we notice that these rotations are caused by the DRM. Whereas the two components of the displacement field of the Rayleigh wave propagate at the same speed (and come from "the same place"), this is not the case for a combination of P and S wave. Therefore, one should be really careful when trying to decompose a signal on different types of waves while using the DRM at the same time. This erratic behavior is illustrated on Figure 20.

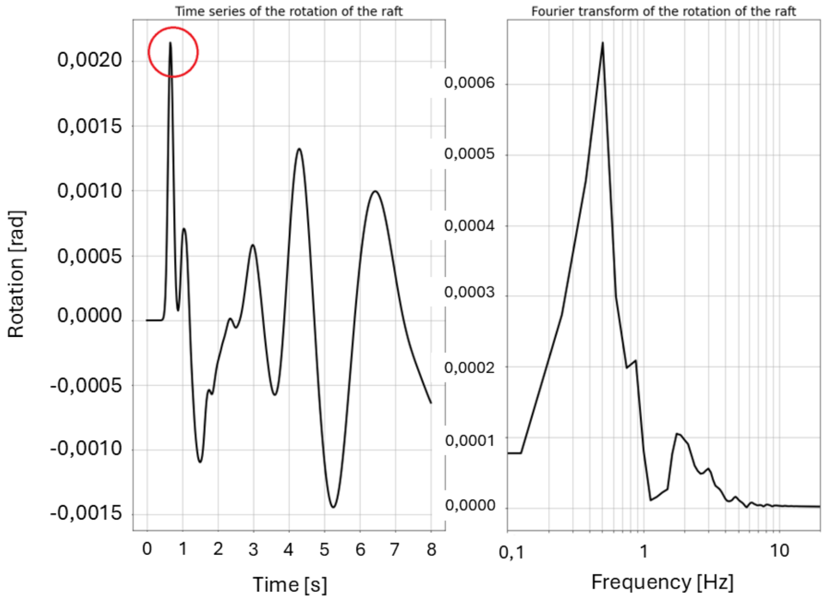

Figure 20 : Rotation of the foundation under P and S combination with DRM in the time domain. The maximal spurious rotation is circled in red.

N.B.: Body waves do not generate any rotation because the structure is superficial. When the structure is embedded in the ground, S waves will generate a rotation.

The case of a superficial foundation is a clear manifestation of the fact that we cannot define "equivalent" body waves to reproduce the *same* response as with surface waves. This observation participates in the validation of our methodology as it is in agreement with other observations in the literature (see for example Abell *et al.*, 2018).

4.2. Results obtained for one sample

In this paragraph, we will present the results when running the analysis in Real-ESSI. We study a foundation subjected to a pure Rayleigh wave in the time domain. The central frequency f is set equal to 1,0 Hz and the length of the raft L_f equal to 100 m.

Figure 21 shows snapshots of the time domain simulation generated with *WarpByVector*. We see the wavelet hitting the foundation. The color gradient represents the magnitude of the displacement field.

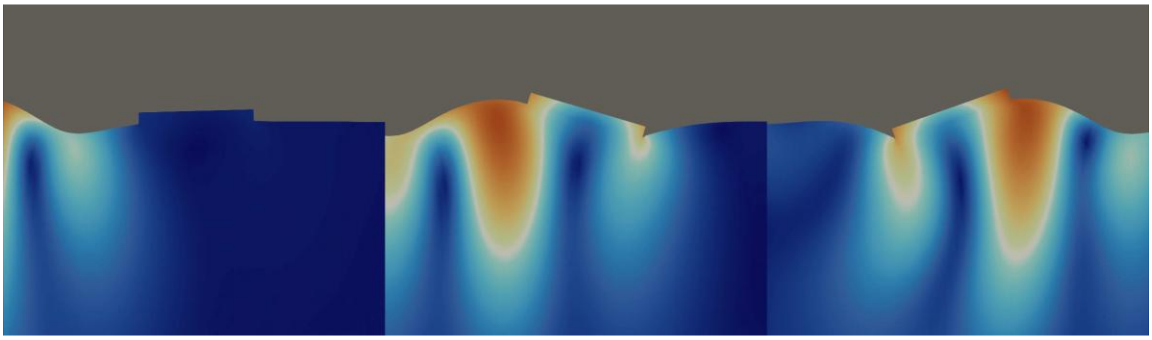

Figure 21 : Snapshots of the time domain simulation.

Figure 22 shows the results when post-processing them with *Glyph*. The movement of the particles is indeed retrograde for shallow depths.

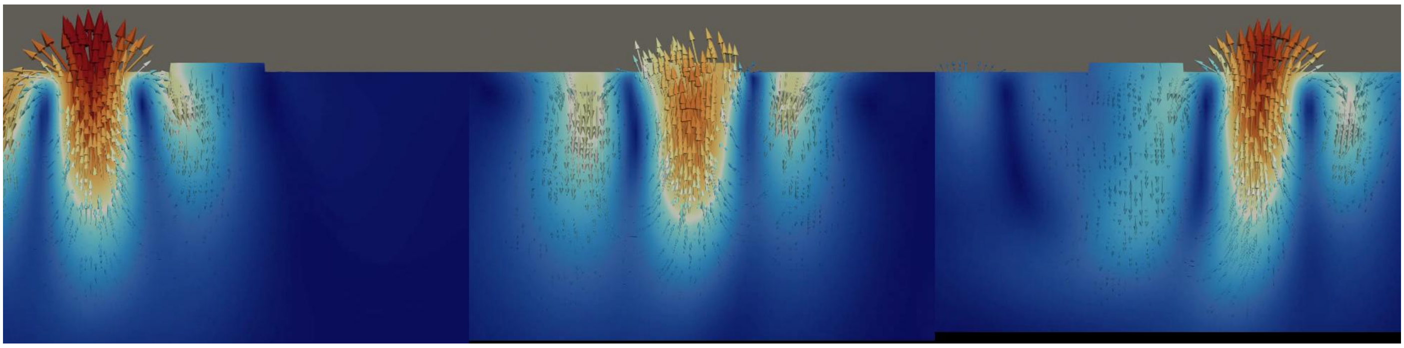

Figure 22 : *Glyph* view of the simulation.

Figure 23 shows the rotation of the foundation in the time domain and its FT. The rotation follows the Ricker wavelet as expected. Also, the spurious reflection generated at the boundary of the model is circled in red. Its amplitude is very low compared to the one of the main wavelet. Moreover, we observed on the different models that the spurious reflection does not mix with the main wavelet. In other words, the main wavelet and the spurious reflection remain separated in time. Therefore, it will not impact on the results of the study, since we are interested in the maximum rotation the foundation is subjected to (see next paragraph).

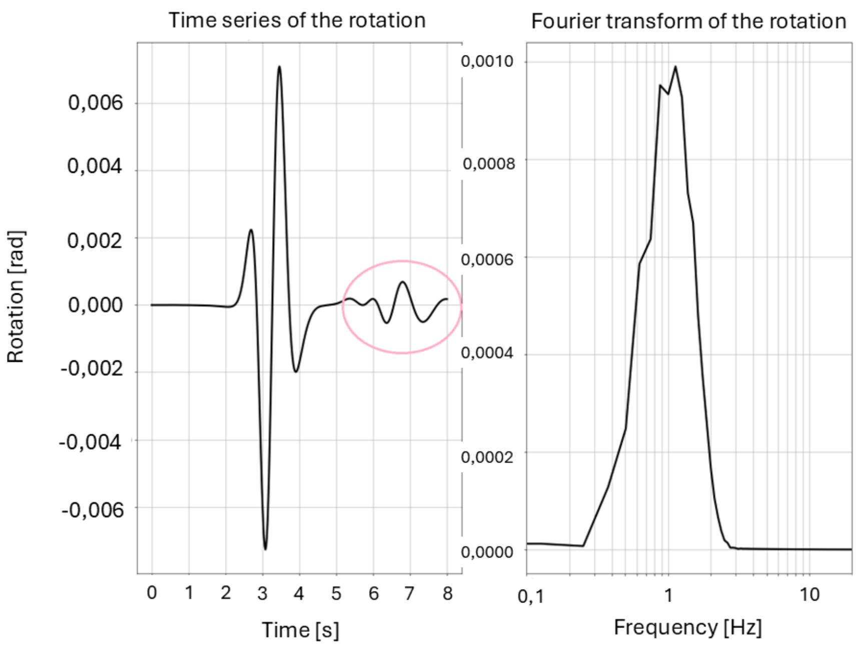

Figure 23 : Rotation of the foundation and stray reflection.

Figure 24 shows the rotation of the soil at the free surface between the foundation and the DRM layer. We notice that the kinematic interaction generated a reflected wave, circled in blue. The frequency of this reflected wave is higher than the one of the wavelet.

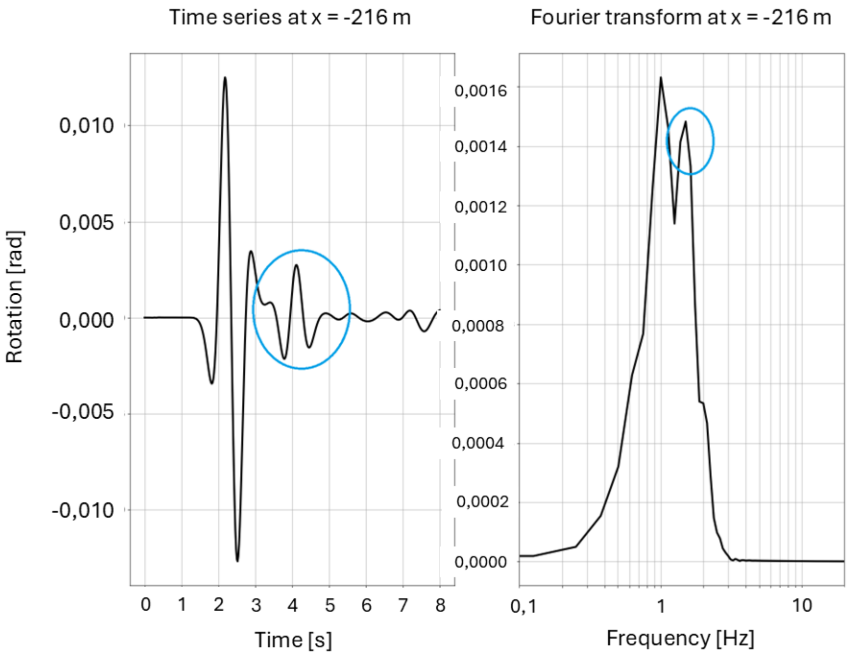

Figure 24 : Rotation at the free surface before the foundation.

Figure 25 shows the rotation at the free surface after the foundation. We notice that its amplitude is greater than the rotation of the foundation but lower than the one before the foundation. Thus, we infer that a part of the energy has been dissipated in the form of a reflected wave at the foundation and that the stiffness resists the movement.

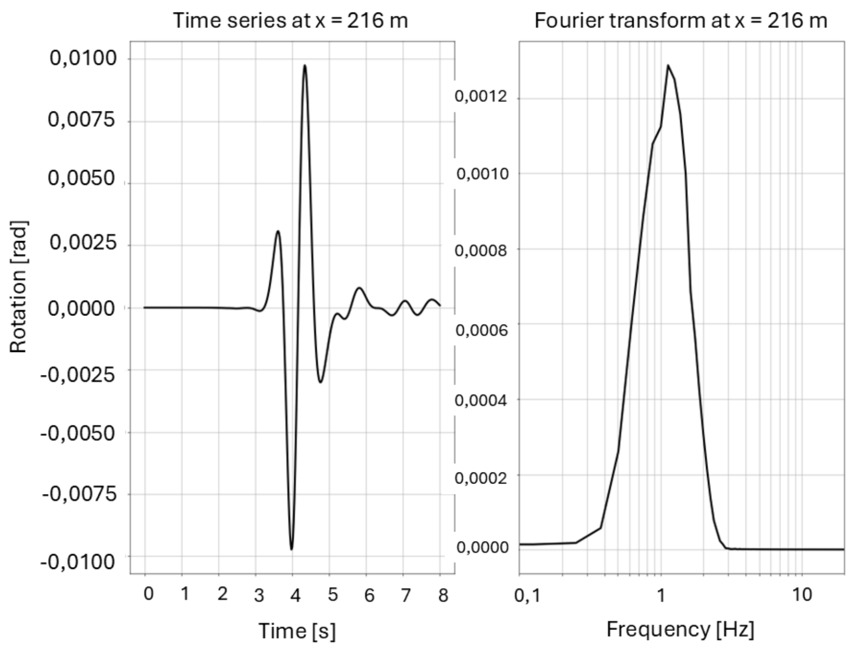

Figure 25 : Rotation at the free surface after the foundation.

The amplitude of the wavelet (i.e. the amplitude of the vertical displacement at ground surface, by the DRM layer) is 5 cm. The rotations measured are approximately 10 mrad.

4.3. Results obtained for a parametric study over the wavelength

4.3.1. Presentation of the parametric study

To assess the effect of the kinematic interaction, we run a parametric study over the central frequency of the Ricker wavelet. This way, we vary the characteristic ratio λ_R/L_f. We then plot the maximum rotation reached by the foundation (absolute and normalized) as a function of this ratio. Following what we have discussed in the introduction, we expect to see a peak at $\lambda_R/L_f = 4$ and then convergence towards 0 as λ_R/L_f goes to 0 and to $+\infty$.

The length of the foundation is $L_f = 100$ m. We consider nine central frequencies of the wavelet ranging from 0,4 Hz to 2,6 Hz. The frequencies under scrutiny and the corresponding characteristic ratio λ_R/L_f is given in Table 2.

Central frequency of the wavelet [Hz]	Characteristic ratio λ_R/L_f
0,4	5,83
0,6	3,88
0,8	2,91
1,0	2,33
1,2	1,94
1,4	1,66
1,8	1,29
2,2	1,06
2,6	0,90

Table 2 : Central frequencies under scrutiny and their corresponding characteristic ratio

N.B.: The frequencies are not evenly spaced. This is because the wavelength is inversely proportional to the frequency. Thus, we opted for a rather regular discretization of characteristic ratios over the range [1; 6] approximately.

The parameter study is run both in time domain (with Real-ESSI) and in frequency domain (with SASSI). We then compare the results of these two computations.

4.3.2. Results in time domain (Real-ESSI) and in frequency domain (SASSI)

Figure 26 shows the maximum rotation (without normalization) as a function of λ_R/L_f in the time domain. As ratio λ_R/L_f decreases (from high values to smaller ones), we see that the foundation rotation is increasing until a peak at $\lambda_R/L_f = 2$ and then it decreases rapidly towards 0.

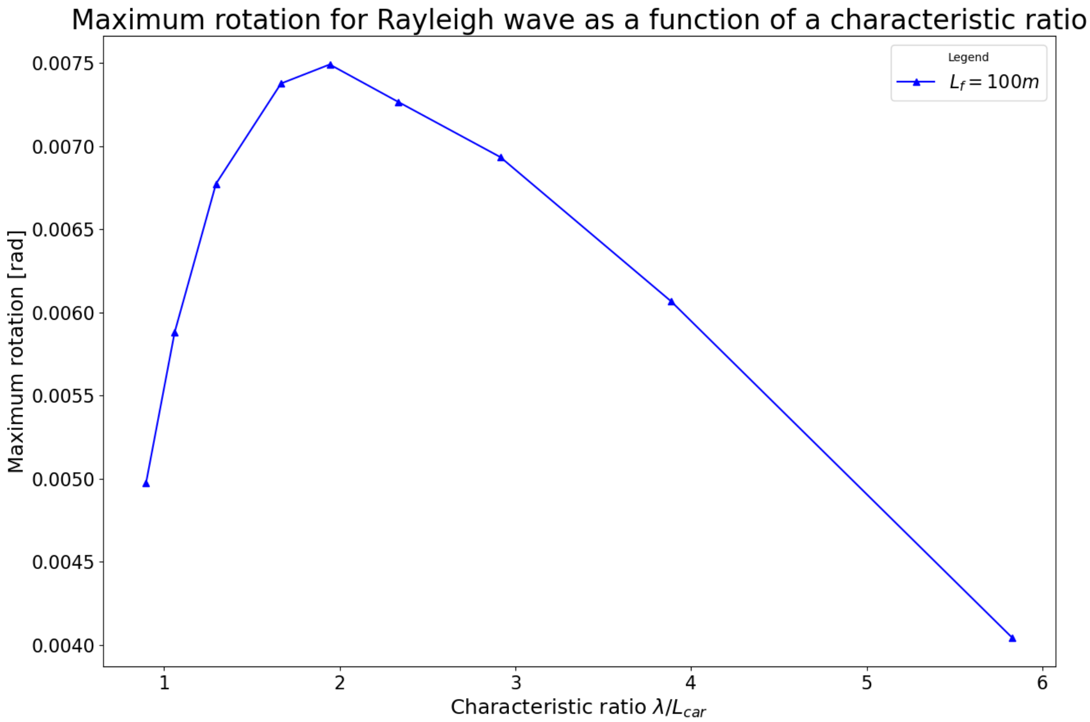

Figure 26 : Maximum rotation of the foundation as a function of λ_R/L_f.

Figure 27 shows the same function but this time each rotation has been normalized by the maximum rotation at the DRM layer for each case. Thus, the change from Figure 26 to Figure 27 is not a mere scaling of the y-axis. This time, we see that the normalized rotations are monotonously increasing with λ_R/L_f. The function is concave and seems to tend to 1. In other words, for high ratios λ_R/L_f (i.e. large wavelength and small foundation) the rotation is close to the one of the free-field. Then, as the ratio λ_R/L_f decreases (i.e. the wavelength becomes smaller or the foundation becomes larger) the normalized rotation decreases as well.

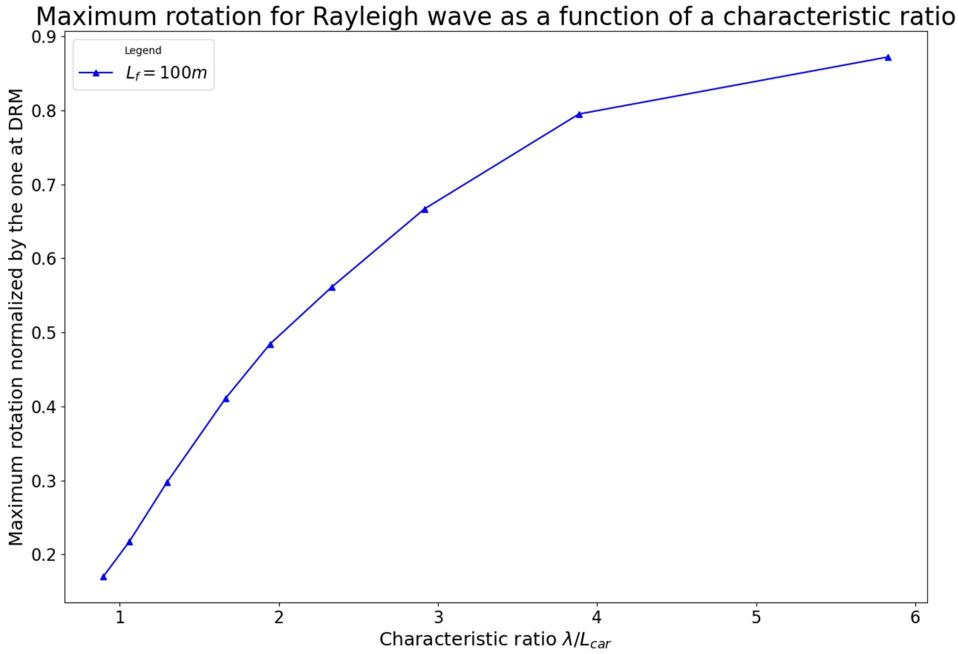

Figure 27 : Normalized maximum rotations of the foundation.

The same analysis is run in frequency domain with SASSI. Comparison of the two results in terms of the maximal rotation is shown on Figure 28. Even though there is a slight difference between the two, it does not interfere with the physical interpretation of the phenomenon as the tendencies are the same in both cases.

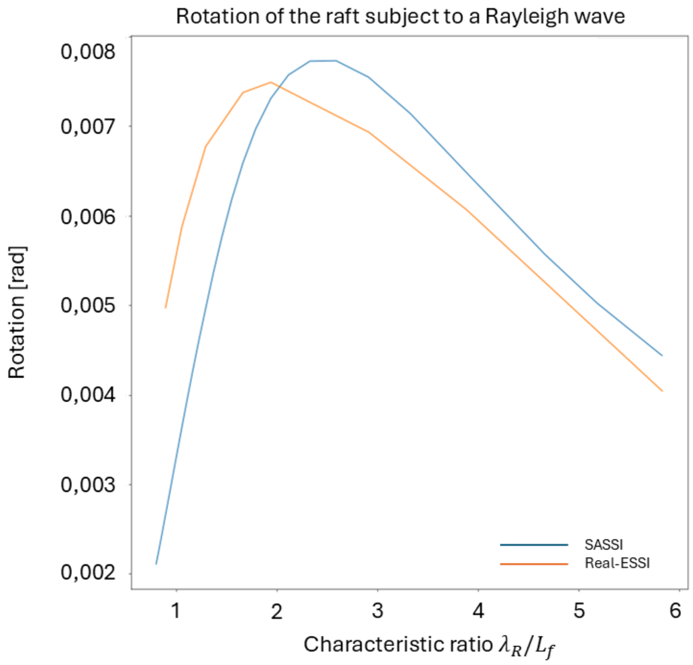

Figure 28 : Comparison between a frequency analysis (blue) and transient analysis (orange).

We plot the same comparison when the rotations are normalized by the free-field in the engineering approach (see paragraph 3.8.2). Results are shown on Figure 29. The gap between the solution of SASSI and the one of Real-ESSI is always less than 10 %. This gap could be explained by the difference in the modeling of the damping, even though we chose a small value. Indeed, damping in Real-ESSI corresponds to Rayleigh damping, whereas damping in SASSI is modeled with complex moduli.

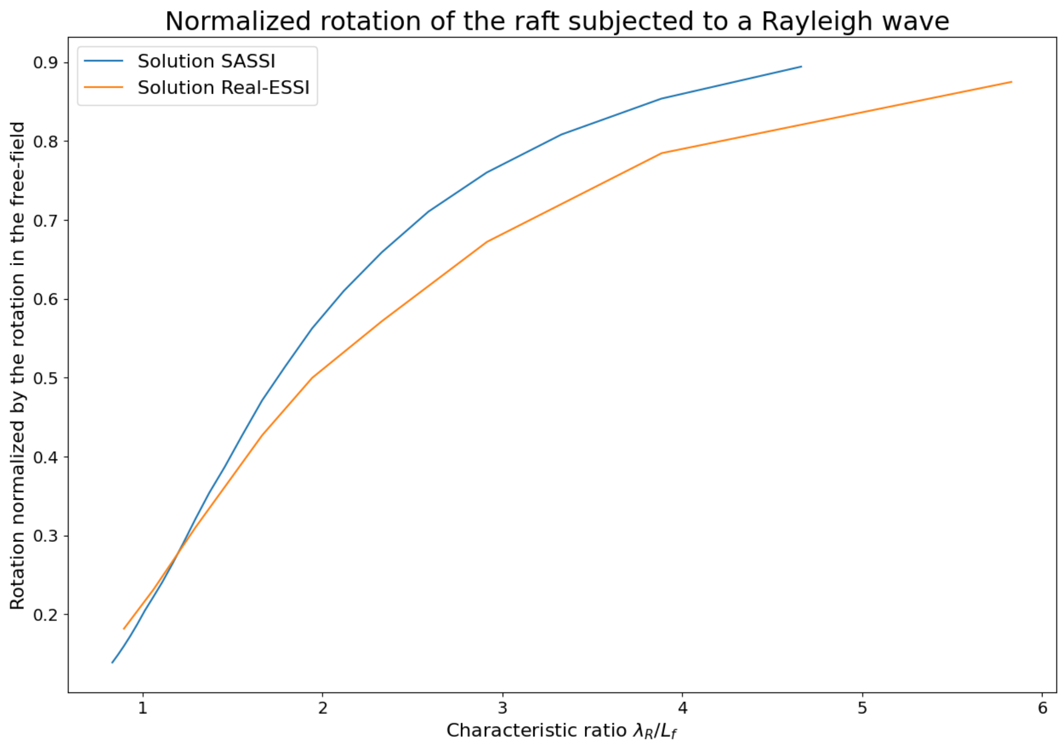

Figure 29 : Comparison between the kinematic interaction coefficients obtained from SASSI and Real-ESSI.

4.3.3. Interpretation of the results

4.3.3.1. Rotation in the free-field

Before interpreting the results, remember that the maximum slope that the foundation is subjected to is proportional to the frequency (i.e. inversely proportional to the wavelength). Figure 30 shows the maximum rotation induced by the Rayleigh wave at the soil surface under free-field conditions. It is seen that, when we reduce the frequency, we actually decrease the rotation imposed on the foundation. This because we decided to follow the figure of the "Tecdoc" and keep constant the amplitude of the vertical displacements. It is essential to keep this in mind when interpreting the results.

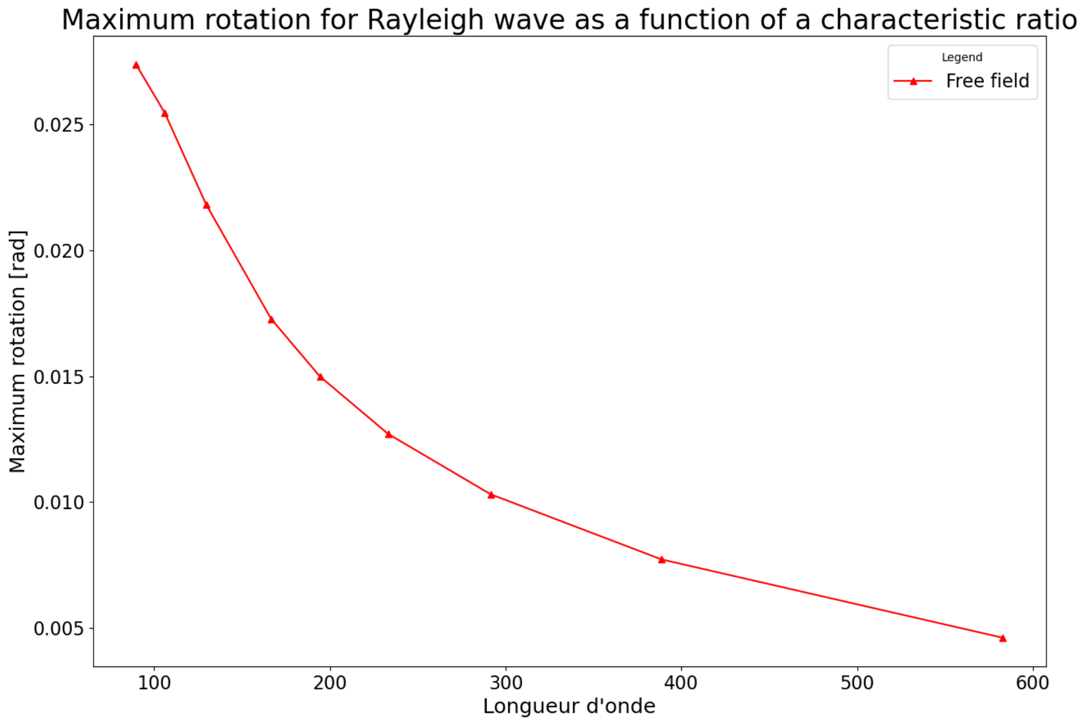

Figure 30 : Maximum rotation at the free surface in free-field conditions.

Figure 30 was obtained by measuring the rotation of the central finite element in a free-field model, i.e. a model without the foundation. The central finite element is the finite element of the free surface over which the center of the foundation would have been in a complete model. This is different from the "free-field rotation" used to normalize the graph in Figure 27. The difference between these two "free-field rotations" refers to the two approaches presented in paragraph 3.8.2.

4.3.3.2. Interpretation from a theoretical perspective

When interpreting the results, it is interesting to confront them with the predictions of the "Tecdoc".

We may conclude from Figure 26 that:

- When λ_R/L_f is small (i.e. when the wavelength is short with respect to the foundation dimension), the rotations are absorbed by the stiffness of the foundation. Indeed, the imposed rotation increases as λ_R/L_f decreases, but the measured rotations also decrease. This is coherent with the prediction of the "Tecdoc"; for short wavelengths, only local deformations may happen.
- When λ_R/L_f is large (i.e. when the wavelength is large with respect to the foundation dimension), the absolute rotations decrease because the amplitude of vertical displacement is kept constant. Thus, the absolute rotations imposed on the structure decrease as the wavelength increases. However, the normalized rotations (Figure 27) actually increase, which means that the capacity of the foundation to "absorb" the rotation via kinematic interaction is reduced. This situation is different

from the one predicted by the "Tecdoc". The movement of the foundation still comprises a rotational component, which must be treated as such and cannot be captured by an equivalent vertical motion.

- There is no amplification phenomenon when $\lambda_R/L_f = 4$. The only visible "peak" occurs around $\lambda_R/L_f = 2$ in the plot of absolute rotations (Figure 26). This peak is actually a change of regime, going from local deformations to global movement.

Figure 27 corroborates our conclusions:

- When λ_R/L_f is small, the normalized rotation (in reference to the one in the free-field) is small too. This means that the imposed motion is absorbed by the foundation.
- When λ_R/L_f is large, the normalized rotation tends towards 1. This means that all the rotation of the imposed motion is actually converted into rotation of the foundation.
- The fact that the function is monotonous proves that there is no amplification phenomenon for $\lambda_R/L_f = 4$.

The key aspect of the kinematic interaction with the surface wave is that the wave is not a standing wave, but it actually propagates on the free surface along the positive x-axis. From a geometric perspective, the foundation "sees" all the slopes of the passing Rayleigh wave. In other words, it sees all the slopes of the profile of vertical displacements at the ground surface. Thus, the bigger the foundation is compared to the wavelength, the less likely it is to "follow" the slope of the imposed motion. On the other hand, the smaller the foundation, the more likely it is to follow the slope of the surface wave. In this study, we find no amplification phenomenon for superficial structures.

In the end, concerning the kinematic interaction with Rayleigh waves, we may distinguish two regimes:

- Local deformations when $\lambda_R/L_f < 2$. The smaller the ratio, the greater the part of the imposed motion converted to local deformations.
- Global movement when $\lambda_R/L_f > 2$. The movement of the foundation is a rotation. The interaction is characterized by the capacity of the foundation to follow the slope of the seismic excitation.

N.B.: In this study, we assumed that the raft had infinite stiffness. Therefore, no local deformations may happen. We choose to keep the name "local deformations regime" nonetheless because it accounts for the physical phenomenon that happens in reality, when the raft has a finite stiffness. In our study, we could call this regime "strong attenuation regime".

Also, as mentioned in the introduction, it is unlikely that one encounters very high rotation attenuation due to kinematic interaction. Indeed, this would require the combination of a soft soil, a very large and rigid foundation and a high central frequency for the surface waves in order for the ratio λ_R/L_f to be sufficiently small.

4.3.3.3. Interpretation from an engineering perspective

In the previous paragraph, the rotation of the foundation has been compared with the one in the free-field, next to the DRM layer. In this paragraph, we compare the rotation with the one in the free-field where the center of the foundation would have been in the full model. We plot the maximum rotation of the foundation, normalized by this type of free-field rotation. The results are shown on Figure 31.

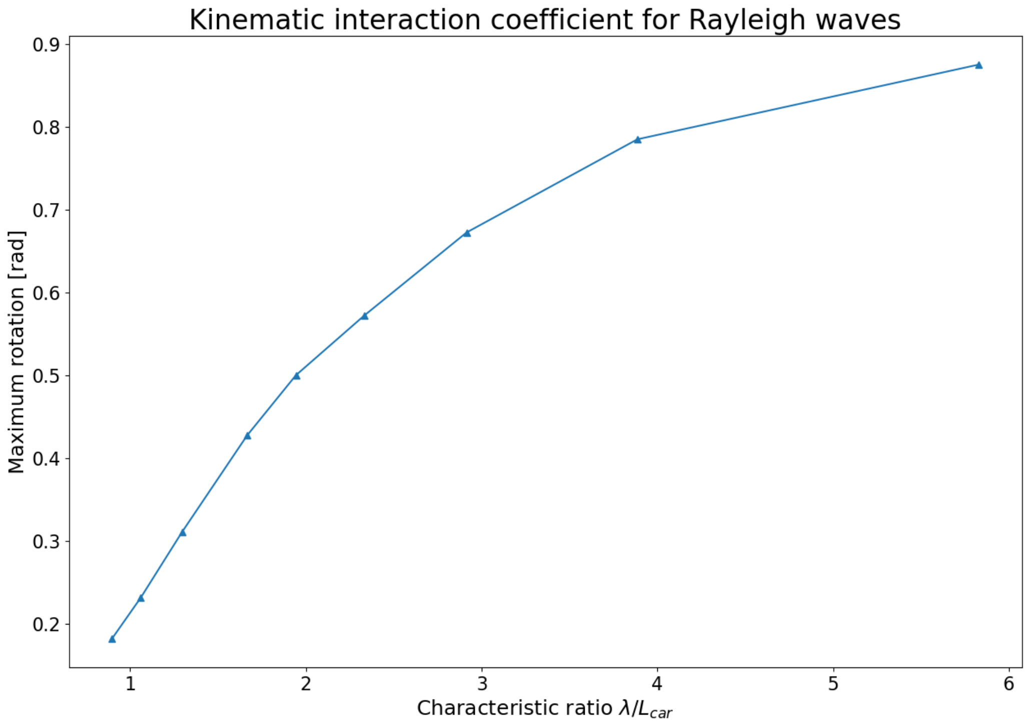

Figure 31 : Kinematic interaction coefficient.

This time, the ratio of the foundation rotation over the free-field rotation may be interpreted as a direct kinematic interaction coefficient, in other words a coefficient that gives the rotation of the foundation when multiplied by one calculated at the exact same point in the free-field. This interpretation is the one that could be used in the engineering practice. Namely, one could establish graphs for the kinematic interaction coefficient (from real or numerical experiments) under different conditions (soil stratification, soil damping, stiffness of the foundation, etc.…) to estimate the response of the structure from the response of the free-field.

In this case, the kinematic interaction coefficient follows the same trend as the normalized rotation in the theoretical perspective. The smaller the characteristic ratio is, the more the rotations are absorbed by the foundation. Also, the fact that the function is monotonous shows that there is no amplification phenomenon.

N.B.: In this study, the soil is uniform linear elastic with almost no damping. This is the reason why results plotted on Figure 27 and Figure 31 are almost identical. However, in the general case, these two graphs would be different.

4.4. Canonical curve for the effects of pure kinematic interaction with Rayleigh waves

4.4.1. Superficial structures

In the previous paragraph, we presented the results for a parametric study over the wavelength for a given length of raft. It is interesting to run the same parametric study for different lengths of the raft for the following reasons:

a. Check if the characteristic ratio λ_R/L_f is the relevant dimensionless group to study SSI problems. Indeed, the pure kinematic interaction could *a priori* be governed by any other dimensionless group (e.g. one that involves the characteristics of the soil and the seismic excitation). If λ_R/L_f is the relevant dimensionless group, we expect to draw curves that share the same trends when we vary either λ_R or L_f.

b. Cover a broader range of characteristic ratios.

Considering the fact that the problem is linear elastic and that we intend to run a lot of computations, we use SASSI for time and storage convenience (see next paragraph for computational aspects). However, calculations with Real-ESSI would give analogous results (as the comparisons of paragraph 4.3.2 show). Moreover, if one wanted to pursue such a study for nonlinear problems, the use of Real-ESSI would be required.

The central frequencies of the Ricker wavelet used range from 0,5 Hz to 2,8 Hz with a frequency step of 1 Hz. We run the computations for the following lengths of the raft: 35 m, 50 m, 65 m, 85 m, 100 m, 125 m.

We will present the results in four different forms, depending on which axes we normalize. This will help gather hindsight on the physical phenomenon. When normalizing the rotations, we use the engineering approach.

The results of this complete study, without any normalization, are shown on Figure 32. The gray dotted line shows a simple interpolation between the maxima of each curve. The conclusions from the previous paragraph are still valid when the length of the raft varies. One may identify two regimes on each curve. We also note that when the length of the raft increases, the amplitudes of rotations decrease. Moreover, the longer the raft is, the "flatter" the maximum rotation curve is. In other words, the variation of the wave regime has less impact on the response of the foundation. This is in good agreement with our geometrical interpretation where the kinematic interaction with surface waves is governed by the ability of the raft to follow the slope of the seismic excitation. Hence, the impact of the surface waves is stronger on small foundations.

The amplitudes of the Ricker wavelet have been chosen in accordance with the time history of the 1971 San Fernando earthquake ($A = 5$ cm). Thus, it is interesting to look at the rotations such a seismic event would induce in the structure. The rotations range between 5 mrad and 20 mrad. Depending on the structure under scrutiny (e.g. on its height, on the presence of sensitive equipment, etc.), such rotations may need to be taken into account to guarantee the safety of the structure.

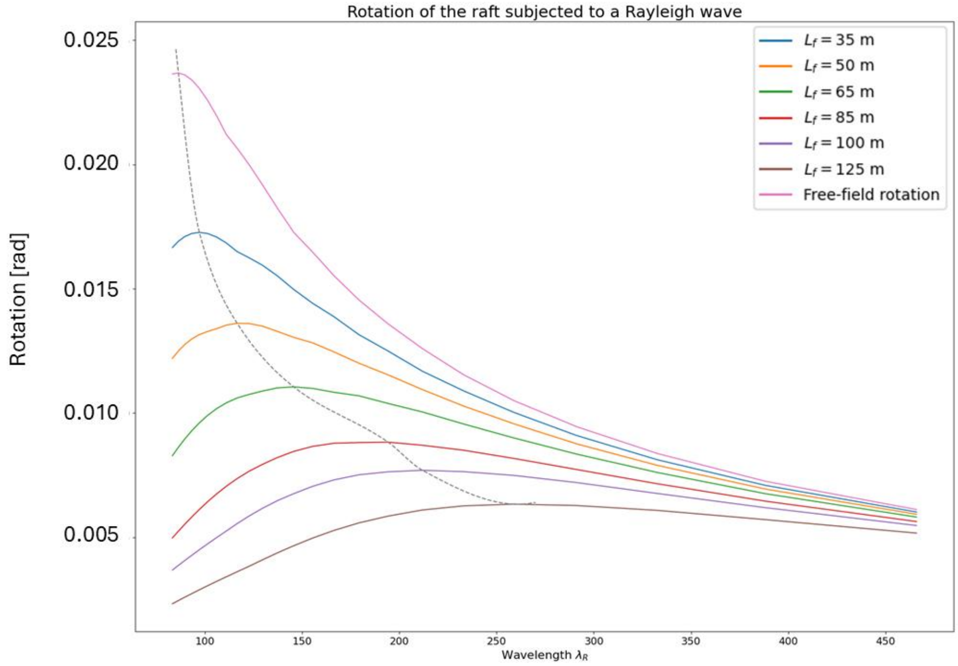

Figure 32 : Maximum rotations of the raft as a function of the characteristic ratio for different values of L_f.

Figure 33 shows the rotations as a function of the characteristic ratio. This graph highlights the fact that the peak of rotations does not occur for the same value of the characteristic ratio, but still depends on the length of the raft. However, the peak always occurs when the characteristic ratio is between 2 and 3.

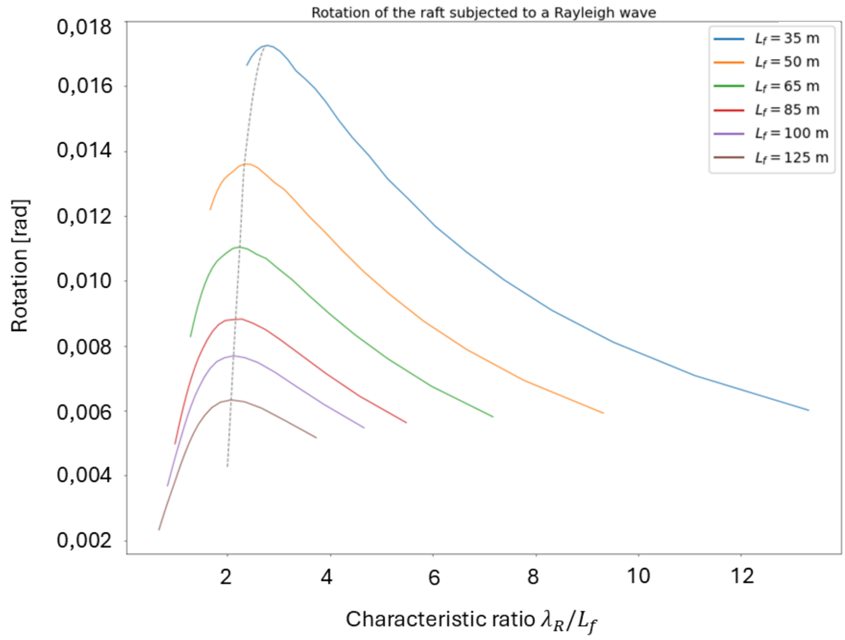

Figure 33 : Maximum rotations as a function of the characteristic ratio for different values of L_f.

Figure 34 shows the normalized rotations as a function of the wavelength. This graph shows that the normalized rotations decrease as the length of the raft increases.

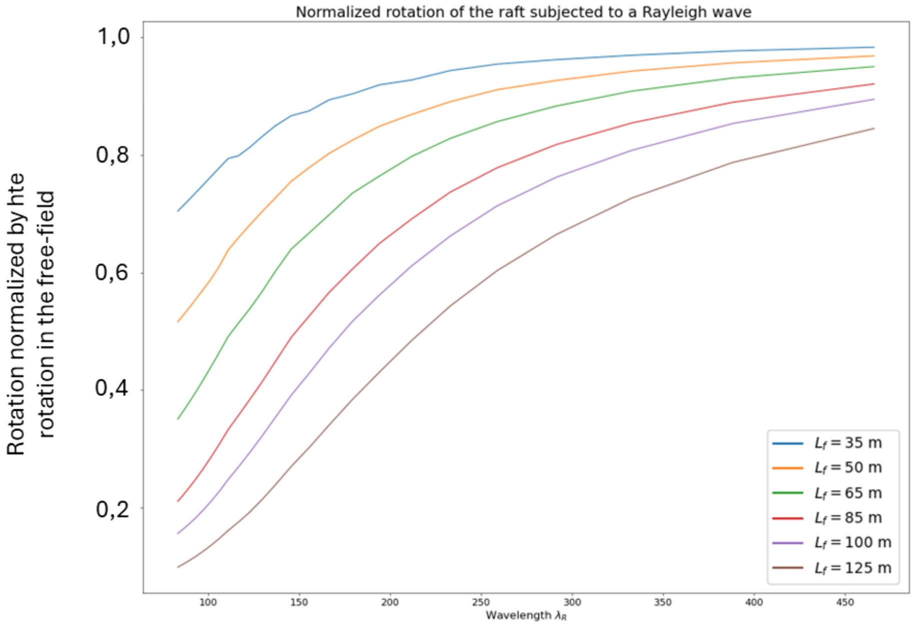

Figure 34 : Normalized rotations as a function of the wavelength.

Finally, when both axes are normalized, we obtain a "canonical" curve for the problem of pure kinematic interaction. This graph is represented on Figure 35. The normalized rotations tend to 0 for small characteristic ratios and they tend to 1 as the ratio increases. Also, there is a point of inflection when the characteristic ratio is approximately equal to 3. The canonical curve allows one to characterize the impact of surface waves on the rotation component for linear elastic SSI problems with uniform soil and a superficial raft with infinite stiffness and no mass. One could easily determine the canonical curve for more complex problems by following the methodology described in this report.

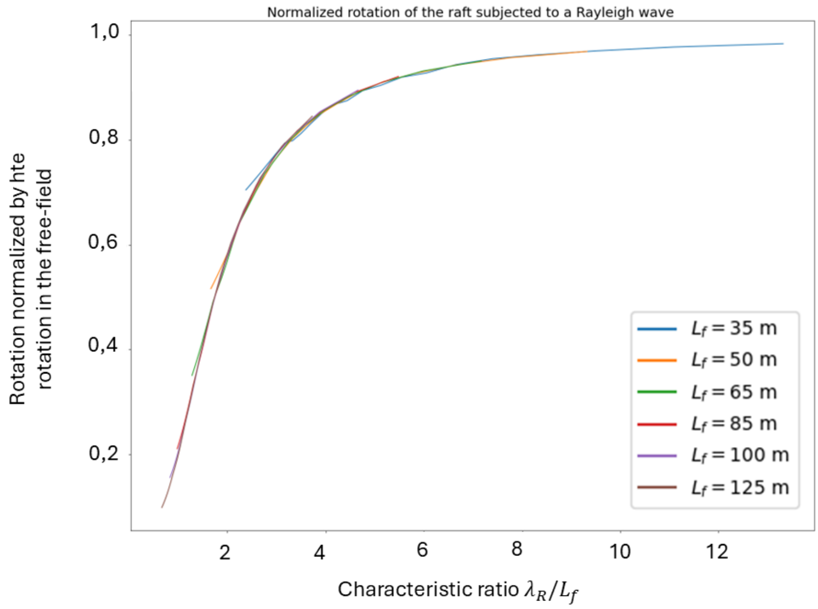

Figure 35 : Canonical curve of rotational kinematic interaction coefficient for rigid massless superficial structures subjected to a Rayleigh wave.

In this engineering practice, one could use this curve to quickly determine the kinematic interaction coefficient for a particular problem. To this end, Figure 36 shows how Figure 35 could be simplified. Indeed, the canonical curve can be approximated as a piecewise function whose branches are linear and affine. We propose the following equations to define the simplified curve:

$$\begin{cases} \theta(x) = 0{,}4x \quad \text{for } x < 2{,}7 \\ \theta(x) = 0{,}013\ (x - 2{,}7) + 0{,}88 \quad \text{for } 2{,}7 < x < 11 \end{cases} \tag{4-1}$$

The coordinates of the point of inflection are: (3,7 ; 0,84).

One could lead a particular study to reduce the imposed rotations. Other than that, the piecewise function is a practical tool to determine the kinematic interaction coefficient.

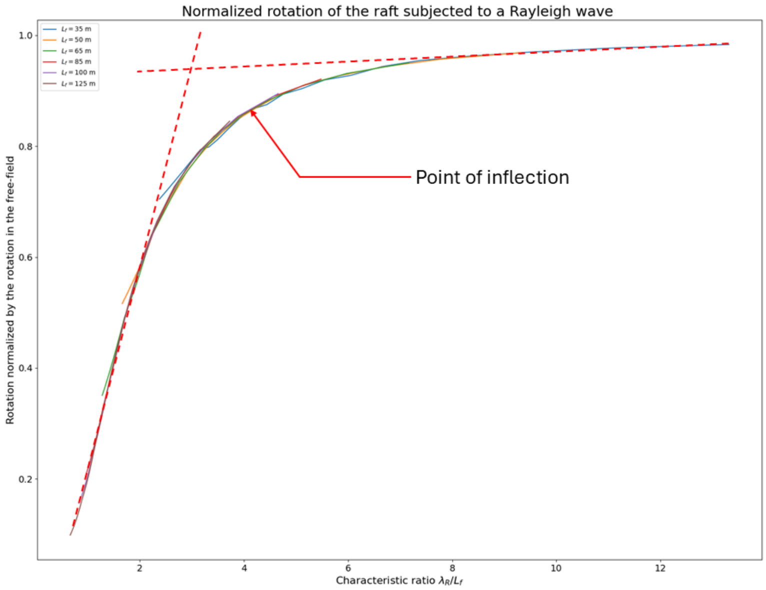

Figure 36 : Canonical kinematic interaction coefficient for the engineering practice.

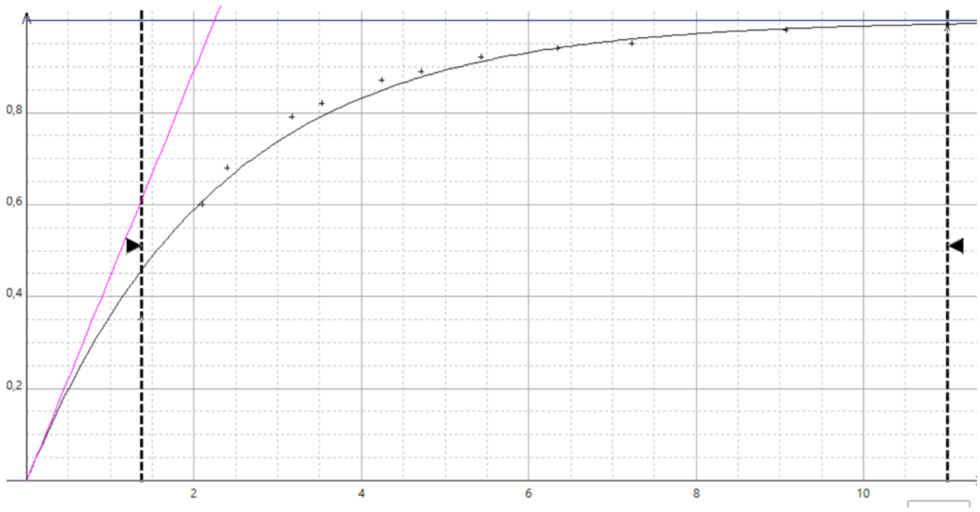

Figure 37 : Exponential approximation to the canonical curve.

To approximate the curve, we may also use an exponential function. Such an approximation is more difficult to calculate by hand but renders the properties of the rotation function in a more satisfying manner. Namely, the function is equivalent to a linear function when $\lambda_R/L_f \to 0$, tends towards 1 when $\lambda_R/L_f \to 0$ and has an inflection point. To determine the appropriate function, we sample some points on the curve and use a regression algorithm to fit the best curve. The results are presented on Figure 37. Formally, the approximation we obtain is:

$$\theta(x) = 1 - e^{-\frac{x}{2,24}} \tag{4-2}$$

The canonical curve gives the rotation of the structure for any linear elastic SSI problem with Rayleigh waves, depending on the frequency of the seismic signal. The engineering applications of this curve are very broad.

4.4.2. Embedded structures

We may draw curves similar to Figure 35 for the case of embedded structures. Indeed, the amplitude of the Rayleigh waves decays along the depth and we expect the impact of the surface waves on the rotation component to be less important. To verify this prediction, we draw canonical curves of the kinematic interaction coefficient for embedded structures. In order to draw a curve, the parameter that needs to be kept constant is the relative embedment η defined by the ratio of the embedment e_f by the length of the structure L_f. Results are presented on Figure 38.

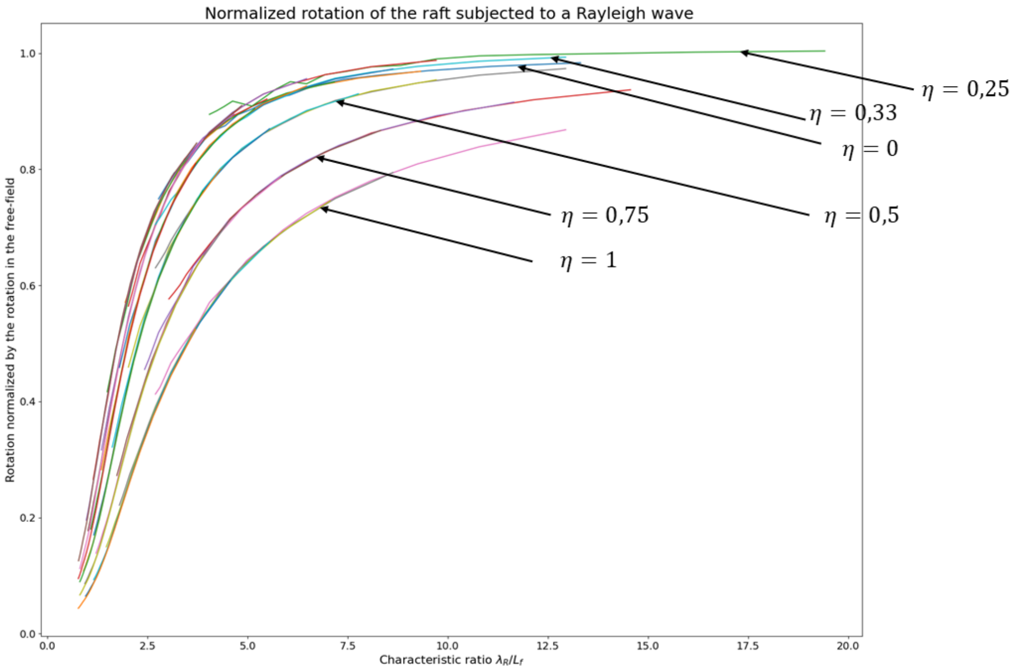

Figure 38 : Abacus for kinematic interaction highlighting the effect of relative embedment.

It is interesting to note that there seems to be a threshold effect for the canonical curves. When the relative embedment is between 0 and $0,33$ approximately, the normalized rotations remain almost the same. In other words, for low relative embedment, the embedment has almost no impact on kinematic interaction. The threshold lies between $0,33$ and $0,5$. Once the threshold is passed, the normalized rotations decrease. The attenuation of rotations through kinematic interaction for Rayleigh waves is more important for embedded structures. Also, the abscissa of the inflection point increases as the relative embedment increases.

4.5. Canonical curve for the effects of pure kinematic interaction with SV waves

As a natural extension of the previous study on Rayleigh waves, we lead the same study for vertically incident SV waves. The methodology we implemented allows one to set up the calculations for it really quickly.

For superficial structures, vertically incident SV waves correspond to a mere horizontal displacement signal. For embedded structures however, the displacements along the wall of the embedded structures vary with the depth. This difference creates a global rocking movement. Thus, we focus on the case of embedded structures. The seismic wave is convolved with a horizontal Ricker wavelet which has the same characteristics as before. The approach is the same as previously thus we only present the canonical curve. However, keep in mind that since the normalization is not the same as for Rayleigh waves (see paragraph 3.8.2), the same name "canonical curve" does not refer to the exact same concept. We study the response of foundations whose relative embedment is kept constant equal to 1 as shown on Figure 39.

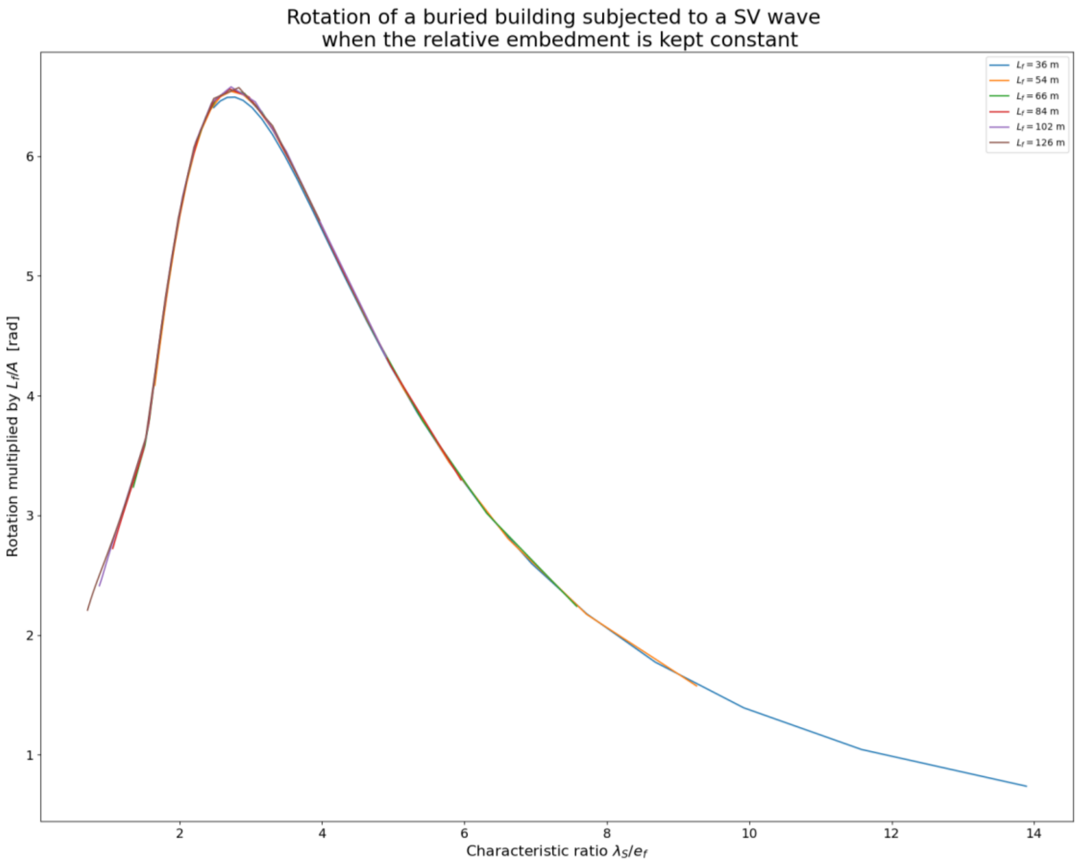

Figure 39 : Canonical curve for SV waves for relative embedment $\eta = 1$.

From Figure 39, we see that the rotations follow the same trends with respect to the characteristic ratio as for Rayleigh waves (in terms of increasing and decreasing but not in terms of rate of increase or decrease). The peak occurs for $\lambda_S/e_f = 2{,}75$.

The interpretation of the canonical curve for SV waves is the same as the one for R waves. When the structure is too large compared to the wavelength (i.e. λ_S/e_f is small), the rotations are attenuated by kinematic interaction. The apparent peak does not necessarily represent an amplification phenomenon, as it may only be caused by the fact that the imposed rotation decreases when the wavelength increases.

As an example, consider a vertically propagating SV wave whose wavelength is $\lambda_S = 250$ m and whose amplitude is $A = 5$ cm and a buried building whose dimensions are $L_f = 25$ m and $e_f = 25$ m. Thanks to the canonical curve, it is straightforward determine the maximum rotation generated by pure kinematic interaction effects : $\theta_{max} =$ 3 mrad.

4.6. Relative contribution of R and SV waves to the rocking motion of embedded structures

Once the canonical curves are obtained, it is interesting to compare the relative contribution of each wave type to the rocking motion. Since it is common to neglect the contribution of R waves to the rotation for embedded structures, one may be interested in knowing what percentage of the rotation has been discarded by doing so. To compare the rotations for the two wave types, we consider a given geometry with relative embedment equal to 1. Then we proceed to the same parametric study as before and we retrieve the maximum of rotation for all the computed frequencies. This corresponds to one point the curves of contributions. Then we repeat this operation for different geometries (by keeping $\eta = 1$) and get several points to draw the curves. We also draw the same contribution curves when the length of the structure is fixed ($L_f = 66$ m) and the embedment e_f progressively increases.

To determine the contribution of each wave type, we define the total rotations as the sum of the maximum rotations caused by each wave type. These total rotations then serve as a reference to calculate the percentage of each wave type. There are two approaches to compare the contribution of each wave type. First, we may consider that the R wave and the SV wave have the same reference signal (they do not share the same direction though), in particular the same amplitude. From a theoretical point of view, this provides insight on the impact these waves have on the response of the structure. Second, we may consider that the two waves do not have the same amplitudes. Indeed, for a real seismic event the amplitude of the SV wave may be several times greater than the one of the Rayleigh wave and vice-versa. We use values of the 1971 San Fernando earthquake. The horizontal amplitude is 11 cm for SV waves and the vertical amplitude is 8 cm for R waves. The results are presented in the form of two graphs, on the left when the relative embedment is kept constant and on the right when the length is kept constant. For theoretical amplitudes, the results are presented on Figure 40. For realistic amplitudes, the results are presented on Figure 41.

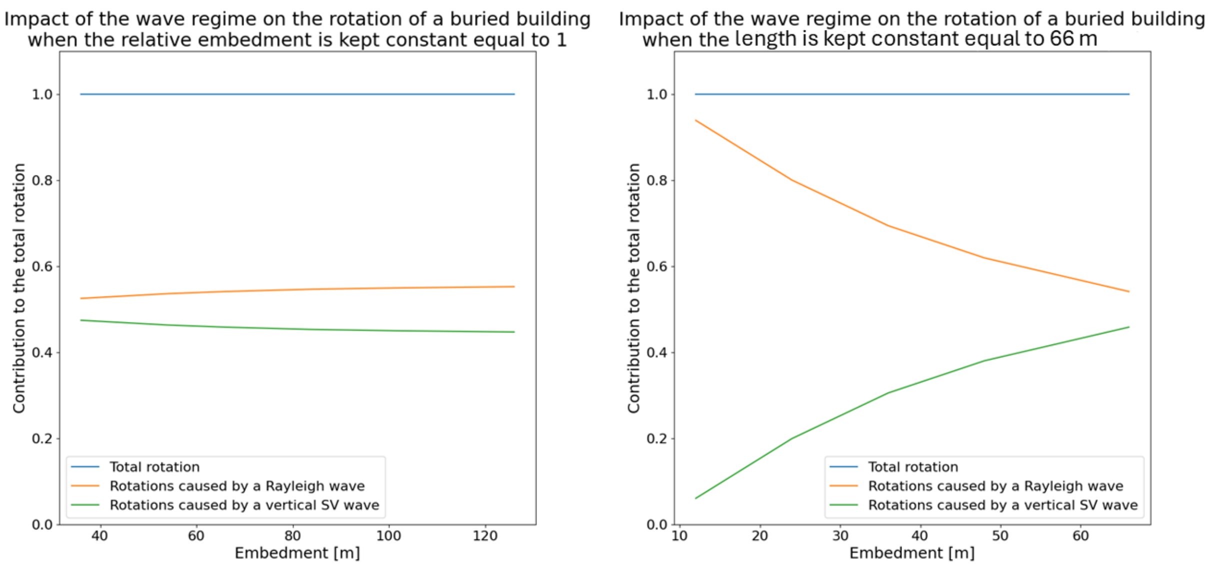

Figure 40 : Relative contributions of the two wave types to the rocking motion for theoretical amplitudes.

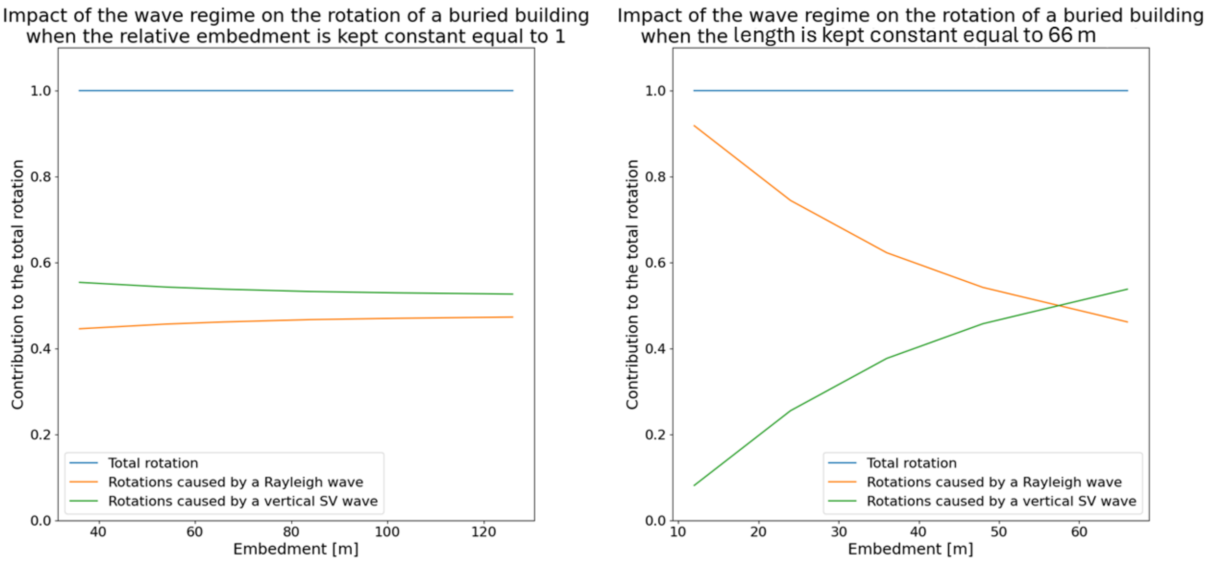

Figure 41 : Relative contributions of the two wave types to the rocking motion for realistic amplitudes.

On these graphs we see that the contribution of the Rayleigh waves to the rotation decreases when the embedment increases and that the contribution of the SV follows the opposite trend. This is in agreement with our expectations. However, we may also see that the contribution of Rayleigh waves is almost always more important than the one of the SV waves. On Figure 41, the two curves intersect when the embedment is 57 m (recall that the length of the structure is $L_f = 66$ m).

In the presence of a surface wave, both superficial and embedded structure will experience induced rocking. As the embedment of the structure increases, the rocking induced by Rayleigh waves tends to decay whereas the rocking induced by SV waves tends to increase. However, depending on the relative amplitudes and frequency content of Rayleigh and SV waves, the rotation due to Rayleigh waves may still be prevalent even for embedded structures.

4.7. Application of the methodology to nonlinear problems

4.7.1. Presentation of the uplift problem

We stated that our methodology was designed to address general (nonlinear nonelastic) problems. However, until now our calculations remained in the linear elastic domain, in order to fully understand the impact of kinematic interaction on the rocking motion. To provide a case study in which our methodology is extended to nonlinear problems, we briefly consider the uplift problem of a raft under seismic excitation, namely a Rayleigh wave injected via the DRM.

For real SSI problems, the contact between the raft and the soil surface is not perfect. Indeed, sliding or uplift may happen. This loss of contact and this movement generate second order effects on the displacement of the structure. Here, we address the problem of uplifting. In essence, this problem is nonlinear because at each time step the response of the foundation is affected by the uplift at the previous one. Our objective is to determine for what amplitude uplift occurs. To do so, we run a parameter study by varying the amplitude of the wavelet and observe when the uplift occurs. We study the uplift ratio, which is defined as the number of open contact elements divided by the total number of contact elements.

The problem is solved in two steps:

1. We solve the problem under self-weight loading (nonlinear static analysis).
2. We introduce the seismic signal at the DRM boundary and solve the nonlinear dynamic problem in time domain.

Some characteristics of the simulations are listed below:

- We replace the perfect contact between the raft and the soil surface (shared nodes) by contact elements that have no resistance to tensile forces and a constant stiffness for compressive forces (equal to 100 times the stiffness of a finite soil element). The two ends of the contact element are coincident nodes.
- We add a vertical rigid beam (a stick) with an oscillating mass on top of it to represent a building.
- The Rayleigh wave is convolved with a Ricker wavelet. The vertical amplitude of the Rayleigh wave varies from 1 cm to 7 cm, by steps of 2 cm.

More details on the parameters of this simulation and on contact elements are given in Appendix 5.

4.7.2. Validation of the initial state

We first need to make sure that the initial state after solving the static self-weight yields satisfying results. The settlement of the model under self-weight is represented on Figure 42. This displacement profile matches our expectations as the soil settles uniformly.

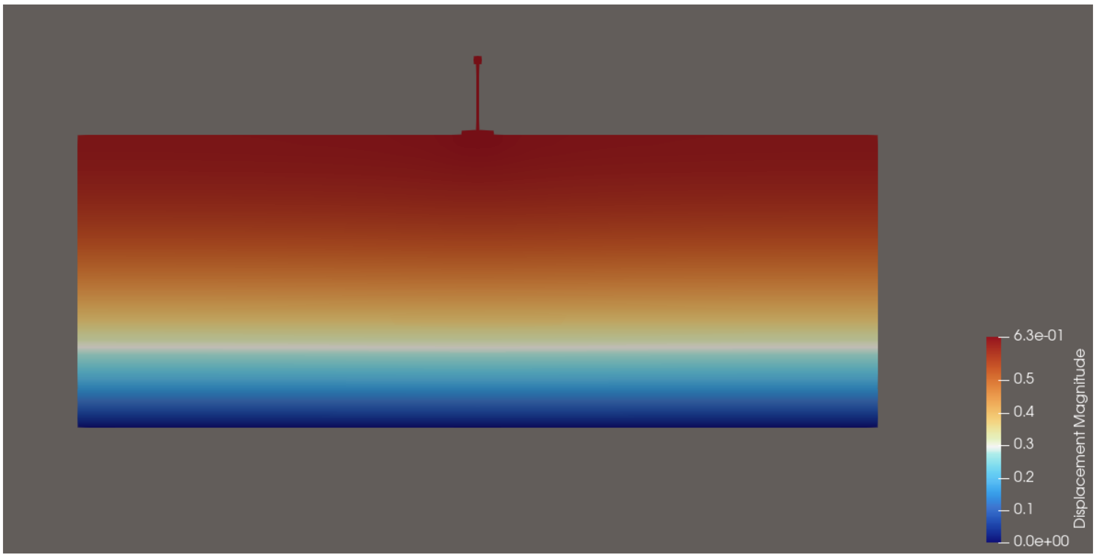

Figure 42 : Settlement under self-weight.

N.B.: Because the boundary conditions are roller supports, the settlement of the soil under self-weight is large. Thus, it is difficult to see on the color map the contribution of the structure to the settlement under self-weight.

We check the stress profile at the interface between the raft and the soil surface. Since the raft is rigid, we may compare our results with one of the analytical solutions provided by Borowicka (1939). The results are presented on Figure 43 and show good agreement between the two stress profiles, given the spatial discretization.

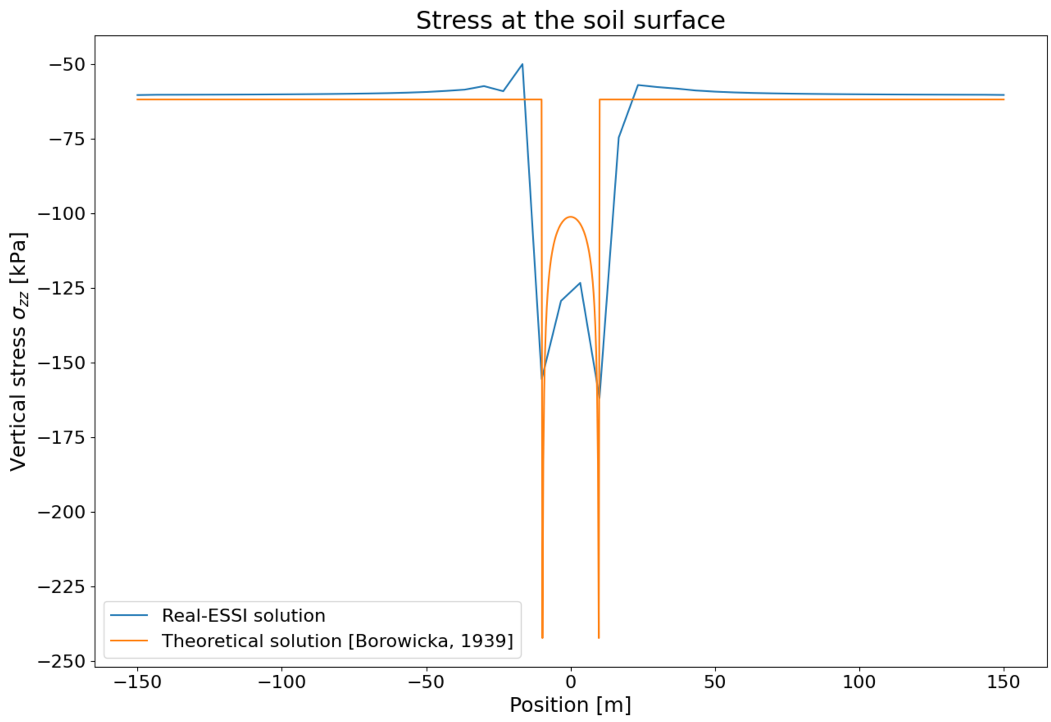

Figure 43 : Stress profile under the raft : comparison between numerical results and the analytical solution.

From these considerations, we conclude that the initial state is valid and that the model is ready for the dynamic analysis.

4.7.3. Results of the time domain analysis

By examining the interpretation of the contact elements, one is able to draw a map of uplifted areas as a function of time. The final results might be saved as a GIF. Snapshots of the uplift map are shown on Figure 44. The gray dotted line has ordinate 0. The contact elements are open when the interpenetration is positive. We notice that the negative interpenetration (contact) is greater (in absolute value) than the positive interpenetration (uplift). This might be an indicator that the penalty stiffness is not high enough (see also Appendix 5).

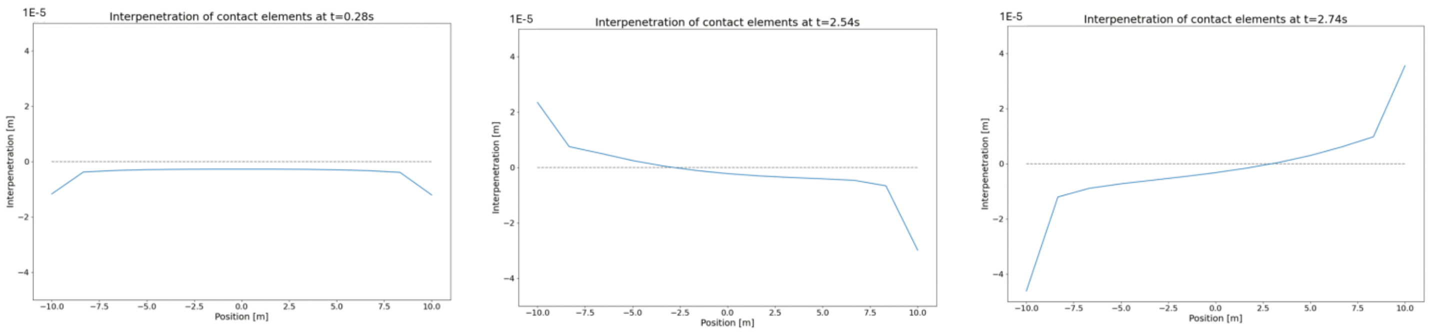

Figure 44 : Snapshots of the uplift map.

We may also plot the uplift ratio as a function of time. The results are presented on Figure 45. The uplift occurs for amplitudes greater than 2 cm. For the Rayleigh wave amplitudes studied here, the uplift ratio remains under 50 %.

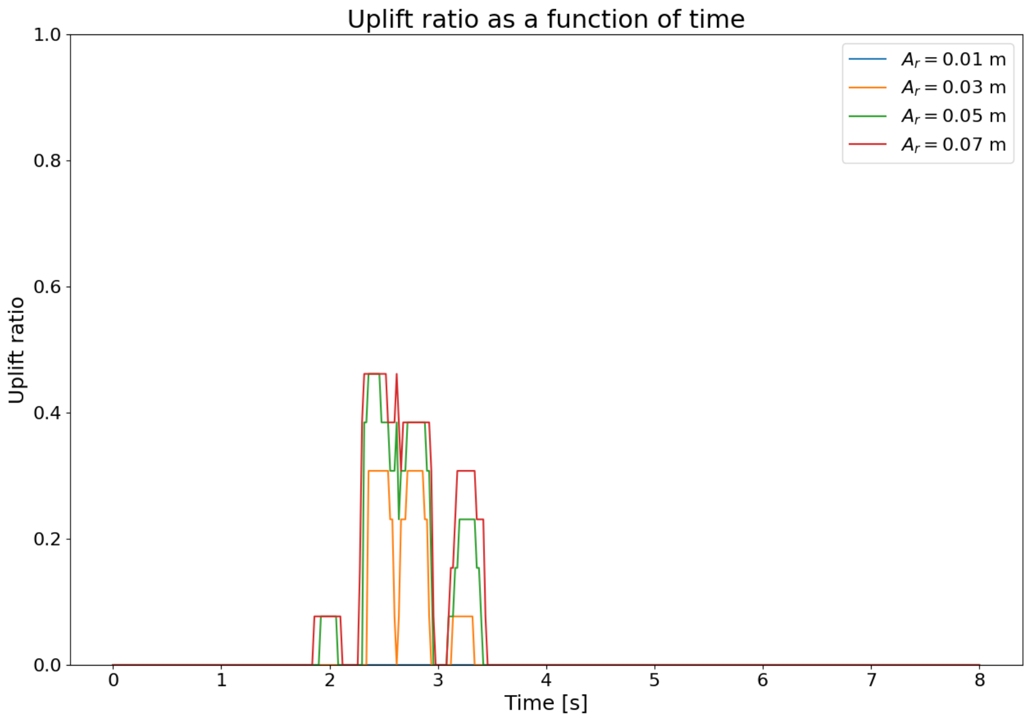

Figure 45 : Uplift ratio as a function of time.

The generation of input files took 8 minutes. This is because one needs to calculate the response in the free-field at every DRM node with SASSI. Then, the software took approximately 1h30 to produce the results (indeed the model is smaller in this case, see Appendix 5). These calculations have demonstrated the relevance of our methodology for nonlinear calculations.

5. Conclusions and prospects

5.1. Conclusions

During this internship, we developed a comprehensive and flexible methodology to study SSI problems. This methodology allows one to study the effect of nonlinear elements along with a full 3D wave regime. In particular, working with the TLM in the frequency domain to determine the free-field response is very efficient. Then, the coupling with the DRM allows us to study the full 3D SSI problem. Furthermore, our methodology does not only apply to isolated cases. On the contrary, one can usually set up parametric studies to deeply understand the impact of each parameter on the SSI problem in a theoretical manner. Moreover, from an engineering perspective, we believe that this methodology allows one to easily plot charts to provide safety coefficients for the SSI problem. Also, thanks to this methodology, one can input pure surface waves, without generating them *via* the reflection of body waves. This allows one to gather fruitful hindsight on the impact of surface waves on the response of structures. Thus, in further studies, one may examine the response of structures subjected to inclined body waves and to surface waves and understand the contribution of each component of the wave regime to the response.

To demonstrate the power of our methodology, we studied the kinematic interaction between a raft with infinite stiffness and no mass and a Rayleigh wave. More precisely, we sought to quantify the rotations induced by the surface wave and compare these results with the predictions of IAEA (2022). Thanks to our results, we were able to provide a clear understanding and a quantification of the physical phenomena that have been qualitatively anticipated in the "Tecdoc".

The key findings of this parametric study are listed below:

- Decomposing arbitrarily a seismic signal onto different components and injecting them into the soil domain might not lead to the expected results.
- The effects of a surface wave cannot be adequately reproduced by "equivalent" body waves.
- Strong attenuation due to the kinematic interaction is almost unattainable in real life.
- The global movement regime of the kinematic interaction is governed by the capacity of the foundation to follow the slope of the seismic excitation.

We were able to plot the canonical curve for elementary kinematic interaction problems with a Rayleigh wave for superficial structures. This canonical curve allows one to determine the kinematic interaction coefficient for any elementary kinematic interaction problem. We were also able to draw such canonical curves when the structure is embedded.

Then, we performed the same study with a vertically incident SV wave. We compared the results with the ones of the R waves and found that even for embedded structures, R waves may contribute more to the rocking motion than vertical SV waves.

Finally, we demonstrated the comprehensiveness of our methodology by tackling a simple nonlinear problem in which we studied the uplifting of a structure founded on a surface raft and subjected to a pure Rayleigh wave.

5.2. Prospects

From a theoretical perspective, the following step would be to further dwell on the kinematic interaction for more complex models. Indeed, we pointed out that the traditional use of body waves is not sufficient to reproduce the response of the structure to a surface wave. Some parameters that it would be interesting to study are:

- Detachment of the foundation (uplift as well as sliding with more complex contact element laws).
- A more complex structure (in order to generate second order rotations and to study the stress inside the structural elements).
- Consideration of nonlinear soil behavior near the structure.
- Obliquely incident body waves.

In complementary calculations not presented in this report, we also observed that the soil profile (i.e. the soil stratification) has a major impact on the propagation of surface waves, and thus on the response of the structure. Quantifying the impact of the soil profile and providing adequate normalization schemes would prove useful in drawing canonical curves for realistic soils.

From an engineering perspective, two major problems need to be addressed. First, it would be useful to determine the canonical curve for more complex situations, in the same spirit as we did for elementary problems. Second, one could systematically assess the gap between the regulations based on equivalent body waves and the real response of the structure with a full 3D wave regime. To achieve this, one could use the time signals from the database of research project MODULATE. Indeed, one could compare the response of the structure when subjected to an equivalent *P+SV* combination and the response for a *R+(P+SV)* combination, with the time history for Rayleigh being given by MODULATE. Our methodology allows one to run such calculations as any type of wave and any combination of waves may be injected into the FEM model thanks to the TLM. This study could also be extended to obliquely incident body waves. The ultimate objective of this approach is to provide charts with margin coefficients to help implement engineering guidelines for realistic situations.

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Appendix

Appendix 1 : Naming and styling conventions

We follow these conventions to write mathematical variables:

- To represent *scalars*, we use a plain italic font: a.
- To represent **vectors**, we use a bold font: $\mathbf{a}$.
- To represent ***tensors*** of order greater than or equal to two, we use a bold italic font: $\boldsymbol{A}$.

The system of axis used is represented on . The waves propagates along the positive x-axis. The z-axis is vertical downwards and the y-axis is the out-of-plane axis.

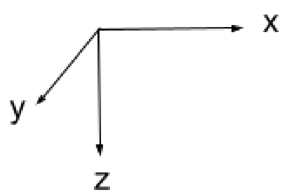


Figure 46 : System of axes used in this internship.

Appendix 2: Excel workbook to input parameters

In this appendix, we present a view of all the sheets used in the workbook to parametrize the SSI problem.

N.B.: It is possible to give two different time discretization on the "SASSI" and the "Numerical" sheets, but this is not recommended as the first one will be scaled to fit the second one during the time domain analysis.

Name of layer	Layer thickness [m]
Layer n°1	300
Layer n°2	0
Layer n°3	0

Total number of layers	1
Total depth [m]	300

Figure 47 : "Soil profile" sheet.

Material name	Mass density [kg/m³]	Velocity of *P*-waves [m/s]	Velocity of S-waves [m/s]	First Lamé coefficient [MPa]	Second Lamé coefficient [MPa]	Young's modulus [MPa]	Poisson's ratio	Shear modulus [MPa]	Bulk modulus [MPa]
Raft	0	N/A	N/A			3500000000	0		
Oscillating mass	770	N/A	N/A			210000	0,2		
Layer n°1	2100	500	250	263	131	350	0,33	131	350
Layer n°2	2100	1500	750	2363	1181	3150	0,33	1181	3150
Layer n°3	2500	2082	1000	5834	2500	6750	0,35	2500	7500

Pre-process materials

Figure 48 : "Materials" sheet.

Name of the parameter	Symbol	Value [m]		
Length of the foundation	L_f	100		
Height of the foundation	h_f	10	Mass of the foundation	0 kt
Length of the soil domain	L_s	600		
Thickness of DRM layer	e_drm	3		
Lateral thickness of absorbing layers	e_al_lat	30		
Thickness of the three deepest absorbing layers	e_al_inf	25		
Width of the post	L_p	5		
Height of the post	h_p	15		
Dimensions of the oscillating mass	r_m	10	Mass of the oscillating mass	0,077 kt
Height of the mass	h_m	10	The center of the oscillating mass is fixed to the top of the pole	
Length for nonlinear area	L_nl	0		
Height of nonlinear are	h_nl	0		
Nodal mass (kt)	m_nod	0	*Automatically distributed by Python	
Mesh size	M_S	6		
Stiffness of contact elements (MN/m)	K_N	0	Recommended stiffness	17500 MN/m
Embedment of the foundation	e_f	0		

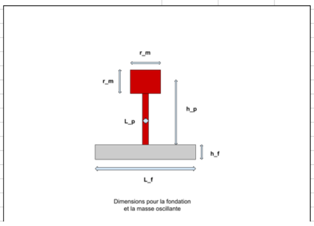
If the stiffness is zero, coincindent nodes will be modeled

Where to inject the wave	Boundary conditions
Partout	Encastrement

Generate journal file

Generate soil profile file

Convert mesh to fei format



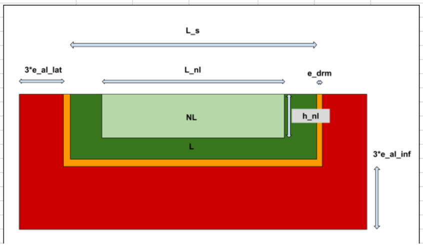


Figure 49 : "Geometry" sheet.

Name of the parameter	Symbol	Value	Units
Number of computed frequencies	NFREQ	50	
Number of points in the frequency domain	NFFT	512	
Time step	dt	0,02	s
Multiplicative coefficient for the frequecies of body waves	A_vol	5	
Multiplicative coefficient for the frequecies of surface waves	A_surf	1	
Point3 radius	r	5	m

Frequency step	df	0,09765625	Hz
Total duration	tf	10,2400	s

Computed frequencies for body waves		
First frequency	0,488	Hz
Last frequency	24,414	Hz

Computed frequencies for surfaces waves		
First frequency	0,098	Hz
Last frequency	4,883	Hz

Wave type to inject		
Rayleigh wave	☐	FAUX
P wave	☑	VRAI
SV wave	☑	VRAI
Incidence angle for body waves	0	

Figure 50 : "SASSI" sheet.

Rayleigh wave input through DRM			
Name of parameter	Symbol	Value	Units
Multiplicative coefficient for amp	A	5	m^2
Frequency	f	0,5	Hz
Oscillation duration	t_e	12	s

Displacements for a harmonic source in depth			
Name of parameter	Symbol	Value	Units
Amplitude	A	0,01	m
Frequency	f	1	Hz
Absciss of source	x	500	m
Ordinate of source	z	-500	m
Wave type		P	
Inclination of wave plane	alpha_p	0	°
Incident angle	alpha_o	45	
Compensation time	t_comp	0	s

What method to use ?
SASSI

Maximum frequency covered by the mesh [Hz]
5,2

Pre-process wave
+ SASSI computation if necessary

Ricker wavelet			
Name of parameter	Symbol	Value	Units
Central frequency	fc	0,7	Hz
Time of peak	t0	2	s
Amplitude	A	0,05	m
Breadth	tp	1,11	s

Wave defined with a potential			
Name of parameter	Symbol	Value	Units
Potential magnitude	A	2	m²
Frequency	f	5	Hz
Wave type		P	
Inclination of wave plane	alpha_p	0	°
Incident angle	alpha_o	80	°
Corresponding displacement amplitude for the first layer	|u|	0,126	m

$$|u| = \frac{\phi\omega}{V}$$

Figure 51 : "Wave" sheet.

General characteristics			
Name of parameter	Symbol	Value	Units
Initial time	t_i	0	s
Final time	t_f	8	s
Time step	dt	0,02	s
Extraction step	N_e	1	

Number of time steps	N	400

Newmark method	
beta	0,25
gamma	0,5

HHT	
alpha	-0,25

What method to use ?
Newmark

Pre-process numerical solution

Generate summary

Figure 52 : "Numerical parameters" sheet.

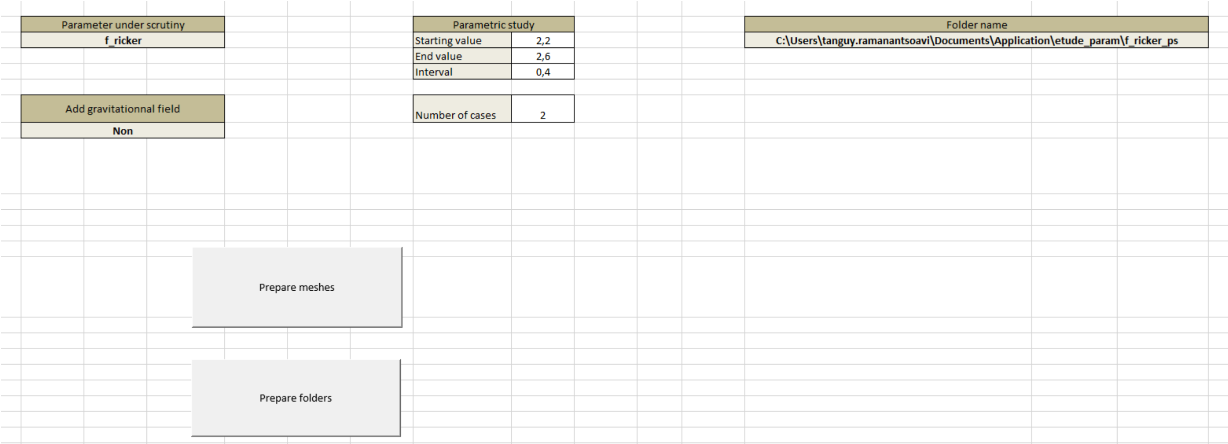


Figure 53 : “Parametric study” sheet.

Appendix 3 : Details on Newmark's integration scheme

This appendix relies on Argyris and Mlejnek (1991).

The objective of transient integration algorithms is to solve the problem:

$$\boldsymbol{M}\ddot{x} + \boldsymbol{C}\dot{x} + \boldsymbol{K}x = \boldsymbol{F} \tag{C-1}$$

The problem is discretized in time with a time step Δt. The original scheme proposed by Newmark (1953) is:

$$\begin{cases} \ddot{x}_{t-\Delta t} = \boldsymbol{M}^{-1}[\boldsymbol{F} - \boldsymbol{C}\dot{x}_{t-\Delta t} - \boldsymbol{K}x_{t-\Delta t}] \\ \dot{x}_t = \dot{x}_{t-\Delta t} + \Delta t[(1-\gamma)\ddot{x}_{t-\Delta t} + \gamma\ddot{x}_t] \\ x_t = x_{t-\Delta t} + \Delta t\dot{x}_{t-\Delta t} + \dfrac{\Delta t^2}{2}[(1-2\beta)\ddot{x}_{t-\Delta t} + 2\beta\ddot{x}_t] \end{cases} \tag{C-2}$$

At each step, we compute the acceleration from the equation of equilibrium. Then the velocity and the displacement are computed thanks to equation (C-2). This procedure is in general implicit (except when $\beta = 0$ and $\gamma = 0{,}5$ and $\boldsymbol{M}$ and $\boldsymbol{C}$ are diagonal). Parameters β and γ control the stability, the numerical dissipation and the precision.

In this formulation, if the mass matrix is zero, the algorithm will fail inevitably since it will not be able to invert the mass matrix. However, this formulation takes the acceleration as the primary variable. We may also use the displacement as the primary variable. Thus, we seek to eliminate $\ddot{x}_t$ and $\dot{x}_t$ from the equations of movement:

$$\begin{cases} \ddot{x}_t = \dfrac{1}{\beta\Delta t^2}(x_t - x_{t-\Delta t}) - \dfrac{1}{\beta\Delta \mathrm{t}}\dot{x}_{t-\Delta t} + (1 - \dfrac{1}{2\beta})\ddot{x}_{t-\Delta t} \\ \dot{x}_t = \dfrac{\gamma}{\beta\Delta t}(x_t - x_{t-\Delta t}) + \left(1 - \dfrac{\gamma}{\beta}\right)\dot{x}_{t-\Delta t} + \Delta t(1 - \dfrac{\gamma}{2\beta})\ddot{x}_{t-\Delta t} \end{cases} \tag{C-3}$$

We then replace these equations in the equation of equilibrium at time t :

$$\begin{aligned} \left[\mathbf{K} + \frac{1}{\beta\Delta t^2}\mathbf{M} + \frac{\gamma}{\beta\Delta t}\mathbf{C}\right]x_t \\ = \mathrm{F_t} + \mathbf{M}\left[\frac{1}{\beta\Delta t^2}x_{t-\Delta t} + \frac{1}{\beta\Delta \mathrm{t}}\dot{x}_{t-\Delta t} - \left(1 - \frac{1}{2\beta}\right)\ddot{x}_{t-\Delta t}\right] \\ + \mathbf{C}[\frac{\gamma}{\beta\Delta t}x_{t-\Delta t} - \left(1 - \frac{\gamma}{\beta}\right)\dot{x}_{t-\Delta t} - \Delta t(1 - \frac{\gamma}{2\beta})\ddot{x}_{t-\Delta t}] \end{aligned} \tag{C-4}$$

In this formulation, the matrix to be inverted is $\left[\mathbf{K} + \frac{1}{\beta\Delta t^2}\mathbf{M} + \frac{\gamma}{\beta\Delta t}\mathbf{C}\right]x_t$ which is non-zero even if $\mathbf{M}$ (and $\boldsymbol{C}$) are zero. This is the formulation used in Real-ESSI (Jeremić *et al.*, 2021).

N.B.: The equation derived is different from the one in Argyris and Mlejnek (1991). Nevertheless, it does not change the conclusion that the mass can be zero.

Appendix 4 : Details on the Ricker wavelet

We compute the velocity of particles subjected to a Ricker wavelet:

$$v_0(t) = A[-4\pi^4 f^4 (t-t_0)^3 + 6\pi^2 f^2 (t-t_0)]e^{-\pi^2 f^2 (t-t_0)^2} \qquad (D\text{-}1)$$

And the acceleration:

$$a_0(t) = A[8\pi^6 f^6 (t-t_0)^4 - 24\pi^4 f^4 (t-t_0)^2 + 6\pi^2 f^2]e^{-\pi^2 f^2 (t-t_0)^2} \qquad (D\text{-}2)$$

Let us calculate the maximum speed. We solve for $t : a_0(t) = 0$. To find the zeros of the acceleration, we use a standard bi-square equation technique:

$$\begin{cases} a_0(t) = A[8\pi^6 f^6 X^2 - 24\pi^4 f^4 X + 6\pi^2 f^2]e^{-\pi^2 f^2 (t-t_0)^2} \\ X = (t-t_0)^2 \end{cases} \qquad (D\text{-}3)$$

The four solutions are:

$$t = t_0 \pm \frac{1}{\pi f}\sqrt{\frac{6 \pm \sqrt{33}}{16}} \qquad (D\text{-}4)$$

(the two $\pm$ signs are independent of one another). The wavelet is symmetrical around t_0, so we may only consider the two following solutions:

$$t = t_0 + \frac{1}{\pi f}\sqrt{\frac{6 \pm \sqrt{33}}{16}} \qquad (D\text{-}5)$$

Then we just need to determine which solution corresponds to a maximum. To this end we compute the velocity for the two solutions:

$$\begin{cases} v_0\left(t = t_0 + \frac{1}{\pi f}\sqrt{\frac{6 + \sqrt{33}}{16}}\right) \approx 3{,}98Af \\ v_0\left(t = t_0 + \frac{1}{\pi f}\sqrt{\frac{6 - \sqrt{33}}{16}}\right) \approx 2{,}32Af \end{cases} \qquad (D\text{-}6)$$

Thus, the maximum is reached for $t = t_0 + \frac{1}{\pi f}\sqrt{\frac{6+\sqrt{33}}{16}}$ and the maximum vertical speed is:

$$v_{0,max} = Af\left[-4\pi\left(\frac{6+\sqrt{33}}{16}\right)^{\frac{3}{2}} + 6\pi\left(\frac{6+\sqrt{33}}{16}\right)^{\frac{1}{2}}\right]e^{-\left(\frac{6+\sqrt{33}}{16}\right)} \qquad (D\text{-}7)$$

Appendix 5 : Details on the modelling of the uplift problem

To run the nonlinear calculations, we slightly adjust the parameters presented in section 3. We reduced the size of the model for the calculations to be faster. Indeed, they are a proof of concept and do not require as much precision as the ones for canonical curves. All the geometric parameters are listed below :

- Length of the foundation : $L_f = 20$ m.
- Height of the foundation : $h_f = 3$ m.
- Embedment of the foundation : $e_f = 0$ m.
- Width of the vertical rigid beam (stick) : $L_p = 2$ m.
- Center of gravity of oscillating mass : $h_p = 40$ m.
- Mass density of oscillating mass : $\rho_p = 770$ kg/m^3.
- Length of the soil domain : $L_s = 300$ m.
- Height of the soil domain : $h_s = 100$ m.

The soil is uniform linear elastic and has the same mechanical properties as presented in section 3. The raft also has the same mechanical properties. The absorbing layers also have the same dimensions.

To solve the static problem, we use Newton's method with 25 loading increments.

The law of the contact element is represented on Figure 54. The penalty stiffness is chosen to be 100 times the one of a finite soil element. Thus, since the mesh size is 6 m and the depth of the model is 1 m, the penalty stiffness of the contact element is $k_c = 17500$ MN/m.

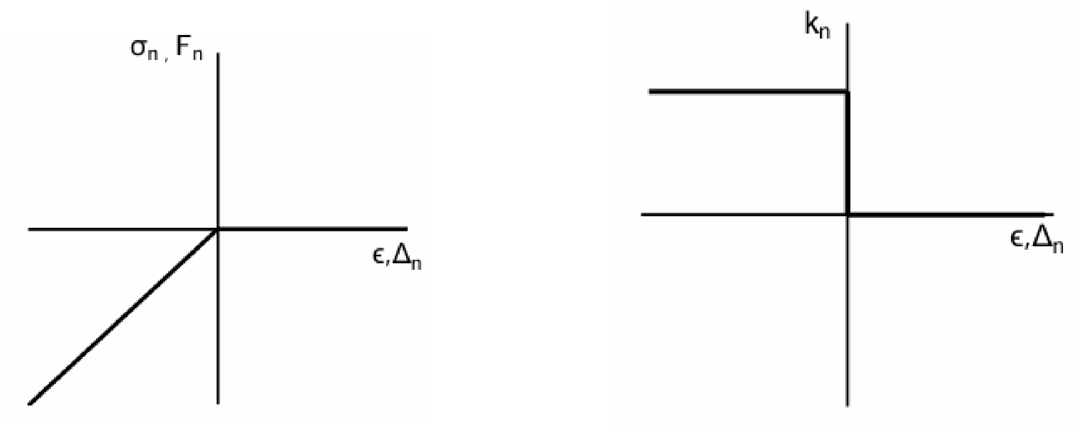


Figure 54 : Behavior law for the contact element – Figure extracted from Jeremić *et al.* (2021).

Appendix 6 : 1971 San Fernando earthquake

This appendix presents the time history of the 1971 San Fernando earthquake, in terms of displacement. Body waves and surface waves have been separated thanks to the methodology of Meza-Fajardo (2015). These time histories are extracted from the database of research project MODULATE.

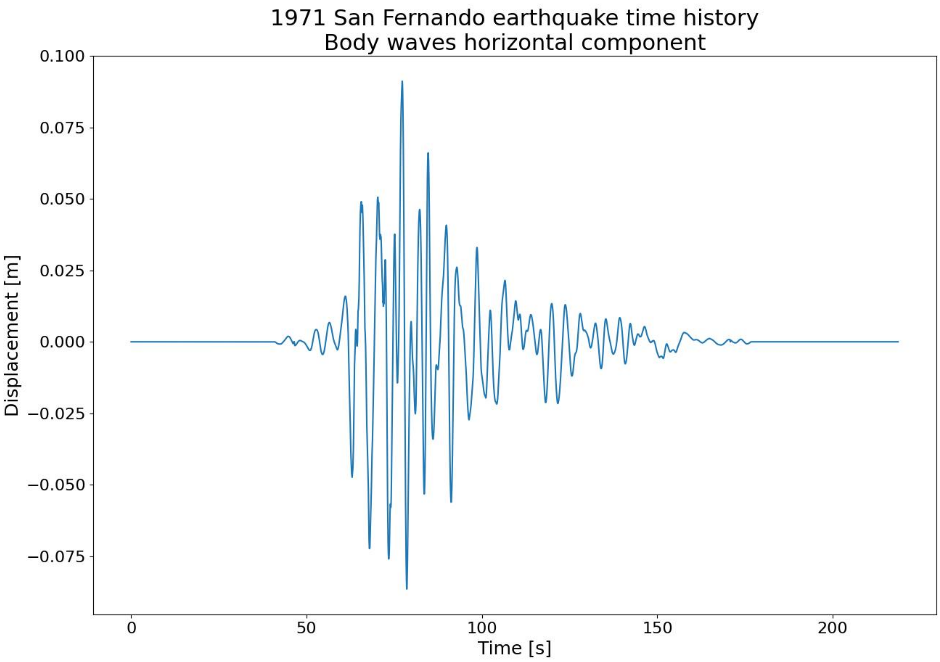


Figure 55 : Body waves horizontal displacement component.

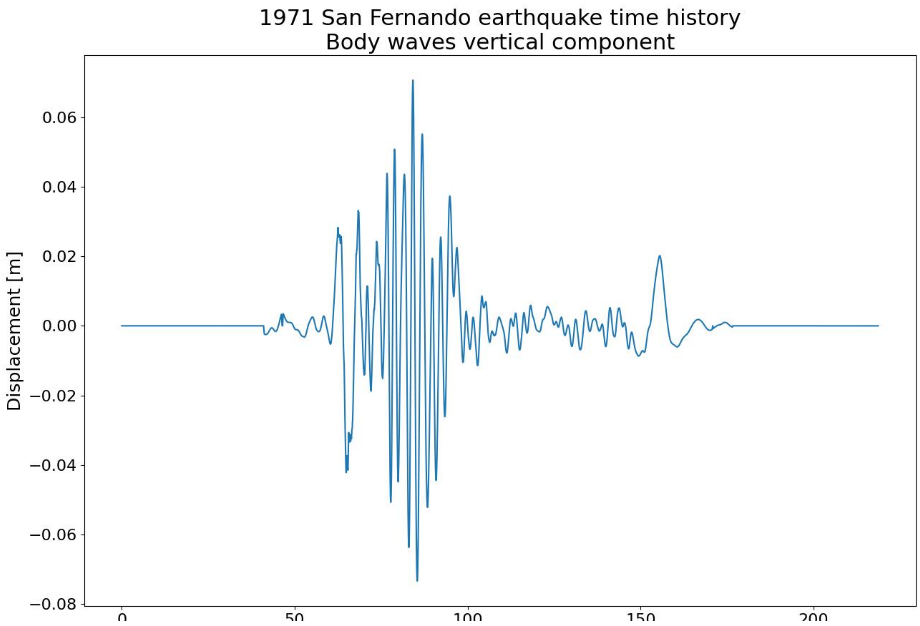


Figure 56 : Body waves vertical displacement component.

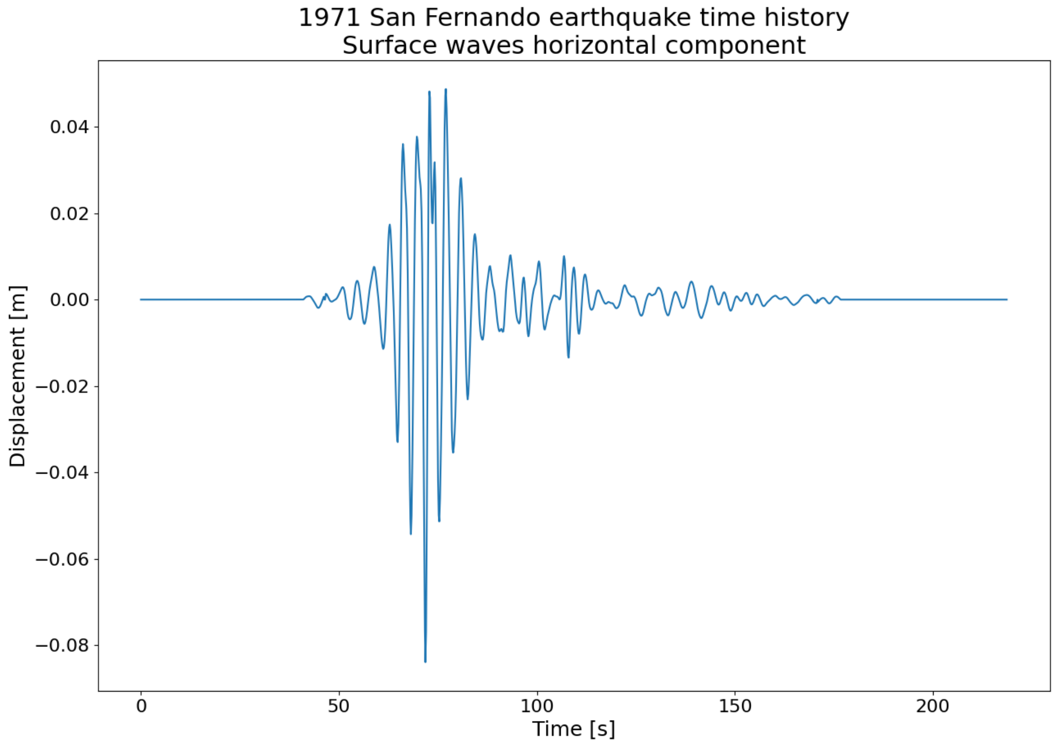


Figure 57 : Surface waves horizontal displacement component.

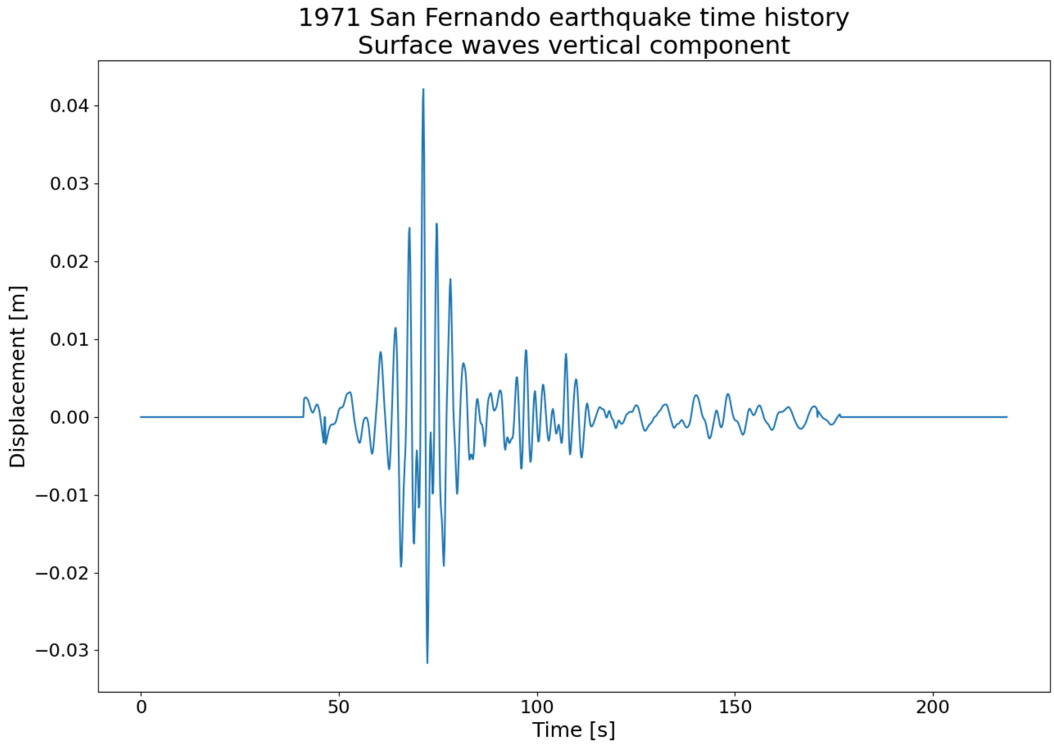


Figure 58 : Surface waves vertical displacement component.